L STRAUSS - MATHEMATICS FOR PHYSICAL SCIENCE

LÖTZ STRAUSS

MATHEMATICS FOR PHYSICAL SCIENCE

MATHEMATICS FOR PHYSICAL SCIENCE

Lötz Strauss
D Sc, THED, L Akad, Sci Nat

Professor, Physics Department
University of Pretoria

Published by the author

Contents

Preface

Mathematics for Physical Science contains all the mathematical tools required for the study of introductory physical science at first year university or college level. It is intended mainly for students who require a practical knowledge of elementary differential and integral calculus and also elementary vector algebra and vector analysis. In writing the book, the requirements for courses in biological sciences were also taken into account.

Although the mathematical concepts are dealt with fairly systematically, the accent is on the development of the student's skill in the use of mathematical techniques in the description of theories in physical science and the solution of problems. In order to attain this goal, the text contains a large number of problems which are worked out in full detail to aid weaker students. At the end of each chapter is a variety of problems of which the answers are supplied at the end of the book.

The prerequisite for the use of this textbook is the successful completion of a mathematics course at high school level. Most of chapter 1 and much of chapter 2 is a revision of work which is normally included in school courses and is intended to stress those topics which are most important for understanding the rest of the course.

Most of this text is presented as an introductory course to Physics 1 at the University of Pretoria. This introductory course is usually covered in about 18 lectures. The students are not required to be able to prove the mathematical deductions which are treated, but they are required to understand the meaning of the concepts and apply them skilfully. In the physics courses for which this textbook is intended, differential and integral calculus and also vector algebra and simple vector analysis are used. The student is required to know definitions and understand physical principles and then use the mathematical tools to solve problems. The final examinations for these courses consist of problems only. Remembering formulas and their derivation do not form a part of the final examination.

Some aspects are treated at a later stage in the physics courses when the need arises. Line integrals are introduced when the concept of work is treated. Experience has shown that students should encounter the concept of a **differential equation** as soon as possible. This enables one to use the techniques of differential and integral calculus at a more elegant level in the solution of problems.

My heartfelt thanks to my dear wife for proofreading at level one (correcting her husband's appallingly poor English spelling and grammar). The most difficult task was the detailed checking of the manuscript which is a translation of the original version in Afrikaans. This immense task was performed by my dear and respected friend Etienne Malherbe. For the elimination of errors missed by the other readers, the final draft was scanned by Danie Steyn, another dear friend who was deeply involved in the preparation of the original Afrikaans version. Sareta van Tonder made the index. My sincere thanks to her for this task. (Now the reader knows who to blame if a concept cannot be traced through the index!)

My thanks to my Maker for the privilege, opportunity and ability to plan and accomplish this task. *Soli Deo Gloria!*

Department of Physics *Lötz Strauss*
University of Pretoria
January 1994

Chapter 1

INTRODUCTION

1.1 Important symbols and constants

$=$	is equal to	$\neq$	is not equal to
$\approx$	is approximately equal to	$\equiv$	is identical to, definition of
$\sim$	is of same order of magnitude as	$\propto$	is proportional to
$>$	is greater than	$\gg$	is much greater than
$<$	is less than	$\ll$	is much less than
$\geq$	is greater than or equal to	$\leq$	is less than or equal to
$\because$	because	$\therefore$	therefore
$\Rightarrow$	this implies, from this follows	$\rightarrow$	tends to
∞	infinitely large		

Δx or δx a change in x. Δx = (final value of x) - (initial value of x)

$\lvert a\rvert$	the absolute value (modulus) of a
	$\lvert a\rvert = a$ if $a > 0$ (a real, positive)
	$\lvert a\rvert = 0$ if $a = 0$
	$\lvert a\rvert = -a$ if $a < 0$ (a real, negative)
$n!$	n-factorial, $n! = n(n-1)(n-2)(n-3)\ldots 5.4.3.2.1$
	$0! = 1$ by definition
π	(circumference of a circle)/(diameter of the circle) = 3,14159265...
e	base of natural (Napierian) logarithms = 2,71828183...
$\sum_i^n x_i$	sum of all the x_i values $= x_1 + x_2 + x_3 + \ldots\ldots + x_n$

1.2 The Greek alphabet

A	α	alpha	N	ν	nu
B	β	beta	Ξ	ξ	xi
Γ	γ	gamma	O	o	omicron
Δ	δ	delta	Π	π	pi
E	ϵ	epsilon	P	ρ	rho
Z	ζ	zeta	Σ	σ	sigma
H	η	eta	T	τ	tau
Θ	θ	theta	Υ	υ	upsilon
I	ι	iota	Φ	ϕ	phi
K	κ	kappa	X	χ	chi
Λ	λ	lambda	Ψ	ψ	psi
M	μ	mu	Ω	ω	omega

1.3 Exponents and logarithms

The following rules are valid for real numbers:

$$\begin{aligned}
a^n &= a \times a \times a \times a \ldots\ldots n \text{ times. } n \text{ is a positive integer.} \\
a^{-n} &= 1/a^n \\
a^{1/n} &= \sqrt[n]{a} \\
a^0 &= 1 \\
(ab)^p &= a^p \times b^p \\
a^p \times a^q &= a^{p+q} \\
a^p \div a^q &= a^{p-q} \\
a^{p/q} &= \sqrt[q]{a^p} \quad = \quad (\sqrt[q]{a})^p \\
(a^p)^q &= a^{pq}
\end{aligned}$$

If $a > 0$ and $a^n = b$, then $b > 0$ for all real values of n. In this relationship a is called the **base**, b is called a power (the n-th power in this case) of a and n is the **exponent** or **logarithm** to which a has to be raised to give b. If n is to be made the subject of the formula, it is written as follows: $n = \log_a b$. Another way in which b may be written as the subject of the formula is $b = \text{antilog}_a n$ which is equivalent to $b = a^n$ in all respects. By writing the relationship $a = b^{1/n}$ the roles of the base and the power are exchanged. Since $1/n = \log_b a$, it follows directly that $\log_a b = 1/(\log_b a)$.

The following are different ways of expressing one and the same relationship between the numbers a (the base), n (the exponent or logarithm) and b (the power or antilogarithm):

$$b = a^n = \text{antilog}_a n \qquad a = b^{1/n} = \text{antilog}_b (1/n) \qquad n = \log_a b$$
$$= n\text{-th power of } a \qquad = n\text{-th root of } b \qquad = (\log_b a)^{-1}$$

Before calculators were available, one had to make use of tables of logarithms and antilogarithms to perform difficult calculations involving multiplication, division, exponentiation and the extraction of roots. For this purpose logarithms to the base 10 were used. To save time, the base was omitted.

$$\text{If } r = \log s, \text{ it means that } s = 10^r$$

Logarithms to the base $e = 2,71828\ldots$ are called **natural logarithms** or **Napierian logarithms**. If $y = \log_e b = \ln b$, then $b = e^y$. The notation $\ln b$ (pronounced "lin-b") is the conventional way for writing natural logarithms.

The rules which express calculations involving logarithms, are just a different way to express those which apply to exponents. With P, Q and a positive numbers and $a \neq 1$, the following rules are valid:

$$\begin{aligned}
\log_a P^n &= n \log_a P \\
\log_a \sqrt[n]{P} &= \tfrac{1}{n} \log_a P \\
\log_a 1 &= 0 \\
\log_a 1/P &= -\log_a P \\
\log_a (P \times Q) &= \log_a P + \log_a Q \\
\log_a P/Q &= log_a P - \log_a Q \\
\log_a P &= (\log_b P) \times (\log_a b) \quad = \quad (\log_b P) \div (\log_b a)
\end{aligned}$$

in which also $b > 0$ and $b \neq 1$.

1.4 The binomial theorem

For small positive integer values of n, the expansion of $(a+b)^n$ may be done directly by the rules which apply to the multiplication of algebraic expressions, e.g.

$$\begin{aligned}
(a+b)^0 &= 1 \\
(a+b)^1 &= a+b \\
(a+b)^2 &= a^2 + 2ab + b^2 \\
(a+b)^3 &= a^3 + 3a^2b + 3ab^2 + b^3 \\
(a+b)^4 &= a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4 \qquad \text{etc.}
\end{aligned}$$

For n not a positive integer (or a large positive integer) the expansion is calculated by using the **binomial theorem**.

For n a **positive integer** we have

$$\begin{aligned}(a+b)^n &= a^n + na^{n-1}b + \frac{n(n-1)}{2!}a^{n-2}b^2 + \frac{n(n-1)(n-2)}{3!}a^{n-3}b^3 + \\ &\quad \frac{n(n-1)(n-2)(n-3)}{4!}a^{n-4}b^4 + \ldots\ldots + b^n \qquad 1.4(1)\end{aligned}$$

which has $n+1$ terms. Equation 1.4(1) may be written in a shorter way as

$$(a+b)^n = \sum_{r=0}^{n} \binom{n}{r} a^{n-r}b^r \qquad 1.4(2)$$

in which

$$\binom{n}{r} = \frac{n!}{r!(n-r)!} \qquad 1.4(3)$$

is known as the **binomial coefficient** of term number $r+1$.

Equation 1.4(1) also holds for negative and fractional values of n, provided that $|b| < |a|$ or $|b/a| < 1$ in which case the expansion has an infinite number of terms which form a convergent series.

In physics the binomial expansion is often used for the case where $a = 1$ and n not necessarily an integer or positive. For the condition that $|b| < 1$, it is an infinite series as follows:

$$(1+b)^n = 1 + nb + \frac{n(n-1)}{2!}b^2 + \ldots$$

If b is much less than unity, the first two or three terms give such a good approximation that the higher powers may by diregarded.

1.5 Factors

The reader should be familiar with the following identities which occur in the study of physics.

$$\begin{aligned}a^2 - b^2 &= (a-b)(a+b) \\ a^4 - b^4 &= (a^2-b^2)(a^2+b^2) = (a-b)(a+b)(a^2+b^2) \\ a^3 \pm b^3 &= (a \pm b)(a^2 \mp ab + b^2) \\ ax^2 + bx + c &= a(x-x_-)(x-x_+) \quad \text{in which} \quad x_\pm = \frac{-b \pm \sqrt{b^2-4ac}}{2a}\end{aligned}$$

1.6 Radian measure

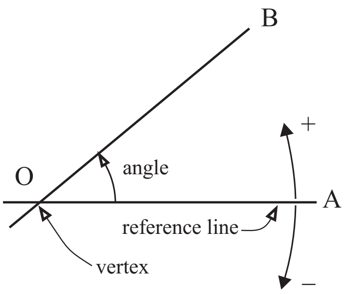


Figure 1.6-1

An angle measures rotation. Consider straight lines OA and OB in Figure 1.6-1. If OA is a fixed line, the angle AOB measures the amount of rotation of OB from the position of OA about the fixed point O to the position indicated in the sketch. The rotation occurs in the plane which contains OA and OB. If the rotation is counter-clockwise, it is taken as positive and if it is clockwise, it is negative. Observers viewing the rotation from opposite sides, will assign different signs to one and the same angle. The possibility of ambiguities is not excluded.

An angle thus specifies the direction of one straight line with respect to another. The two lines are called the **arms of the angle** and their point of intersection, the **vertex**.

The simplest unit in which an angle may be measured, is the revolution. A popular unit is the **degree**. In one revolution there are 360 degrees (360°). A quarter of a revolution (90°) is called a **right angle**. The degree is subdivided into 60 **minutes** (60′) and each minute into 60 seconds (60″). Most pocket calculators have a function for the conversion of degrees, minutes and seconds to degrees expressed in decimals and vice versa. (*Comment: Since the same factors apply, this function may also be used for the conversion of time units.*)

$$\begin{aligned}
\text{1 revolution} &= 360^\circ \\
\text{1 straight angle} &= \frac{1}{2}\,\text{revolution} = 180^\circ \\
\text{1 right angle} &= \frac{1}{4}\,\text{revolution} = 90^\circ
\end{aligned}$$

For scientific and mathematical work, these units are often not suitable. The preferred unit is the **radian**.

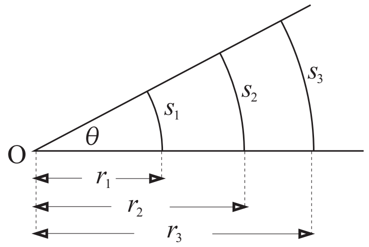


Figure 1.6-2

Consider the sketch in Figure 1.6-2. If r_1, r_2 and r_3 are the radii of three circles with centre O and s_1, s_2 and s_3 the corresponding arcs between the arms of the angle, then

$$s_1/r_1 = s_2/r_2 = s_3/r_3$$

The ratio (arc length)/(radius) depends on the magnitude of the angle only. This allows a new unit for the measurement of angles to be defined.

$$\theta = \frac{\text{arc length}}{\text{radius}} = \frac{s}{r} \qquad 1.6(1)$$

If s and r are measured in the same units, the answer is in **radians** (abbreviation **rad**).

It follows from Equation 1.6(1) that the length of the arc s is given by

$$s = \text{radius} \times \text{angle} = r\theta \qquad \theta \text{ in radians} \qquad 1.6(2)$$

$$\begin{aligned} 1 \text{ revolution} &= \frac{\text{circumference of a circle}}{\text{radius of the circle}} = \frac{2\pi r}{r} = 2\pi \text{ rad} \\ 1 \text{ right angle} &= \frac{\frac{1}{4}\text{ circumference of a circle}}{\text{radius of the circle}} = \frac{\frac{1}{4} \times 2\pi r}{r} = \frac{\pi}{2} \text{ rad} \end{aligned}$$

It is recommended that the reader remember the relationship $180^\circ = \pi$ rad and that this be used to convert degrees to radians and vice versa. From this it follows that

$$\begin{aligned} 1 \text{ rad} &= 180/\pi = 57,29577\ldots \approx 57,3^\circ \\ 1^\circ &= \pi/180 = 0,01745\ldots \text{rad} \end{aligned}$$

Since a calculator will be used for conversions, it will not be necessary to remember these two values. Some calculators can do the conversions directly by the use of one or two keys only. If such a facility does not exist, the following procedure may be used:

$$\begin{aligned} \text{angle in radians} &= (\text{angle in degrees}) \times \pi/180 \\ \text{angle in degrees} &= (\text{angle in radians}) \times 180/\pi \end{aligned}$$

Some calculators make provision for yet another unit which is called **grad**. This is an attempt to decimalise angles. There are 100 grad in a right angle but these units will not be used in this book.

1.7 Useful goniometric relationships

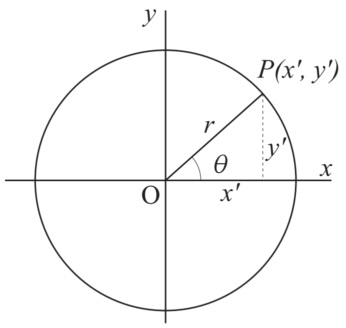


Figure 1.7-1

To indicate the position of a point on a plane surface, a **Cartesian** frame of reference, also referred to as a rectangular frame of reference, is used. Two straight lines which intersect at right angles, are called the axes. It is convenient to think of the axes as rulers. Their point of intersection is called the origin which is usually the common zero on both scales. It is conventional (though not necessary) to refer to the axes as the x-axis and y-axis respectively. On the x-axis (see Figure 1.7-1) the values to the left of the origin are taken as negative and those to the right as positive. On the y-axis the values below the origin are taken as negative, and those above as positive. The position of any point on the plane surface is given unambiguously by the specification of two ordered real numbers, $(x',\, y')$, which are known as its **co-ordinates**. In chapter 4, frames of reference will be dealt with in greater depth.

Consider the Cartesian frame of reference in Figure 1.7-1. Describe a circle, radius r and centre at the origin. θ is the angle between the positive x-axis and radius vector OP. If the angle is measured counter-clockwise, it is taken as positive and when it is measured clockwise, it is negative. In the following goniometric ratios (also called trigonometric ratios), r is always a positive number but the signs of x' and y' are in accordance with the quadrants in which they lie.

$$
\begin{aligned}
\sin\theta &= y'/r & \csc\theta &= (\sin\theta)^{-1} = r/y' \\
\cos\theta &= x'/r & \sec\theta &= (\cos\theta)^{-1} = r/x' \\
\tan\theta &= y'/x' & \cot\theta &= (\tan\theta)^{-1} = x'/y'
\end{aligned}
\qquad 1.7(1)
$$

These definitions are valid for all values of θ, excluding those for which the denominator which appears in the definition is equal to zero. At such values of θ the corresponding trigonometric ratio is undefined. The problem occurs with all the ratios excepting $\sin\theta$ and $\cos\theta$. The graphs of $\sin\theta$ and $\cos\theta$ are shown in Figure 1.7-2.

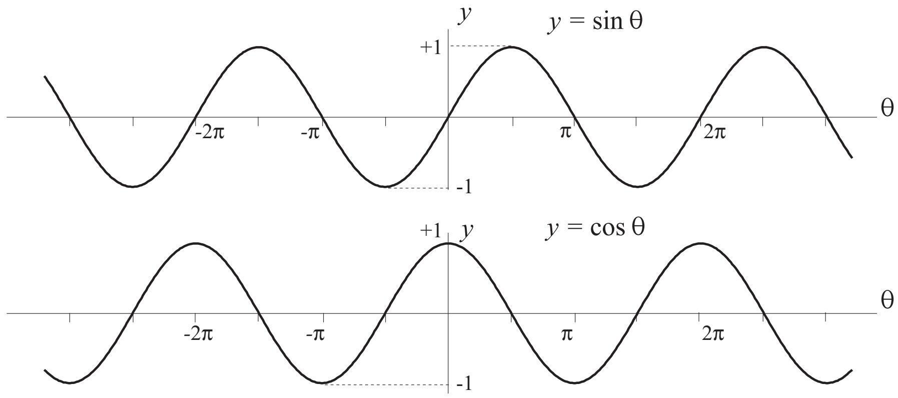


Figure 1.7-2

From the definitions of the goniometric ratios the following identities may be derived. They are often utilised in the mathematics which is used in physics and the reader should be quite familiar with them.

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \qquad 1.7(2)$$

$$\sin^2\theta + \cos^2\theta = 1 \qquad 1.7(3)$$

$$\sin(\theta \pm \phi) = \sin\theta\cos\phi \pm \cos\theta\sin\phi \qquad 1.7(4)$$

$$\cos(\theta \pm \phi) = \cos\theta\cos\phi \mp \sin\theta\sin\phi \qquad 1.7(5)$$

From Equations 1.7(3) to 1.7(4) the following may be deduced quite easily:

$$\cos 2\theta = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta \qquad 1.7(6)$$

$$\sin 2\theta = 2\cos\theta\sin\theta \qquad 1.7(7)$$

From Equation 1.7(6) we have

$$\sin^2\theta = \frac{1}{2}(1 - \cos 2\theta) \qquad 1.7(8)$$

$$\cos^2\theta = \frac{1}{2}(1 + \cos 2\theta) \qquad 1.7(9)$$

The following identities also follow from 1.7(4) and 1.7(5):

$$\cos\theta\cos\phi = \frac{1}{2}[\cos(\theta + \phi) + \cos(\theta - \phi)] \qquad 1.7(10)$$

$$\sin\theta\sin\phi = \frac{1}{2}[\cos(\theta - \phi) - \cos(\theta + \phi)] \qquad 1.7(11)$$

$$\sin\theta\cos\phi = \frac{1}{2}[\sin(\theta+\phi)+\sin(\theta-\phi)] \qquad 1.7(12)$$

$$\cos\theta\sin\phi = \frac{1}{2}[\sin(\theta+\phi)-\sin(\theta-\phi)] \qquad 1.7(13)$$

$$\cos\theta+\cos\phi = 2\cos\frac{1}{2}(\theta+\phi)\cos\frac{1}{2}(\theta-\phi) \qquad 1.7(14)$$

$$\cos\theta-\cos\phi = -2\sin\frac{1}{2}(\theta+\phi)\sin\frac{1}{2}(\theta-\phi) \qquad 1.7(15)$$

$$\sin\theta+\sin\phi = 2\sin\frac{1}{2}(\theta+\phi)\cos\frac{1}{2}(\theta-\phi) \qquad 1.7(16)$$

$$\sin\theta-\sin\phi = 2\cos\frac{1}{2}(\theta+\phi)\sin\frac{1}{2}(\theta-\phi) \qquad 1.7(17)$$

By using differential calculus, the following infinite series may be derived:

$$\sin\theta = \frac{\theta}{1!}-\frac{\theta^3}{3!}+\frac{\theta^5}{5!}-\frac{\theta^7}{7!}+\ldots(\theta \text{ in rad}) \qquad 1.7(18)$$

$$\cos\theta = 1-\frac{\theta^2}{2!}+\frac{\theta^4}{4!}-\frac{\theta^6}{6!}+\ldots(\theta \text{ in rad}) \qquad 1.7(19)$$

The higher-order terms in these infinite series are small compared to those which occur early in the series. It will often be sufficient to make an approximation in which a number of the lower-order terms only are used. Calculators make use of these series when calculating the numerical values of trigonometric ratios of different angles. From the series it should be apparent that $\cos\theta$ is symmetrical about $\theta = 0$ and $\sin\theta$, antisymmetrical. For these reasons, $\cos\theta$ is known as an **even function** and $\sin\theta$ as an **uneven function**.

By means of Figure 1.7-1 $\sin\theta$ was defined as $\sin\theta = y'/r$. The **inverse** goniometric function may be written as

$$\theta = \arcsin(y'/r) \qquad \text{or} \qquad \theta = \sin^{-1} y'/r \qquad 1.7(20)$$

which is equivalent to the statement

$$\theta = \text{the angle of which the sine is equal to } y'/r$$

This is known as the **arc sine** of the given quantity. The second way of writing this function often appears on calculators and should not be confused with the reciprocal of the sine function. Similarly, with reference to Figure 1.7-1, we have

$$\theta = \arctan(y'/x') \qquad \text{and} \qquad \theta = \arccos(x'/r)$$

By means of a calculator the numerical values of arcsin, arccos and arctan may be determined directly. For reasons which will become apparent when the relevant mathematics is studied in depth, these functions exist only between the limits as indicated.

For	arcsin:	$-\pi/2$	and	$\pi/2$
	arccos:	0	and	π
	arctan:	$-\pi/2$	and	$\pi/2$

Answers which are read from a calculator will lie within these limits. The reader is advised to read the instruction booklet supplied with his/her pocket calculator and become familiar with the use of these functions. An important point to be aware of, is whether an answer is required in degrees or radians. The calculator has to be adjusted beforehand and it is indicated on the read-out panel. Verify the following answers as an exercise to become familiar with your calculator.

$\arcsin 0,1234$	$=$	$7,088°$	$\arcsin 0,8660$	$=$	$60,00°$
$\arcsin 0,5000$	$=$	$30,00°$	$\arcsin 1,1234$	does not exist	
$\arcsin 0,5000$	$=$	0,524 rad	$\arccos 0,5236$	$=$	1,020 rad
$\arcsin(-0,5000)$	$=$	$-30,00°$	$\arccos(-0,5000)$	$=$	$120,0°$
$\arctan(-0,5000)$	$=$	$-26,57°$	$\arctan 123,0$	$=$	$89,53°$
$\arctan 0$	$=$	0	$\arctan 1,000$	$=$	$45,00°$
$\arccos 0$	$=$	$90,00°$	$\arccos 1,000$	$=$	0

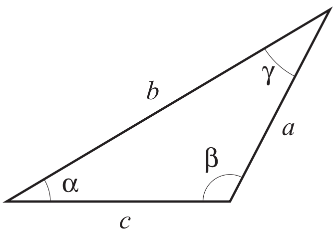


Figure 1.7-3

When a calculator is switched to degrees, numerical answers of the inverse functions will be in degrees and decimal parts of degrees, not degrees, minutes and seconds. Most calculators will perform this conversion by the use of one or two keys. If a trigonometric ratio of an angle which is specified in degrees, minutes and seconds is to be found by means of a calculator, it must first be converted to degrees and decimal parts of a degree. This can also be accomplished directly by means of a calculator.

If certain dimensions of a triangle (sides and angles) are specified, they may define its shape and size unambiguously. If that is the case, the unspecified dimensions may be calculated by using the cosine rule or the sine rule. The cosine rule is of prime importance in the study of physics since it forms the base of that branch of mathematics which is known as vector algebra.

Consider the triangle which is shown in Figure 1.7-3.

The sides opposite to angles α, β and γ are a, b and c respectively. If any two sides and the included angle are known, the third side may be calculated by means of the cosine rule.

$$\begin{aligned} a^2 &= b^2 + c^2 - 2bc\cos\alpha \\ b^2 &= c^2 + a^2 - 2ca\cos\beta \\ c^2 &= a^2 + b^2 - 2ab\cos\gamma \end{aligned} \qquad 1.7(21)$$

For the special case where $\gamma = 90°$, the cosine rule for the calculation of c may be rewritten as

$$c^2 = a^2 + b^2 \qquad 1.7(22)$$

which is known as the theorem of Pythagoras.

The sine rule supplies two more relationships for the calculation of unknown quantities. For the triangle which is shown in Figure 1.7-3 the following is valid:

$$\frac{a}{\sin\alpha} = \frac{b}{\sin\beta} = \frac{c}{\sin\gamma} \qquad 1.7(23)$$

The sides of the triangles in Figure 1.7-4, may be calculated by means of the theorem of Pythagoras. If one of the angles of a triangle is equal to zero, the two adjacent sides will coincide and the opposite side will be equal to zero. By means of these facts, the goniometric ratios of the angles given in Table 1.7-1 may be calculated.

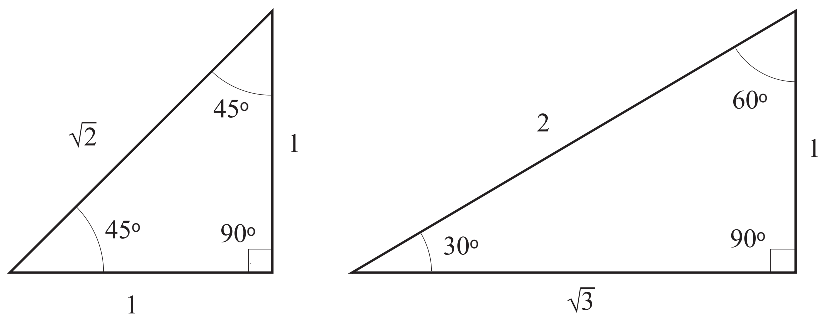


Figure 1.7-4

θ	$\sin\theta$	$\cos\theta$	$\tan\theta$
$0°$	0	1	0
$30°$	$1/2 = 0{,}5000$	$\sqrt{3}/2 = 0,8660$	$1/\sqrt{3} = 0,5774$
$45°$	$1/\sqrt{2} = 0,7071$	$1/\sqrt{2} = 0,7071$	1
$60°$	$\sqrt{3}/2 = 0,8660$	$1/2 = 0{,}5000$	$\sqrt{3} = 1,7920$
$90°$	1	0	undefined

Table 1.7-1

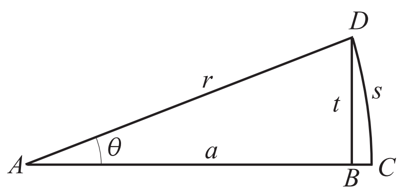


Figure 1.7-5

Figure 1.7-5 shows a triangle of which angle θ is very small. Angle $ABD = 90°$. $AD = r$ ($= AC$) is the hypotenuse of the triangle. $AB = a$ is the adjacent arm of the angle θ. $BD = t$ is the side opposite angle θ. $DC = s$ is the arc length which the angle θ subtends at a distance of r. From the sketch it can be seen that $s \approx t$ and $a \approx r$ when θ is small. With these approximations the following are valid:

$$\begin{aligned}
\sin\theta &= \frac{t}{r} \approx \frac{s}{r} = \theta \qquad (\theta \text{ in radians})\\
\tan\theta &= \frac{t}{a} \approx \frac{s}{r} = \theta \qquad (\theta \text{ in radians})\\
\cos\theta &= \frac{a}{r} \approx \frac{r}{r} = 1
\end{aligned}$$

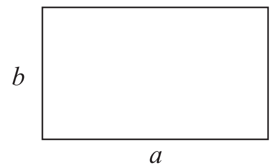
Summary: For small θ: $\sin\theta \approx \tan\theta \approx \theta$ (θ in radians) and $\cos\theta \approx 1$ 1.7(24)

The approximations for $\sin\theta$ and $\cos\theta$ also follow directly from the series given in Equations 1.7(18) and 1.7(19). The approximations are often used in physics and they represent one of the advantages of using radians as units for measuring angles.

1.8 Geometric formulae

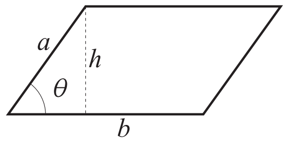
■ Rectangle, length a and width b

Perimeter $= 2a + 2b$
Area $= ab$



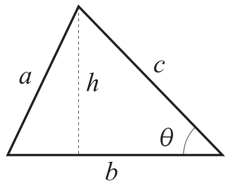
■ Parallelogram, base b and height h

$$\begin{aligned} \textit{Perimeter} &= 2a + 2b \\ \textit{Area} &= bh = ab\sin\theta \end{aligned}$$



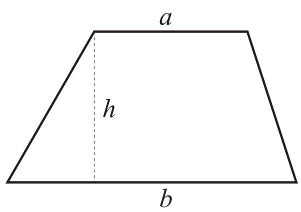
■ Triangle, base b and height h

$$\begin{aligned} \textit{Perimeter} &= a + b + c \\ \textit{Semi-perimeter} &= s = \tfrac{1}{2}(a + b + c) \\ \textit{Area} &= \tfrac{1}{2}bh = \tfrac{1}{2}bc\sin\theta \\ &= \sqrt{s(s-a)(s-b)(s-c)} \end{aligned}$$



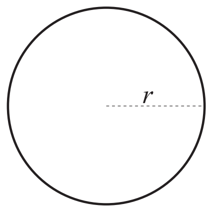
■ Trapezium with parallel sides a and b and height h

$$\textit{Area} = \tfrac{1}{2}(a + b)h$$



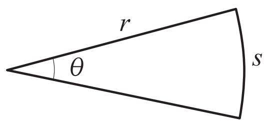
■ Circle, radius r

$$\begin{aligned} \textit{Diameter } D &= 2r \\ \textit{Circumference} &= \pi D = 2\pi r \\ \textit{Area} &= \tfrac{1}{4}\pi D^2 = \pi r^2 \end{aligned}$$



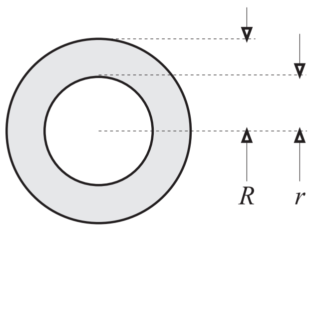
■ Sector of a circle with radius r

$$\begin{aligned} \textit{Arc length} &= s = r\theta && (\theta \text{ in rad}) \\ \textit{Perimeter} &= 2r + r\theta && (\theta \text{ in rad}) \\ \textit{Area} &= \tfrac{1}{2}rs = \tfrac{1}{2}r^2\theta && (\theta \text{ in rad}) \end{aligned}$$



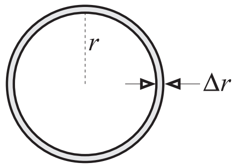
■ Annulus, inner radius r, outer radius R

$$\begin{aligned} \textit{Area} &= \pi(R^2 - r^2) \\ &= \pi(R - r)(R + r) \end{aligned}$$



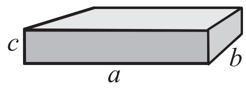
■ Narrow annulus, radius r and width Δr

$$\begin{aligned} \textit{Area} \quad &\approx \text{circumference} \times \text{width} \\ &= 2\pi r \Delta r \end{aligned}$$



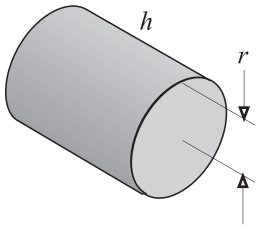
■ Rectangular parallelepiped, sides a, b and c

$$\begin{aligned} \textit{Area} \quad &= 2(ab + bc + ca) \\ \textit{Volume} \quad &= \text{abc} \end{aligned}$$



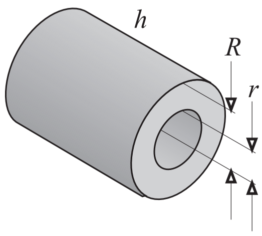
■ Solid right circular cylinder
with radius r and length h

$$\begin{aligned} \textit{Area} \quad &= 2\pi r^2 + 2\pi r h \\ &= 2\pi r(r + h) \\ \textit{Volume} \quad &= \pi r^2 h \end{aligned}$$



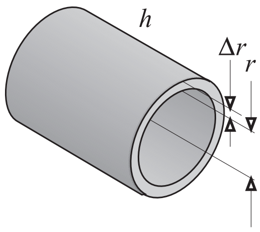
■ Hollow right circular cylinder,
inner radius r, outer radius R, length h

$$\begin{aligned} \textit{Area} \quad &= 2\pi(R^2 - r^2) + 2\pi h(R + r) \\ &= 2\pi(R + r)(R - r + h) \\ \textit{Material volume} &= \pi(R^2 - r^2)h \end{aligned}$$



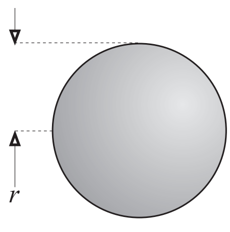
■ Hollow right circular cylinder,
radius r, length h and thin wall Δr

$$\begin{aligned} \textit{Area} \quad &\approx 4\pi r h \\ \textit{Volume of material} &\approx 2\pi r h \Delta r \end{aligned}$$

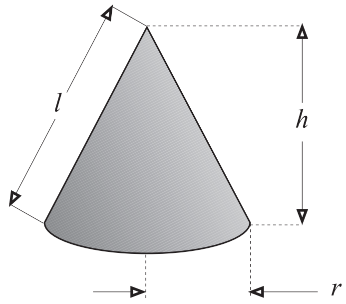


■ Sphere, radius r

$$\begin{aligned} \textit{Area} \quad &= 4\pi r^2 \\ \textit{Volume} \quad &= \tfrac{4}{3}\pi r^3 \end{aligned}$$



■ Right circular cone, radius r, height h



$$\begin{aligned} \textit{Area} &= \pi r l + \pi r^2 \\ \textit{Volume} &= \tfrac{1}{3}\pi r^2 h \end{aligned}$$

1.9 PROBLEMS CHAPTER 1

1. Calculate the following without the use of calculating aids:

(a) $64^{1/6}$ (b) $8^{2/3}$ (c) $27^{-2/3}$
(d) $2^3 \times 3^2$ (e) $\sqrt[7]{2 \times 8^2}$ (f) $\sqrt[3]{8^{-2}}$
(g) $6^4/3^3$ (h) $10^6 \times 10^4$ (i) $10^6/10^4$
(j) $10^6/10^{-4}$ (k) $9^{2,5}$ (l) $16^{1,5}$
(m) $8^{-0,6667}$ (n) $\left(\frac{32\times 10^5}{2^3\times 10^3}\right)^{\frac{1}{2}}$ (o) $[10 \times \left(\frac{1,08\times 10^4}{300}\right)^{\frac{1}{2}} + 4]^{\frac{1}{2}}$

2. Calculate the following without the use of calculating aids:

(a) $\log_2 32$ (b) $\log_3 27^{\frac{2}{3}}$ (c) $\log_2 (1/32)$
(d) $\log_5 1$ (e) $\log_5 (-1)$ (f) $\ln (1/e^2)$
(g) $\log 100$ (h) $\log 0$ (i) $\log 10^{-5}$

3. Use a calculator to calculate the following:

(a) $12,34^2$ (b) $12,34^3$ (c) $12,34^{2,56}$
(d) $12,34^{-2}$ (e) $12,34^{\frac{1}{2}}$ (f) $12,34^{\frac{1}{3}}$
(g) $12,34^{-\frac{1}{3}}$ (h) $(12,34^3)^{\frac{1}{5}}$ (i) $1/(12,34)^{\frac{1}{3}}$

4. Use a calculator to calculate the following:

(a) $\log (12,34^2)$ (b) $(\log 12,34)^2$ (c) $\log \sqrt{12,34}$
(d) $\log_8 12,34$ (e) $\log_2 12,34$ (f) $\log_5 12,34$
(g) $\log_{12,34} 10$ (h) $\log (12,34/10)$ (i) $(\log 12,34)/10$
(j) $\log (12,34/100)$ (k) $(\log 12,34)/(\log \sqrt{12,34})$

5. Use the binomial theorem to calculate the following:

(a) $(a + b)^5$ (b) $(x + 2)^4$ (c) $(2y - 3)^4$
(d) $(1 - v^2/c^2)^{-\frac{1}{2}}$ (e) $(1 + 0,002)^2$ (f) $(1 - 0,005)^{\frac{1}{2}}$
(g) $(1,003)^{\frac{1}{2}}$ (h) $(0,995)^{\frac{1}{2}}$ (i) $(0,995)^{-\frac{1}{2}}$

6. Calculate the following angles in radians without the use of a calculating aid. Give the answers in terms of π.

(a) $30°$ (b) $60°$ (c) $45°$
(d) $270°$ (e) $216°$ (f) $720°$
(g) $450°$ (h) $630°$ (i) $72°$

7. Calculate the following angles in radians:

(a) $85,34°$	(b) $123,4°$	(c) $815,6°$
(d) $89°39'42''$	(e) $219°54'29''$	(f) $356°0'56''$

8. The following angles are in radians. Express them in degrees without making use of a calculator.

(a) $3\pi/2$	(b) $\pi/2$	(c) $2\pi/7$
(d) $4\pi/5$	(e) $2,5\pi$	(f) 4π
(g) 3π	(h) $\pi/6$	(i) $4\pi/3$

9. Use a calculator to express the following angles in degrees:

(a) 7,345 rad	(b) 0,1234 rad	(c) 0,5678 rad
(d) -5,568 rad	(e) $0,5678\pi$ rad	(f) $14,56\pi$ rad

10. Use a calculator to calculate the following:

(a) $\sin 56,78°$	(b) $\cos 56,78°$	(c) $\tan 56,78°$
(d) $\cot 56,78°$	(e) $\csc 56,78°$	(f) $\sec 56,78°$
(g) $\sin 98°34'45''$	(h) $\cos 125°28'48''$	(i) $\tan 135°45'38''$
(j) $\tan(2,65 \text{ rad})$	(k) $\csc(0,567 \text{ rad})$	(l) $\cos(2,345\pi \text{ rad})$
(m) $\cot(-27°5'27'')$	(n) $\sin(-1,23\pi \text{ rad})$	(o) $\sec(-0,56\pi \text{ rad})$

11. Use a calculator to calculate the following angles, first in degrees and then in radians:

(a) $\arctan 1,234$	(b) $\arctan(-1,234)$	(c) $\arccos(-0,8660)$
(d) $\arcsin 1,234$	(e) arccsc1,234	(f) arccot(-1,234)
(g) $\arcsin 0,0012$	(h) $\arctan 0,0012$	(i) arcsec2,3456
(j) $\arccos(-1,543)$	(k) $\arccos 1,543$	(l) $\arcsin 0,8660$

12. Write down, without the use of a calculator, the sine, cosine and tangent of the following angles. Check your answers by means of Table 1.7-1.
0°, 30°, 45°, 60°, 90° and 180°.

13. The following apply to the triangle in Figure 1.7-3. Calculate in each case the unspecified quantities if the following are known:

(a) $a = 20$ mm, $c = 30$ mm, $\beta = 120°$	(b) $c = 12,5$ mm, $\alpha = 40°$, $\beta = 60°$
(c) $c = 63$ m, $\alpha = 62°$, $\gamma = 55°$	(d) $a = 25$ m, $b = 35$ m, $c = 45$ m
(e) $c = 50$ m, $a = 30$ m, $\alpha = 20°$	

14. Calculate the areas of the triangles specified in Question 13. If possible, avoid using quantities which had to be calculated.

15. Calculate the area of an equilateral triangle with side length 2 m by (i) using the magnitude of one of its angles, (ii) using only the lengths of the sides.

16. A circular race track has an internal circumference of 2 km and a width of 6 m. Calculate the area of the paving on the track in square metres.

17. The sides of a trapezium are 1 m, 1 m, 3 m and 4 m respectively. Calculate its area.

18. Calculate the area of a circle with diameter 12,34 m.

19. Calculate the diameter of a circle with area 4 m^2.

20. Calculate the area of a right circular cylinder with radius 20 mm and length 40 mm.

21. An empty cylindrical pipe has an outer diameter of 30 mm and an inner diameter of 25 mm. Calculate the material volume of 1 m of the pipe and also the area exposed to the atmosphere. Calculate the volume of water that would fill 2 m of this pipe.

Chapter 2

DIFFERENTIAL CALCULUS

2.1 Functions

2.1.1 Introduction

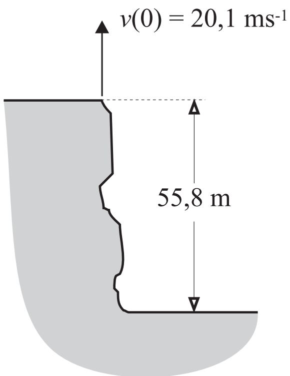


Figure 2.1-1

Figure 2.1-1 shows a precipice which is 55,8 m deep. Standing on the edge, a person throws an object vertically upwards at an initial speed of 20,1 m s^{-1}. The height of the object is measured from the edge and it is indicated by x. Values of x which are measured upwards are taken as positive and those in the opposite direction, negative. As time progresses, x will first increase and then decrease until the object hits the bottom of the precipice.

At each instant, t, there exists one value of x. It should be clear that the object cannot be at two different positions at one instant and also that x may only assume values which lie between two limits, the maximum height which the object reaches and the bottom of the cliff. The complete motion of the object lasts a finite interval of time, and therefore the possible values of t

also lie between two limits. It is said that x is a **function** of t. Since one and the same value of x may correspond to different values of t, t is not a function of x in this time interval.

This relationship between x and t which is called a **function**, may be represented by

$$x = x(t) \qquad 2.1(1)$$

In this representation, t is known as the **independent variable** and x the **dependent variable**.

The quantitative information which is gathered during an experiment is known as **data**. The data is used to determine the **functional relationship** between the variables which describes a process. There are three ways in which a function may be represented.

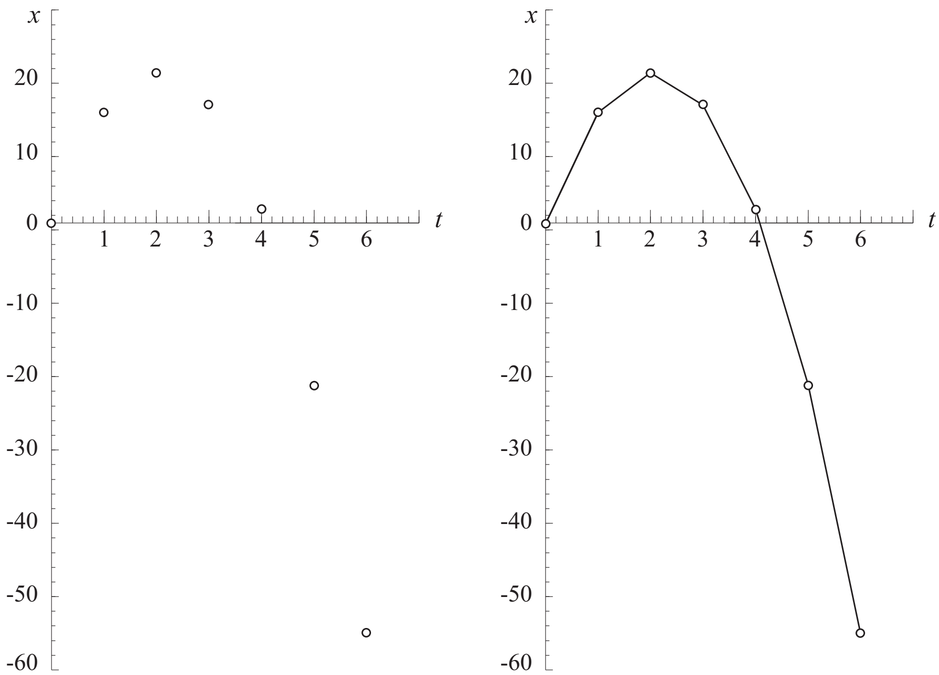
(i) a table of numbers
(ii) a graph
(iii) an equation

For the example of the object which is thrown vertically upwards from the edge of a cliff, Table 2.1-1 gives the relationship between x and t for the duration of the motion. The value of x is given at the beginning and thereafter at the end of each second while the motion exists. At instant $t = 6$ s, the motion stops. Although this table gives valuable information about the motion, it has certain shortcomings. The first is that it can never be complete, which means that values of x at unspecified times, cannot be known. A superficial inspection might lead to the conclusion that the maximum height is 20,6 m. This is incorrect and more information is required before the correct maximum value can be determined.

Time, t, in seconds	Position, x, in metres
0,0	0,0
1,0	15,2
2,0	20,6
3,0	16,2
4,0	2,0
5,0	-22,0
6,0	-55,8

Table 2.1-1

A graphic representation of this data makes it more informative. The independent variable is usually represented by a suitable scale along a straight line from left to right. In this case it will be t and the line is called the **abscissa axis**. The dependent variable, x in this case, is represented on a line perpendicular to the first and which is known as the **ordinate axis**. The pairs of corresponding numbers are called co-ordinates. Each pair of co-ordinates gives one point on the graphic representation. Figure 2.1-2(a) gives a graphic representation of the data in Table 2.1-1.



(a) *(b)*

Figure 2.1-2

If this graph is to be used to obtain information about values of x at values of t which lie between the points which are known, the technique of **interpolation** is used. This term applies to any reasonable method by which points that lie between those which are known, are determined. If the known points are joined with straight lines as shown in Figure 2.1-2(b), they do not describe the function correctly, but are an aid to facilitate interpolation. By means of drawing aids such as curve stencils or flexible drawing guides, a smooth curve may be drawn through the points. Such a curve gives more accurate values when interpolating. Figure 2.1-3 was drawn from the data in Table 2.1-1 with the aid of a computer program. The best technique to use is that of curve fitting by means of the method of **least squares of deviations**. That was the technique used by the computer program which was used to produce the graph in Figure 2.1-3. The technique of curve fitting will not be treated in this book but readers are advised to gain access to computer programs which may be used for this purpose.

Known data may be used in some cases to predict the behaviour of a system in regions which lie outside the known domain. This technique is known as

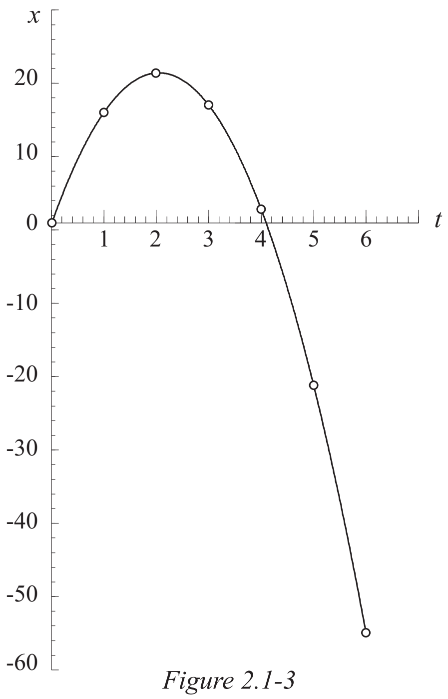


Figure 2.1-3

extrapolation. Graphic extrapolation implies the extension of a curve to a region where no known data points exist. The shape of the portion of the curve which is known to be correct is used as a guide. In some cases graphic extrapolation is fairly easy but in others it might be very difficult or impossible.

In the study of physics, the aim is mostly to describe a function by means of an equation. With the techniques at our disposal this may be done quite readily in most cases. The equation which describes the motion of the object in question is

$$x = 20,1t - 4,9t^2 \qquad 2.1(2)$$

in which the position, x, is measured in metres and the time, t, in seconds. By means of this equation, the value of x at any admissible value of t, may be calculated and it describes the behaviour of the function for the complete motion.

Having considered a practical example, it is now possible to give a more formal definition of a function. If two variables x and y are related in such a way that for each value of x in a set of admissible values which is called the *domain* of x, there exists only one value of y from another set of admissible values which is called the **range** of y, then y is said to be a function of x. In physics the following notation is used to indicate a function:

$$y = y(x) \qquad 2.1(3)$$

It is convenient to interpret this notation as follows: A variable y depends on x of which it is a function. In the notation, x is called the independent variable (abscissa) and y the dependent variable (ordinate). Each co-ordinate pair, (x, y), represents one point on the graph which represents it. In Figure 2.1-2, x was used to represent the ordinate whereas in a graph of Equation 2.1(3) it would be the abscissa. This was done on purpose to impress upon the reader that a symbol may be used for any purpose.

The symbolic notation $y = y(x)$ contains no information for the calculation of the pairs of co-ordinates. That is possible only when the equation for the function is known as is shown in the example below.

$$
\begin{array}{llcl}
\text{Consider} & y = y(x) & = & 3x^2 - 5x + 2 \\
\text{Then} & y(0) & = & 3(0)^2 - 5(0) + 2 = 2 \\
 & y(1) & = & 3(1)^2 - 5(1) + 2 = 0 \\
 & y(-1) & = & 3(-1)^2 - 5(-1) + 2 = 10 \\
 & y(2) & = & 3(2)^2 - 5(2) + 2 = 4 \qquad \text{etc.}
\end{array}
$$

There are also functions of more than one variable as shown in the following examples:
(i) The length, c, of the hypotenuse of a variable right angled triangle, depends on both the other two sides, a and b.

$$c = (a^2 + b^2)^{\frac{1}{2}}$$

The symbolic notation $c = c(a, b)$ indicates that c is a function of both variables a and b.
(ii) The volume, V, of an ideal gas, depends on the number of moles, n, of the gas, its pressure, p, and the temperature, T.

$$V = V(n, p, T,) = RnT/p \qquad \text{in which } R \text{ is constant}$$

In this book, mostly functions of a single variable will be considered.

2.1.2 A number of important functions of a single variable and their graphic representations

(1) $y \propto x$ or $y = mx$ in which m is a constant. In this relationship y is said to be **directly proportional** to x and m is known as the **proportionality constant**. The graph is a straight line which always passes through the origin. The graph in Figure 2.1-4 represents a direct proportionality for which $m > 0$. In this chapter the meaning of m will be of prime importance.

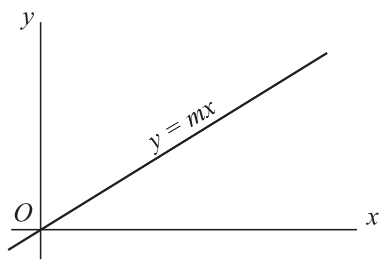


Figure 2.1-4

(2) $\Delta y \propto \Delta x$ or $y = mx + c$ in which m and c are constants. In this case the *change* in y is directly proportional to the *increase* in x. The graph is also a straight line but it does not generally pass through the origin. In the equation c is the **intercept** on the ordinate axis. The sketch in Figure 2.1-5(a) shows the graphic representation of such a proportionality for which $m < 0$ and $c > 0$.

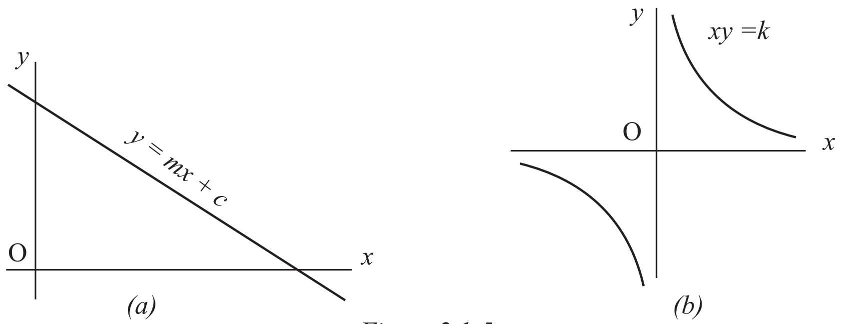


Figure 2.1-5

(3) $y \propto 1/x$ or $x \propto 1/y$ which may also be written as $y = k/x$ or $xy = k$. The relationship is an **inverse proportionality** and its graphic representation is called a **rectangular hyperbola**. The sketch in Figure 2.1-5(b) represents an inversely proportional relationship with $k > 0$. In the region visible in the sketch, the graph, which consists of two branches, is curved. When x becomes very small or very large, it is difficult to distinguish the graph from a straight line and the co-ordinate axes become tangents to it. This behaviour is known as **asymptotic behaviour** and the axes are called its **asymptotes**. Not all graphs have asymptotes.

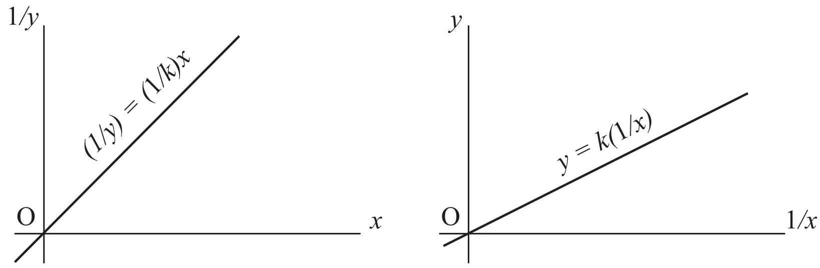


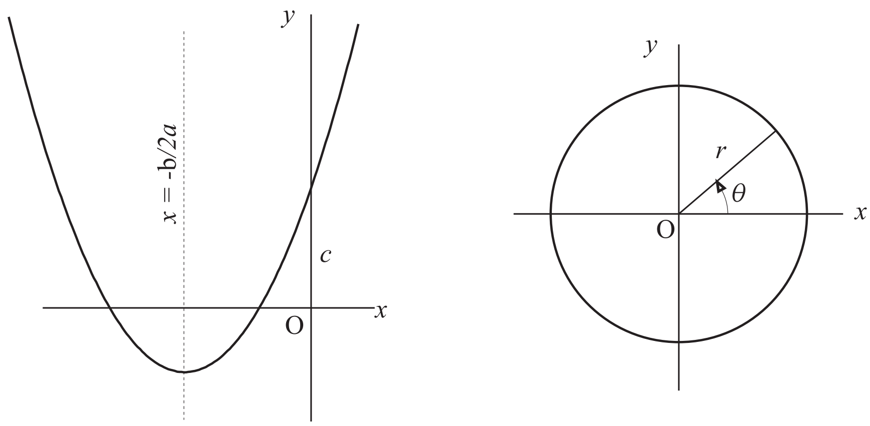
Figure 2.1-6

An inverse proportionality may be represented by a straight line by plotting the value of $1/y$ along the ordinate axis instead of y, or $1/x$ along the abscissa axis instead of x. Figure 2.1-6 shows these two possibilities. The straight-line representation is often used in physics since the graphic extrapolation of a hyperbola is difficult.

(4) $y = ax^2 + bx + c$. The graph of this function is called a **parabola** and its **axis of symmetry**, which is given by $x = -b/2a$, is parallel to the y-axis. If $b^2 > 4ac$, the parabola intersects the x-axis at two positions given by

$x = (-b \pm \sqrt{b^2 - 4ac})/2a$. If $b^2 = 4ac$, the x-axis is a tangent to it at position $x = -b/2a$. If $b^2 < 4ac$, the graph does not intersect or touch the x-axis. The graph shown in Figure 2.1-7(a) represents a parabola for which a, b and c are positive and $b^2 > 4ac$. c is the intercept on the ordinate axis.

(5) $y^2 + x^2 = r^2$ in which r is constant, is the equation for a **circle** with radius r and centre at the origin. The relationship $y^2 = r^2 - x^2$ is not a function because two values of y (or x) correspond to each value of x (or y). The circle consists of two functions namely $y = +\sqrt{r^2 - x^2}$ and $y = -\sqrt{r^2 - x^2}$ in which the domain of x is given by $-r \leq x \leq r$. The circle may also be described by means of the *two* **parametric equations** $x = r\cos\theta$, $y = r\sin\theta$. The **parameter** θ is shown in figure 2.1-7(b).



(a) *(b)*

Figure 2.1-7

(6) $x^2/a^2 + y^2/b^2 = 1$ is the equation of an **ellipse** of which the axes of symmetry have lengths of $2a$ and $2b$ respectively and which coincide with the co-ordinate axes. The ellipse consists of the two functions $y = (b/a)\sqrt{a^2 - x^2}$ and $y = -(b/a)\sqrt{a^2 - x^2}$ in which the domain of x is given by $-a \leq x \leq a$. The parametric equations of an ellipse are $x = a\cos\theta$ and $y = b\sin\theta$. It should be clear that an ellipse becomes a circle if $a = b$. Figure 2.1-8(a) shows an ellipse for which $a > b$. a is the **semi major axis** of the ellipse and b the **semi minor axis**.

The functions which have been discussed up to this stage are related in a wonderful way. They are generated by the intersection of a plane surface with a right circular cone. For this reason they are known as **conic sections**. They are of prime importance in physics because they describe the orbits of electri-

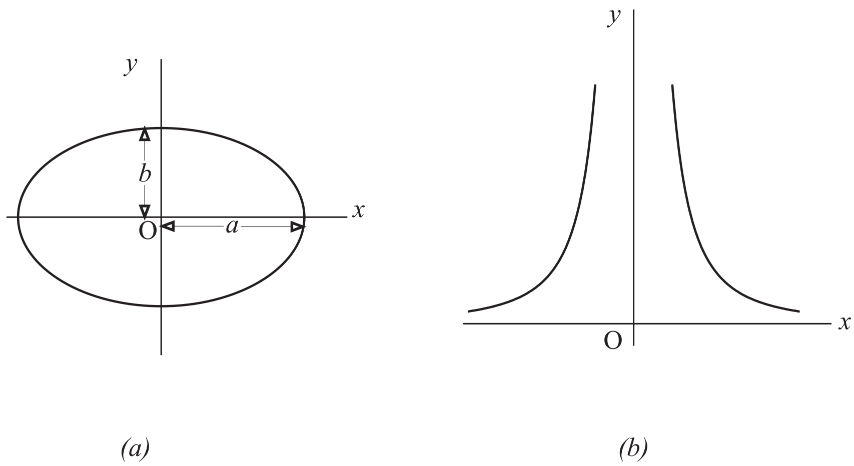


Figure 2.1-8

cally charged particles, planets, comets, etc., under the influence of an electrical and/or a gravitational force.

(7) $y \propto 1/x^2$ or $y = k/x^2$. The graph of this function consists of two branches and has no special name. The law which governs the magnitude of y, is known as the **inverse square law**. This law describes an important property of the three-dimensional space in which we live: It determines the way in which electric point charges and point masses exert forces on one another respectively. Figure 2.1-8(b) shows a graph of this function in which the constant k is positive.

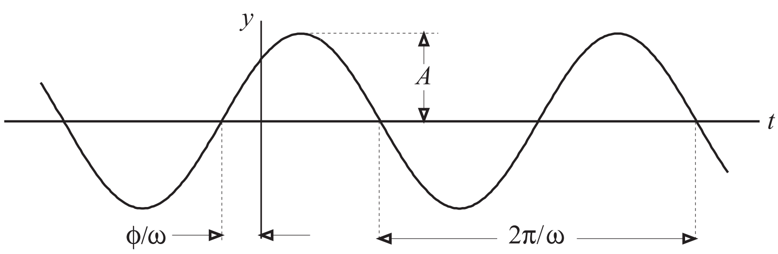


Figure 2.1-9

(8) $y = A \sin(\omega t + \phi)$, in which A, ω and ϕ are constants. When this function occurs in physics, it usually involves time and for this reason the time, t is used as independent variable. In this function A is called the **amplitude** and $(\omega t + \phi)$ the **phase angle** or **phase** in short. ω is the **angular frequency** and ϕ the **phase constant** or **initial phase**. The function $y = A\cos(\omega t + \phi)$ has exactly the same shape as that shown in Figure 2.1-9 but as if this curve has been moved a distance $\pi/2\omega$ to the left.

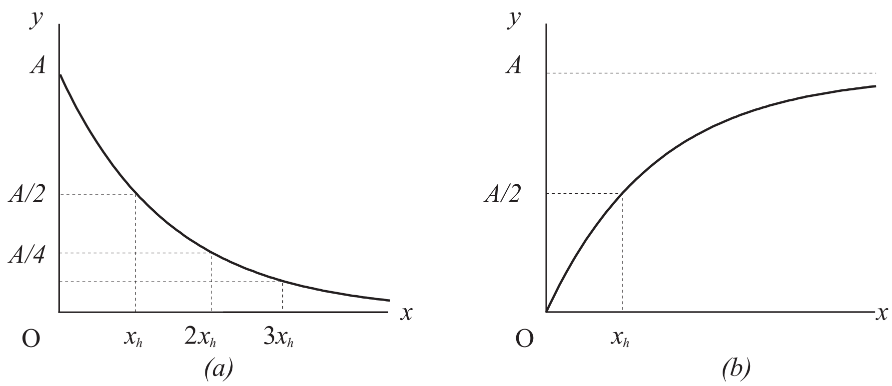


Figure 2.1-10

(9) $y = Ae^{-kx}$ in which A and k are positive constants, is known as an **exponential decay curve** with **decay constant** k. It has the interesting property that its value is A when $x = 0$ and after interval $\Delta x = x_h = (\ln 2)/k$, it has half this value, namely $A/2$. After every increment of $\Delta x = x_h$ the value of the function is half of what it is at the beginning of the interval. When $x \to \infty$, y tends to zero. The graph is shown in Figure 2.1-10(a).

(10) $y = A(1 - e^{-kx})$ in which A and k are positive constants, is known as **bounded or limited exponential growth** with **growth constant** k. The way in which this function grows, is a mirror image of the decay of that in (9). The graph of this function is shown in Figure 2.1-10(b).

In studying physics, the reader will also encounter other functions. There are, however, few phenomena in nature in which one or more of the functions mentioned here, do not play a part.

2.2 Calculations with zero and infinity. Limits

2.2.1 Calculations involving zero and numbers tending to infinity

For any finite number a we have

$$a + 0 = a, \qquad a - 0 = a, \qquad a \times 0 = 0 \qquad 2.2(1)$$

"Division by zero" illustrates the fact that the arithmetic operations are *defined*, and that the definitions are chosen so as to be consistent with previously stated rules of arithmetic.

Assume that $a/0 = b$, a finite number. It follows from the definition of a quotient and 2.2(1) that $a = b \times 0 = 0$. But this last equation cannot be valid, because $b \times 0 = 0$, no matter what number we choose for b, and we have assumed specifically that $a \neq 0$. Since we cannot assign any number b to the symbol $a/0$ that will be consistent with the method of assigning numbers to other quotients, we simply do not define the symbol $a/0$. Therefore $a/0$ is meaningless, that is, no meaning has been assigned to division of a finite number by zero.

A somewhat different problem is the case of the quotient $0/0$. Assume that $0/0 = b$. It follows from the definition of a quotient that $0 = b \times 0$ But this last equation is true for any number b, and there is no particular reason to prefer one number over any other. Again we avoid the difficulty by leaving $0/0$ undefined.

Consider the function $y = y(x) = 1/x$. As x tends to 0 from the positive side (written $x \to 0^+$), $1/x$ increases without bound. That is, $1/x$ becomes greater than any finite number we may think of. In mathematical notation we write $1/x \to \infty \quad \text{as} \quad x \to 0^+$.

If x tends to zero from the negative side (written $x \to 0^-$), $1/x$ decreases without bound and we may write $1/x \to -\infty \quad \text{as} \quad x \to 0^-$.

If x increases or decreases without bound, $|1/x|$ becomes **infinitesimally** small, that is, it tends to zero. We may write $1/x \to 0 \quad \text{as} \quad x \to \pm\infty$. In this case a limit value exists (see section 2.2.2) which is expressed as

$$\lim_{x \to \pm\infty} \frac{1}{x} = 0$$

When we say that n *"tends to ∞"*, we simply mean that n is supposed to assume a series of values which increase beyond all limit. **There is no number "infinity"**. An equation such as

$$n = \infty$$

as it stands, is *meaningless*. The reader will always have to bear in mind that ∞ *by itself* means nothing, although *phases containing it*, sometimes mean something, e.g. the phrase "tends to ∞".

Problems in which factors tend to zero and infinity will be understood well once the techniques discussed in the theory of *limits* have been mastered. It is of prime importance that $a/0$, $0/0$, ∞/∞ and $0 \times \infty$ are not defined and therefore have no meaning. Calculators do not have a button for infinity and cannot give it as a read-out, but the reader might try some undefined calculations involving zero to observe the calculator's response.

2.2.2 Limits

Consider the function $y = 3x/(x+1)$, $x \neq -1$.

In order to illustrate an interesting property, the value of y is calculated for different values of $x \geq 0$.

$$
\begin{array}{lll}
x = 0 & y = 0/1 & = 0 \\
x = 1 & y = 3/2 & = 1{,}500000000 \\
x = 2 & y = 6/3 & = 2{,}000000000 \\
x = 3 & y = 9/4 & = 2{,}250000000 \\
x = 5 & y = 15/6 & = 2{,}500000000 \\
x = 10 & y = 30/11 & = 2{,}727272727 \\
x = 50 & y = 150/51 & = 2{,}941176470 \\
x = 100 & y = 300/101 & = 2{,}970297029 \\
x = 10^3 & y = 3000/1001 & = 2{,}997002997 \\
x = 10^6 & y = 3000000/1000001 & = 2{,}999997000
\end{array}
$$

When $x \to \infty$, y takes on the form ∞/∞ which has no meaning since it is undefined. From the calculation it would seem to be a reasonable conclusion that if x becomes very much larger than any of the values used in the calculations, the function tends towards a **limit value** of 3.

This result may also be obtianed by changing the procedure of the calculation. The numerator and denominator of the fraction which constitute the function may be divided by any non-zero number without affecting a change in its value. Dividing numerator and denominator by x $(x \neq 0)$, the function becomes

$$y = \frac{3x}{x+1} = \frac{3}{1+1/x}$$

From this follows that the contribution of the term $1/x$ in the denominator becomes less significant as the magnitude of x increases and the following statement becomes valid:

The limiting value of y when $x \to \infty$, is $\frac{3}{1+0}$. In short this may be written as

$$\lim_{x\to\infty} y = \lim_{x\to\infty} \frac{3x}{x+1} = \lim_{x\to\infty} \frac{3}{1+1/x} = \frac{3}{1+0} = 3$$

Consider also the following function:

$$y = \frac{x^2-9}{x-3}, \quad x \neq 3$$

If x is assigned the value 3, we obtain the form 0/0 which is meaningless. This means that the function $(x^2 - 9)/(x - 3)$ is not defined at $x = 3$. That is why $x = 3$ has been excluded from the definition of the function under consideration. It would, however, be of interest to investigate the behaviour of the function when $x \to 3$. Since $x \neq 3$, $x - 3 \neq 0$ and we may divide numerator and denominator by $x - 3$ and investigate the result when $x \to 3$ as shown below.

$$\begin{aligned}
\lim_{x\to 3} y &= \lim_{x\to 3} \frac{x^2 - 9}{x - 3} = \lim_{x\to 3} \frac{(x+3)(x-3)}{x-3} \\
&= \lim_{x\to 3} (x+3) = 6
\end{aligned}$$

For the purpose of this study, the following example is the most important since it will be needed to develop a fundamental concept in calculus.

Consider the function $y = 2x^2 - 3x + 4$. Calculate the value of

$$\lim_{h\to 0} \frac{y(x+h) - y(x)}{h}$$

Direct substitution of $h = 0$ raises the problem of zero divided by zero which is undefined. It is, however, permissible to calculate the value of the expression for a very small value of h and investigate the behaviour of the function when h tends to zero.

$$\begin{aligned}
\lim_{h\to 0} \frac{y(x+h) - y(x)}{h} &= \lim_{h\to 0} \frac{[2(x+h)^2 - 3(x+h) + 4] - [2x^2 - 3x + 4]}{h} \\
&= \lim_{h\to 0} \frac{4xh + 2h^2 - 3h}{h} = \lim_{h\to 0} (4x + 2h - 3) \\
&= 4x - 3
\end{aligned}$$

In this section the concept of a limit was treated very superficially. It is assumed that the reader will also follow a course in mathematics in which the topic will be treated in a more satisfactory way. The present treatment is sufficient to deal with the scope of this work.

2.3 The gradient of a straight line

In section 2.1.2(2) it was shown that the equation of a straight line is $y = mx+c$ in which m and c are constants. The **gradient** or **slope** of the straight line which represents this function, is the **rate** at which y changes with increasing x and may be defined as follows: The gradient (slope) of the straight line $y = mx + c$ is the *change* in y between two points on it divided by the corresponding *increase* in x.

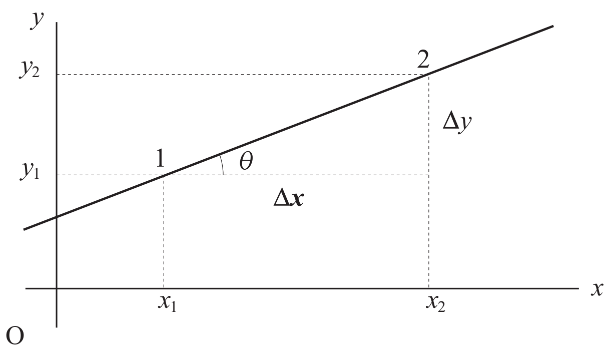


Figure 2.3-1

Consider the graph of $y = mx + c$ which is shown in Figure 2.3-1. Points 1 and 2 have co-ordinates (x_1, y_1) and (x_2, y_2) respectively and $x_2 > x_1$. According to the definition of the gradient, it follows that

$$\begin{aligned}
\text{gradient} &= \frac{\text{change in } y}{\text{increase in } x} = \frac{\Delta y}{\Delta x} \quad (\Delta x > 0) \\
&= \frac{y_2 - y_1}{x_2 - x_1} \quad (x_2 > x_1 \text{ because of } \textit{increase}) \\
&= \frac{y(x_2) - y(x_1)}{x_2 - x_1} \\
&= \frac{(mx_2 + c) - (mx_1 + c)}{x_2 - x_1} \\
&= \frac{m(x_2 - x_1)}{x_2 - x_1} \\
&= m
\end{aligned}$$

In Figure 2.3-1 the angle θ between the x-axis and the graph is called the **angle of inclination** or **slope angle**. In the solution of physics problems, it is often helpful to make use of the fact that the tangent of the angle of inclination is equal to the gradient.

$$\boxed{m = \text{gradient} = \Delta y / \Delta x = \tan\theta} \qquad 2.3(1)$$

Consider the straight lines in Figure 2.3-2.

In sketch (a) the function increases if x increases ($y_2 > y_1; \Delta y = y_2 - y_1 > 0$), and the gradient is a positive number. For the function $y = mx + c$ it means that $m > 0$.

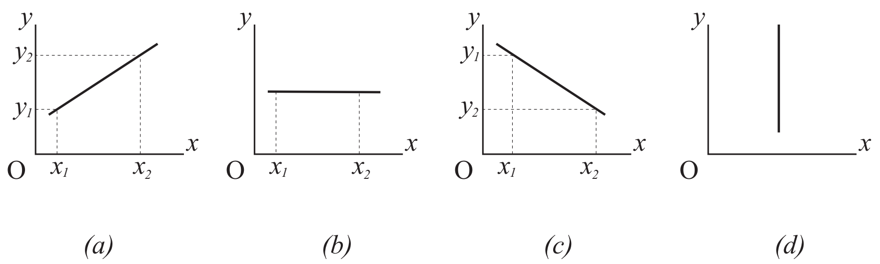


Figure 2.3-2

In sketch (b), y remains unchanged if x increases ($y_2 = y_1; \Delta y = 0$) and the gradient is zero. For the function $y = mx + c$ it means that $m = 0$ so that y = c (constant).

In sketch (c) y decreases with increase in x ($y_2 < y_1; \Delta y < 0$) and the gradient is negative ($m < 0$).

In sketch (d) the straight line is *not* the representation of a function since an infinite number of y-values correspond to one and the same value of x. Strictly, the gradient of this line is undefined since it would imply dividing by zero. For some problems in physics it is useful to refer to the "gradient" of the line as $+\infty$ or $-\infty$.

From the preceding material it should be clear that the gradient or slope of a straight line is a measure of its *inclination*. If the direction of the x-axis is taken as horizontal and that of the y-axis as vertical and the inclination of the line is examined in the direction in which x increases, the following will be valid:

$m > 0$ means that the slope of the line is "uphill". If $m_1 > m_2$, it means that the line with gradient m_1 is steeper uphill than one with a gradient of m_2.

$m = 0$ means that the line is horizontal.

$m < 0$ means that the slope of the line is "downhill".

$m \to \pm\infty$ refers to a vertical line.

Spoornet uses the notation 1:200 to indicate the incline of a stretch of rail. This notation refers to the sine instead of the tangent of the angle of incline. Can you think of a reason why this is used? Is the difference in definitions of any practical importance?

2.4 The gradient of a curve

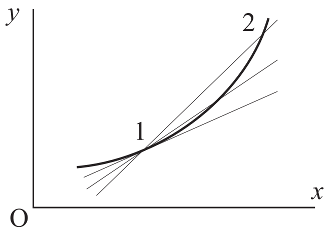


Figure 2.4-1

If a graph is said to be curved in a given region, it simply means that its gradient is not constant and then we refer to the graph as a **curve**. The procedure described for the determination of the gradient of a straight line, is not directly applicable to a curve for two simple reasons. Firstly, the gradient differs from one position to the next and secondly, the answer for a given position (x_1, y_1) will be influenced by the magnitude of the interval Δx.

A new definition is required to specify the gradient of the curve at a given position unambiguously. The problem is solved if the gradient of a curve at a given position is defined as the gradient of the tangent to the curve at that position. Strictly, this definition will be valid only if a single tangent exists at the specified point.

The gradient of the tangent can be calculated by using the procedure for a straight line. Although the definition is correct, it does not contain information about a method to calculate the gradient of the curve at the required position. A beginner might try a graphic method by drawing a tangent at the given position and base the calculation on measurement. The accuracy of this method is not very reliable.

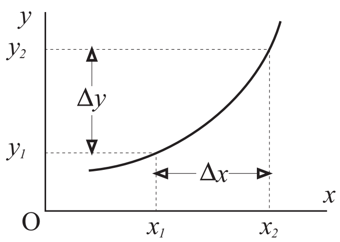


Figure 2.4-2

Figure 2.4-1 not only gives a slightly different approach to the definition, but also supplies information for the calculation of the gradient of the curve at a given position. Suppose that the gradient is required at position 1 of which the co-ordinates are (x_1, y_1). Position 2 is a second point on the curve to the right of 1 $(x_2 > x_1)$. The gradient of the cord (the straight line joining points 1 and 2) will generally differ from the gradient of the tangent at point 1. From the sketch it should be clear that the gradient of the cord will differ less from the gradient of the tangent if position 2 is chosen nearer to 1. It may also be stated in the following way: As position 2 tends to position 1, the gradient of the cord tends to the gradient of the tangent, which is the gradient of the curve at the required position.

To state that point 2 tends to point 1, is the same as to state that $x_2 \to x_1$, or $\Delta x \to 0$. If $\Delta x \to 0$, then $\Delta y \to 0$ and the gradient apparently becomes $\Delta y/\Delta x = 0/0$ which is undefined. The sketch in Figure 2.4-1 shows that the gradient of the cord tends to a limit (that of the tangent) when $\Delta x \to 0$, and the technique discussed in section 2.2.2 may be applied to calculate it. This allows for a better definition for the gradient of a curve which also incorporates a method for its calculation.

$$\text{gradient} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} \qquad 2.4(1)$$

Other notations which are preferred by some, are as follows:

$$\text{gradient} = \lim_{h \to 0} \frac{y(x_1 + h) - y(x_1)}{h} = \lim_{h \to 0} \frac{f(x_1 + h) - f(x_1)}{h} \qquad 2.4(2)$$

Example: Calculate the gradient of the function $y = x^2 - 6x + 10$ at the positions where (a) $x = 4$, (b) $x = 1,5$. The graph is shown in Figure 2.4-3.

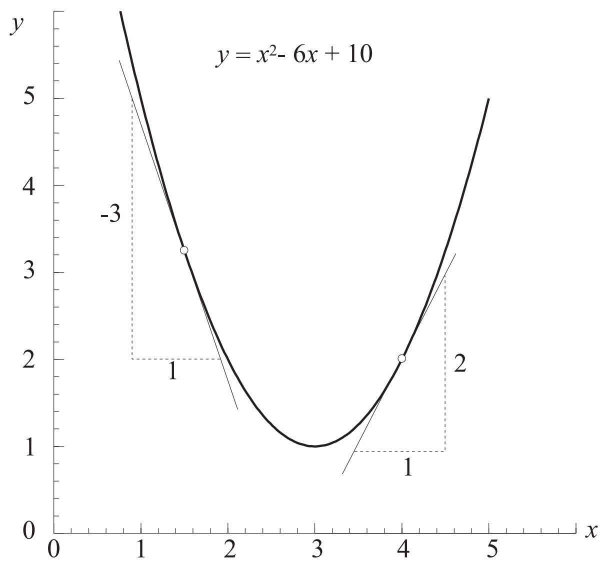


Figure 2.4-3

$$\begin{array}{llcl}
\text{(a)}\quad \text{At} \quad x_1 = 4, & y_1 & = & y(4) = 4^2 - 6(4) + 10 = 2 \\
\quad\text{At} \quad x_2 = 4 + \Delta x, & y_2 & = & y(4 + \Delta x) \\
 & & = & (4 + \Delta x)^2 - 6(4 + \Delta x) + 10 \\
 & & = & (16 + 8\Delta x + [\Delta x]^2) - (24 + 6\Delta x) + 10 \\
 & & = & 2 + 2\Delta x + [\Delta x]^2
\end{array}$$

From the definition, Equation 2.4(1), it follows

$$\begin{aligned}
\text{gradient} &= \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \frac{y_2 - y_1}{x_2 - x_1} \\
&= \lim_{\Delta x \to 0} \frac{(2 + 2\Delta x + [\Delta x]^2) - (2)}{(4 + \Delta x) - (4)} = \lim_{\Delta x \to 0} \frac{2\Delta x + [\Delta x]^2}{\Delta x} \\
&= \lim_{\Delta x \to 0} (2 + \Delta x) = 2
\end{aligned}$$

(b) In the same way as in (a) above, the gradient may be calculated at $x = 1,5$.

$$\begin{aligned}
\text{gradient} &= \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \frac{y(1,5 + \Delta x) - y(1,5)}{\Delta x} \\
&= \lim_{\Delta x \to 0} \frac{([1,5 + \Delta x]^2 - 6[1,5 + \Delta x] + 10) - (1,5^2 - 6[1,5] + 10)}{\Delta x} \\
&= \lim_{\Delta x \to 0} \frac{(2,25 + 3\Delta x + [\Delta x]^2 - 9 - 6\Delta x + 10) - (2,25 - 9 + 10)}{\Delta x} \\
&= \lim_{\Delta x \to 0} \frac{-3\Delta x + [\Delta x]^2}{\Delta x} = \lim_{\Delta x \to 0} (-3 + \Delta x) = -3
\end{aligned}$$

Excluding cases where the behaviour of the function is such that its gradient is undefined and does not exist at a given position, the method may be used in principle to calculate the gradient of a function at any position.

2.5 Derived functions or derivatives

From the work done in the previous section, it should be obvious that the gradient of a function changes if the abscissa changes. The gradient is indeed also a function of the independent variable. If it is possible to determine the function which describes the gradient (i.e. an equation or a formula by means of which the gradient may be calculated at any position), it will not be necessary to follow the procedure which was illustrated in section 2.4 for the calculation of the gradient at a given position.

The function which expresses the gradient in terms of x is called the **gradient function**, the **derived function**, or **derivative** and it will allow the calculation of the gradient at each position where it exists.

For the calculation of the derivative a similar procedure is followed as that in section 2.4. The difference is that instead of using a fixed value x_1 at which to calculate the gradient, an unspecified value x is used. We use x instead of x_1 and $x + \Delta x$ in stead of x_2. The derivative may be defined as follows:

$$\text{derivative} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \frac{y(x + \Delta x) - y(x)}{\Delta x}$$

Unfortunately there are a large number of different notations to indicate the derivatives of $y = y(x)$. Although each notation has advantages, this variety may initially confuse one who is new to the study of calculus. A number of different notations are shown.

$\dfrac{dy}{dx}$ or dy/dx : This is read *d-y-d-x* and *not dy* divided by dx. It is a symbolic notation for the derivative, i.e. a function by means of which the gradient of $y = y(x)$ may be calculated. In this book preference will be given to this notation

$\dfrac{d}{dx}(y)$: This notation is read *d-d-x* of y. It is seldom used symbolically since the position of y is usually occupied by the expression for y in terms of x. The portion d/dx is known as an **operator** which may be interpreted as a procedure to be applied to y in order to determine its derivative.

$D(y)$ or $D_x y$: The same as above with operators D and D_x instead of d/dx. They are read *d*-of-*y* and *d*-*x*-of-*y* respectively.

$y', f'(x), \dot{y}$: They are read y-prime, f-prime-of-x and y-dot respectively. They are all symbolic notations for the derivative of y.

By means of the notation which is preferred in this book, the definition for the derivative of the function $y = y(x)$ may be written as follows:

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \frac{y(x + \Delta x) - y(x)}{\Delta x} \qquad 2.5(1)$$

This definition is now applied to determine the derivative of the function of which the graph is shown in Figure 2.4-3, namely $y = x^2 - 6x + 10$.

$$\begin{aligned} \frac{dy}{dx} &= \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \frac{y(x + \Delta x) - y(x)}{\Delta x} \\ &= \lim_{\Delta x \to 0} \frac{([x + \Delta x]^2 - 6[x + \Delta x] + 10) - (x^2 - 6x + 10)}{\Delta x} \end{aligned}$$

$$
\begin{aligned}
&= \lim_{\Delta x \to 0} \frac{(x^2 + 2x[\Delta x] + [\Delta x]^2 - 6x - 6\Delta x + 10) - (x^2 - 6x + 10)}{\Delta x} \\
&= \lim_{\Delta x \to 0} \frac{2x[\Delta x] - 6\Delta x + [\Delta x]^2}{\Delta x} \\
&= \lim_{\Delta x \to 0} (2x - 6 + \Delta x) \\
&= 2x - 6
\end{aligned}
$$

The formula $dy/dx = 2x-6$ enables one to calculate the gradient of the function $y = x^2 - 6x + 10$ at any value of x.

$$\left.\frac{dy}{dx}\right|_{x=-1} = 2(-1) - 6 = -8 \qquad \left.\frac{dy}{dx}\right|_{x=1,5} = 2(1,5) - 6 = -3$$

$$\left.\frac{dy}{dx}\right|_{x=0} = 2(0) - 6 = -6 \qquad \left.\frac{dy}{dx}\right|_{x=4} = 2(4) - 6 = 2$$

The two values on the right agree with those calculated previously with much more effort. It should be clear that the determination of the derivative of a function and then using it to calculate the gradient of the function at any desired position, is much easier than calculating the gradient at each position by using the definition.

The procedure by which the derivative is determined is called **differentiation** and the verb is **differentiate**. An assignment to *differentiate a given function*, means that its derivative is to be determined. The only means at our disposal at this stage is the procedure which has been followed in the above example.

2.6 Useful differentiation formulas

A number of functions often occur in the study of physics. For this reason it has definite advantages to differentiate them by using the definition and then remembering the results. The results are then simply used as a formulae to write down the answers when such functions are to be differentiated.

2.6.1 $y = x^n$ in which n is any real finite constant

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \frac{y(x + \Delta x) - y(x)}{\Delta x} = \lim_{\Delta x \to 0} \frac{(x + \Delta x)^n - x^n}{\Delta x}$$

But $(x+\Delta x)^n = x^n + nx^{n-1}\Delta x + [n(n-1)/2!]x^{n-2}[\Delta x]^2$ + higher powers of Δx. This result follows directly from the binomial theorem which is explained in section 1.4. Using this result, the derivative becomes

$$\begin{aligned}\frac{dy}{dx} &= \lim_{\Delta x\to 0}\frac{(x^n + nx^{n-1}\Delta x + \text{ higher powers of } \Delta x) - x^n}{\Delta x}\\ &= \lim_{\Delta x\to 0}(nx^{n-1} + \text{ powers of } \Delta x) = nx^{n-1} \qquad 2.6(1)\end{aligned}$$

2.6.2 $y = ax^n$ in which a and n are real finite numbers

The derivative is calculated in exactly the same way as that in 2.6.1. It is left to the reader as an exercise.

$$\frac{d}{dx}(ax^n) = anx^{n-1} \qquad 2.6(2)$$

Equation 2.6(1) is the special case of 2.6(2) where $a = 1$. Since the latter represents a more general case it is the one that needs to be remembered.

2.6.3 $y = \sin ax$ in which a is any real finite number

$$\begin{aligned}\frac{dy}{dx} &= \lim_{\Delta x\to 0}\frac{\sin a(x+\Delta x) - \sin ax}{\Delta x}\\ &= \lim_{\Delta x\to 0}\frac{\sin ax\cos a\Delta x + \cos ax \sin a\Delta x - \sin ax}{\Delta x}\end{aligned}$$

In the last step, Equation 1.7(4), namely $\sin(\theta+\phi) = \sin\theta\cos\phi + \cos\theta\sin\phi$, was used. Since Δx and also $a\Delta x$ may be chosen to be as small as desired, the following approximations may be applied: $\sin a\Delta x \approx a\Delta x$ (x in radians) and $\cos a\Delta x \approx 1$. The derivative becomes

$$\begin{aligned}\frac{dy}{dx} &= \lim_{\Delta x\to 0}\frac{\sin ax + a\Delta x\cos ax - \sin ax}{\Delta x}\\ &= a\cos ax \qquad 2.6(3)\end{aligned}$$

2.6.4 $y = \cos ax$ in which a is any real finite number

The procedure is the same as that used in 2.6.3 and it is left to the reader as an exercise.

$$\frac{d}{dx}(\cos ax) = -a\sin ax \qquad 2.6(4)$$

Examples:

Problems (a) to (f) are applications of the result in 2.6.2

(a) $d/dx(x^3) = 3x^{3-1} = 3x^2$

(b) $d/dx(7x^3) = 21x^2$

(c) $d/dx(7) = 0$ The graph is horizontal. See Fig. 2.3-2(b).

(d) $d/dx(1/x^3) = d/dx(x^{-3}) = -3x^{-3-1} = -3x^{-4}$

(e) $d/dx(x^{1/3}) = (1/3)x^{1/3-1} = (1/3)x^{-2/3}$

(f) $d/dx(\sqrt{x}) = d/dx(x^{1/2}) = (1/2)x^{-1/2} = 1/(2\sqrt{x})$

Problems (g) to (i) are applications of the results in 2.6.3 and 2.6.4.

(g) $d/dx(\sin 5x) = 5\cos 5x$

(h) $d/dx(\sin x) = \cos x$

(i) $d/dt(\cos 7t) = -7\sin 7t$

2.7 Useful differentiation rules

2.7.1 The sum rule

The functions for which the derivatives were calculated in 2.6, all consist of a single term. The sum rule shows how a function consisting of the sum of a number of terms may be differentiated.

Let $u = u(x)$ and $v = v(x)$, i.e. both u and v are functions of x. Form a new function by the addition of $u(x)$ and $v(x)$: $y = u(x) + v(x)$.

If x increases by Δx, then $u(x)$ and $v(x)$ will generally change and that will also be the case for $y = y(x)$ so that

$$\begin{aligned} u(x+\Delta x) &= u + \Delta u \qquad \text{and} \qquad v(x+\Delta x) = v + \Delta v \\ \text{and} \qquad y(x+\Delta x) &= y + \Delta y \\ \text{so that} \quad y + \Delta y &= (u+\Delta u) + (v + \Delta v) \\ \Rightarrow \qquad \Delta y &= \Delta u + \Delta v \end{aligned}$$

For $\Delta x \neq 0$, the left and right-hand sides may be divided by Δx as follows:

$$\begin{aligned} \frac{\Delta y}{\Delta x} &= \frac{\Delta u}{\Delta x} + \frac{\Delta v}{\Delta x} \\ \text{so that} \quad \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} &= \lim_{\Delta x \to 0} \left(\frac{\Delta u}{\Delta x} + \frac{\Delta v}{\Delta x} \right) \end{aligned}$$

There is a theorem about the limit of a sum which will not be proved here. It is as follows: *The limit of the sum of a number of expressions is equal to the sum of the limits of the individual expressions.* Applied to the above, it follows that

$$\begin{aligned} \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} &= \lim_{\Delta x \to 0} \frac{\Delta u}{\Delta x} + \lim_{\Delta x \to 0} \frac{\Delta v}{\Delta x} \\ \text{which, by definition, is} \quad \frac{dy}{dx} &= \frac{du}{dx} + \frac{dv}{dx} \qquad 2.7(1) \end{aligned}$$

This relationship is known as the sum rule and may be formulated in words as follows: *The derivative of the sum of a number of expressions is equal to the sum of the derivatives of the individual expressions.* Since u and v may each consist of more terms than one, the sum rule applies to the sum of any number of terms.

Examples:

(a) $d/dx(3x^2 - 4x^5) = 6x - 20x^4$

(b) $d/dx(5x^4 - 3\sin 4x) = 20x^3 - 12\cos 4x$

(c) $d/dx(2x^{-3} + 3\cos 2x - 7) = -6x^{-4} - 6\sin 2x$

(d) $d/dr(\sqrt{r^3} - 4\cos 2r - 4r) = 1,5r^{1/2} + 8\sin 2r - 4$

2.7.2 The product rule

The product rule is used to differentiate the product of two functions. If $u = u(x)$ and $v = v(x)$ and a new function is formed by their product $y = y(x) = u(x) \times v(x)$, then

$$\begin{aligned} y(x + \Delta x) &= u(x + \Delta x) \times v(x + \Delta x) \\ \text{i.e.} \quad y + \Delta y &= (u + \Delta u) \times (v + \Delta v) \\ &= uv + u\Delta v + v\Delta u + \Delta u\Delta v \\ \Rightarrow \quad \Delta y &= u\Delta v + v\Delta u + \Delta u\Delta v \end{aligned}$$

$$\begin{aligned}
\text{so that} \qquad \frac{\Delta y}{\Delta x} &= u\frac{\Delta v}{\Delta x} + v\frac{\Delta u}{\Delta x} + \Delta u\frac{\Delta v}{\Delta x} \\
\text{and} \qquad \lim_{\Delta x\to 0}\frac{\Delta y}{\Delta x} &= \lim_{\Delta x\to 0}\left(u\frac{\Delta v}{\Delta x} + v\frac{\Delta u}{\Delta x} + \Delta u\frac{\Delta v}{\Delta x}\right) \\
&= \lim_{\Delta x\to 0} u\frac{\Delta v}{\Delta x} + \lim_{\Delta x\to 0} v\frac{\Delta u}{\Delta x} + \lim_{\Delta x\to 0} \Delta u\frac{\Delta v}{\Delta x}
\end{aligned}$$

Another theorem about limits which is needed and which will not be proved here, is as follows: *The limit of the product of two expressions is equal to the product of the limits of the two individual expressions.* The last term on the right-hand side of the above equation thus becomes

$$\lim_{\Delta x\to 0}\Delta u\frac{\Delta v}{\Delta x} = (\lim_{\Delta x\to 0}\Delta u)\left(\lim_{\Delta x\to 0}\frac{\Delta v}{\Delta u}\right) = 0\times\frac{dv}{dx} = 0$$

Finally it follows that

$$\frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx} \qquad 2.7(2)$$

which is known as the product rule.

Examples:

(a) $y = (3x^2 - 5)(2x^3 + 7x)$. Calculate dy/dx.

$$\begin{aligned}
\text{Let}\ u &= 3x^2 - 5. \qquad \text{Then}\ du/dx = 6x \\
\text{and}\ v &= 2x^3 + 7x. \qquad \text{Then}\ dv/dx = 6x^2 + 7
\end{aligned}$$

$$\begin{aligned}
\text{But}\ \frac{dy}{dx} &= u\frac{dv}{dx} + v\frac{du}{dx} \\
&= (3x^2 - 5)(6x^2 + 7) + (2x^3 + 7x)(6x) \\
&= 18x^4 - 9x^2 - 35 + 12x^4 + 42x^2 \\
&= 30x^4 + 33x^2 - 35
\end{aligned}$$

The answer may be calculated directly by the sum rule after the expansion of the product of the two factors.

$y = (3x^2 - 5)(2x^3 + 7x) = 6x^5 + 11x^3 - 35x$, so that $dy/dx = 30x^4 + 33x^2 - 35$

(b)

$$\begin{aligned}
y &= (3x^4 - 7x^2)\sin 3x \\
\frac{dy}{dx} &= (3x^4 - 7x^2)\frac{d}{dx}(\sin 3x) + (\sin 3x)\frac{d}{dx}(3x^4 - 7x^2) \\
&= (3x^4 - 7x^2)(3\cos 3x) + (12x^3 - 14x)(\sin 3x)
\end{aligned}$$

(c)

$$\begin{aligned} y &= \sin^2 x = (\sin x)(\sin x) \\ \frac{dy}{dx} &= (\sin x)(\cos x) + (\sin x)(\cos x) \\ &= 2\sin x\cos x = \sin 2x \end{aligned}$$

2.7.3 The quotient rule

If a function consists of the quotient of two expressions, it may be differentiated by means of the quotient rule. If $u = u(x)$ and $v = v(x)$ and a new function is formed by their quotient as follows: $y = y(x) = u(x)/v(x)$, then

$$\begin{aligned} y + \Delta y &= \frac{u + \Delta u}{v + \Delta v} \\ \Delta y &= \frac{u + \Delta u}{v + \Delta v} - y = \frac{u + \Delta u}{v + \Delta v} - \frac{u}{v} \\ &= \frac{v(u + \Delta u) - u(v + \Delta v)}{v(v + \Delta v)} = \frac{v\Delta u - u\Delta v}{v^2 + v\Delta v} \\ \text{so that} \quad \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} &= \lim_{\Delta x \to 0} \frac{v(\Delta u/\Delta x) - u(\Delta v/\Delta x)}{v^2 + v\Delta v} \end{aligned}$$

The following theorem regarding the limit of a quotient is stated without proof: *The limit of the quotient of two expressions is equal to the quotient of the limits of the two expressions.* From this follows that

$$\begin{aligned} \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} &= \frac{\lim_{\Delta x \to 0} v(\Delta u/\Delta x) - \lim_{\Delta x \to 0} u(\Delta v/\Delta x)}{\lim_{\Delta x \to 0}(v^2 + v\Delta v)} \\ \text{so that} \quad \frac{dy}{dx} &= \frac{v(du/dx) - u(dv/dx)}{v^2} \qquad 2.7(3) \end{aligned}$$

which is known as the quotient rule.

Examples:

(a)

$$\begin{aligned} y &= \frac{x^3 + 8}{x + 2}, \quad x \neq -2 \\ \text{Let} \quad u &= x^3 + 8. \qquad \text{Then } du/dx = 3x^2 \\ \text{and} \quad v &= x + 2. \qquad \text{Then } dv/dx = 1 \\ \text{but} \quad \frac{dy}{dx} &= \frac{v(du/dx) - u(dv/dx)}{v^2} \\ &= \frac{(x + 2)(3x^2) - (x^3 + 8)(1)}{(x + 2)^2} \\ &= \frac{3x^3 + 6x^2 - x^3 - 8}{(x + 2)^2} = \frac{2x^3 + 6x^2 - 8}{(x + 2)^2} \\ &= 2x - 2 \end{aligned}$$

The final answer may be obtained by factorising the numerator or using long division.

For the calculation of this derivative the use of the quotient rule could be avoided since it could be written directly that

$$y = \frac{x^3+8}{x+2} = \frac{(x+2)(x^2-2x+4)}{x+2} = x^2 - 2x + 4$$

Application of the sum rule gives $dy/dx = 2x - 2$. This control shows that the quotient rule was used correctly.

(b) $y = \dfrac{3x-5}{7x-2}, \quad x \neq 2/7$

Let $u = 3x - 5 \;\Rightarrow\; du/dx = 3$ and $v = 7x - 2 \;\Rightarrow\; dv/dx = 7$, so that

$$\frac{dy}{dx} = \frac{(7x-2)(3)-(3x-5)(7)}{(7x-2)^2} = \frac{29}{(7x-2)^2}$$

(c) $y = \tan x$. Since $\tan x = (\sin x)/(\cos x)$, the quotient rule may be used.

$$\begin{aligned}\frac{dy}{dx} &= \frac{(\cos x)(\cos x)-(\sin x)(-\sin x)}{\cos^2 x}\\ &= \frac{\cos^2 x + \sin^2 x}{cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x\end{aligned}$$

(d) $y = (2x^3-3)^{-1} = 1/(2x^3-3)$
Let $u = 1 \;\Rightarrow\; du/dx = 0$ and $v = 2x^3 - 3 \;\Rightarrow\; dv/dx = 6x^2$

$$\frac{dy}{dx} = \frac{(2x^3-3)(0)-(1)(6x^2)}{(2x^3-3)^2} = -\frac{6x^2}{(2x^3-3)^2}$$

By means of the chain rule which is treated in section 2.7.4, this differentiation may be done in a much simpler way.

(e) $y = \csc x = 1/\sin x$.

$$\frac{dy}{dx} = \frac{(\sin x)(0)-(1)(\cos x)}{\sin^2 x} = -\frac{\cos x}{\sin^2 x} = -\csc x \cot x$$

In this calculation the chain rule would also have supplied the result by a simpler method.

2.7.4 The chain rule

The chain rule is a most powerful tool in the calculation of derivatives and is used mostly in differentiating **a function of a function**. We call $\cos x$ a function of x and although $\cos(2x^2-1)$ is also a function of x, it may be taken to be a function of $(2x^2-1)$ which in itself is also a function of x. $\cos 3x$ may be taken to be a function of $3x$, but its derivative, $(d/dx)(\cos 3x)$ may be calculated directly by the result in Equation 2.6(4). That is not the case for $(d/dx)(\cos[2x^2-1])$. If, however, the substitution $u = 2x^2-1$ is made, the function becomes $y = \cos u$ for which Equation 2.6(4) is applicable if dy/du is to be calculated. For the determination of dy/dx a different approach will have to be found.

If $y = y(u)$ and $u = u(x)$, it means that y is a function of u, which is a function of x and it may be said that if x increases by Δx, u will change by Δu. This change will cause y to change by an amount equal to Δy. For any *finite* values of Δx, Δu and Δy, we may write that

$$\frac{\Delta y}{\Delta x} = \frac{\Delta y}{\Delta u}\frac{\Delta u}{\Delta x}$$

If $\Delta x \to 0$, both Δu and Δy also tend to zero and from this follows that

$$\begin{aligned}
\lim_{\Delta x\to 0}\frac{\Delta y}{\Delta x} &= \lim_{\Delta x\to 0}\left(\frac{\Delta y}{\Delta u}\frac{\Delta u}{\Delta x}\right)\\
&= \left(\lim_{\Delta x\to 0}\frac{\Delta y}{\Delta u}\right)\left(\lim_{\Delta x\to 0}\frac{\Delta u}{\Delta x}\right)\\
\text{so that}\qquad \frac{dy}{dx} &= \frac{dy}{du}\frac{du}{dx} \qquad\qquad 2.7(4)
\end{aligned}$$

This rule is known as the chain rule and may be extended for a function of a function of a function, etc. to contain as many factors ("links") as may be necessary. Such an extension will be demonstrated in the examples.

Examples:

(a) $y = (2x^2-3)^2$. Calculate dy/dx.

$$\begin{aligned}
\text{Let}\quad u &= 2x^2-3 \quad\Rightarrow\quad du/dx = 4x\\
\text{Then}\quad y &= u^2 \quad\Rightarrow\quad dy/du = 2u = 2(2x^2-3)\\
\text{and}\quad \frac{dy}{dx} &= \frac{dy}{du}\frac{du}{dx} = 2(2x^2-3)(4x) = 16x^3-24x
\end{aligned}$$

This answer may be verified in the following way:

$$y = (2x^2-3)^2 = 4x^4-12x^2+9 \quad\Rightarrow\quad dy/dx = 16x^3-24x$$

(b) $y = \sqrt{3x^5-2x^2}$. Calculate dy/dx.

$$\text{Let}\qquad p = 3x^5-2x^2 \quad\Rightarrow\quad dp/dx = 15x^4-4x$$

$$\begin{aligned}
\text{Then} \quad y &= \sqrt{p} = p^{1/2} \quad \Rightarrow dy/dp = \frac{1}{2}p^{-1/2} = \frac{1}{2}(3x^5 - 2x^2)^{-1/2} \\
\text{and} \quad \frac{dy}{dx} &= \frac{dy}{dp}\frac{dp}{dx} = (15x^4 - 4x) \times \frac{1}{2}(3x^5 - 2x^2)^{-1/2} \\
&= (15x^4 - 4x)/(2\sqrt{3x^5 - 2x^2})
\end{aligned}$$

(c) $y = \sin(3x^2 - 2x)$. Calculate dy/dx.

$$\begin{aligned}
\text{Let} \quad s &= 3x^2 - 2x \quad \Rightarrow ds/dx = 6x - 2 \\
\text{Then} \quad y &= \sin s \quad \Rightarrow dy/ds = \cos s = \cos(3x^2 - 2x) \\
\text{and} \quad \frac{dy}{dx} &= \frac{dy}{ds}\frac{ds}{dx} = (6x - 2)\cos(3x^2 - 2x)
\end{aligned}$$

Comment: With practice this answer may be written down without showing the substitution in writing. The reader should work towards attaining this goal. In the preceding problem

$dy/db =$ the derivative of the sine function with respect to the entire angle
$\quad =$ the cosine of the same angle as shown in example (h) on page 39.

$db/dx =$ the derivative of the angle $(3x^2 - 2x)$ with respect to x and that is equal to $6x - 2$ which follows from the application of the sum rule.

The final answer is the product of these two factors.

(d) $y = \cos^3(3x^3 - 2x)$. Calculate dy/dx.

In this calculation it will be necessary to make more that one substitution and the "chain" will consist of more than two "links".

$$\begin{aligned}
\text{Let} \quad v &= 3x^3 - 2x \quad \Rightarrow dv/dx = 9x^2 - 2. \\
\text{Let} \quad u &= \cos(3x^3 - 2x) = \cos v \Rightarrow du/dv = -\sin v = -\sin(3x^3 - 2x) \\
\text{then} \quad y &= u^3 \quad \Rightarrow dy/du = 3u^2 = 3\cos^2(3x^3 - 2x) \\
\text{and} \quad \frac{dy}{dx} &= \frac{dy}{du}\frac{du}{dv}\frac{dv}{dx} = (9x^2 - 2) \times [-\sin(3x^3 - 2x)] \times [3\cos^2(3x^3 - 2x)] \\
&= -3(9x^2 - 2)\sin(3x^3 - 2x)\cos^2(3x^3 - 2x)
\end{aligned}$$

2.8 Expansion in series and exponential functions

2.8.1 Expansion in series

Subject to a few conditions which will not be mentioned or treated here, a large variety of functions of one variable may be expressed by a **power series** of the independent variable as follows:

$$f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 \ldots\ldots \qquad 2.8(1)$$

in which the coefficients a_0, a_1, a_2, etc. are constants which may be positive, negative or equal to zero. They may be calculated by a method which will be illustrated later in this book.

This phenomenon is known as the **theorem of Maclaurin**. The expansions of $\sin x$ and $\cos x$ which are given in Equations 1.7(18) and 1.7(19) respectively, and also the binomial theorem in section 1.4, are examples of such power series.

Expansion in a series is important in the study of physics. One of its many applications, is curve-fitting when experimental data is to be described by means of a smooth curve.

2.8.2 The derivative of an exponential function

The function $y = a^x$ in which the base, a, is a positive constant number, is an **exponential function**. In an exponential function the exponent is variable. It must not be confused with a power function such as $y = x^n$ in which the base is variable and the exponent a constant number. To calculate its derivative, the definition of a derivative is used.

$$\begin{aligned} dy/dx &= \lim_{h\to 0} \frac{y(x+h) - y(x)}{h} = \lim_{h\to 0} \frac{a^{x+h} - a^x}{h} \\ &= \lim_{h\to 0} \frac{a^x a^h - a^x}{h} \qquad = \lim_{h\to 0} \frac{a^x(a^h - 1)}{h} \\ &= a^x \lim_{h\to 0} \frac{a^h - 1}{h} \end{aligned}$$

In this result the factor a^x is independent of h and may be written outside the limit. Since the rest of the expression contains no x, it must be a constant which can depend on a only (h tends to zero). Provisionally we represent this constant about which very little is known, by $k(a)$. The derivative of an exponential function can thus be written as follows:

$$\frac{d}{dx}(a^x) = [k(a)]a^x \qquad 2.8(2)$$

It is possible to gain information about the magnitude of $k(a)$ by means of an experiment. A number of different values of a are chosen and a graph of $y = a^x$ is drawn in each case. By means of construction and measurement, the gradient of each graph is determined at one position. By substituting these values in equation 2.8(2), it will be possible to calculate the value of $k(a)$ for each value of a.

When $x = 1$, the value of the function, $y = 1^x$, is always equal to 1. The graph is a straight line parallel to the x-axis of which the gradient is zero at all positions as shown in Figure 2.8-1. Substitution in Equation 2.8(2) gives

$$0 \;=\; [k(1)] \times 1^x \;=\; [k(1)] \times 1$$

so that

$$k(1) \;=\; 0$$

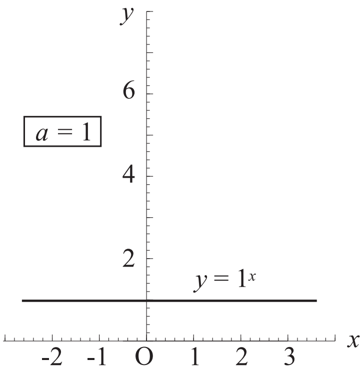


Figure 2.8-1

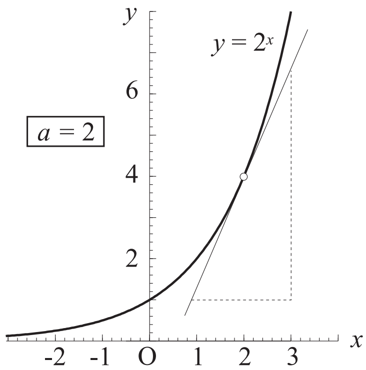


Figure 2.8-2

The next function is that for which $a = 2$ so that $y = 2^x$. The graph is shown in Figure 2.8-2. Its gradient is different at each position. The tangent at $x = 2$ is drawn as well as it is possible by means of a ruler and its gradient determined by measurement and calculation. According to the construction shown in the sketch, the gradient is approximately equal to 2,8. By substituting the relevant values in Equation 2.8(2), it follows that

$$2,8 \;\approx\; [k(2)] \times 2^2 = 4k(2)$$

$$\text{so that} \qquad k(2) \approx 0,7$$

Since it is based on a construction of questionable accuracy, this result is only an approximation. It is, however, quite clear that $k(2) < 1$ and about this result there can be little doubt. It may be verified by using other points on the graph in Figure 2.8-2.

If the same procedure is followed for $y = 3^x$, it will follow that $k(3) \approx 1,1$. Similarly $k(4) \approx 1,4$. It is quite clear that a value of a exists between 2 and 3 for which $k(a) = 1$. This value of a is indicated by the symbol e. At this stage it is known that $k(e) = 1$ and $2 < e < 3$. From this and Equation 2.8(2) follows that

$$\frac{d}{dx}(e^x) = [k(e)] \times e^x = 1 \times e^x = e^x \qquad 2.8(3)$$

Experimentally we have discovered a number which has the strange property that if it is used as the base of an exponential function, the function is the same as its derivative. The expansion of this function as a power series, allows one to calculate the numerical value of e. From 2.8(1) it follows that

$$e^x = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \ldots$$

The coefficients a_0, a_1, a_2, etc. may be calculated in the way which is illustrated below.

$$\begin{array}{lrcl}
 & e^x & = & a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \ldots \\
\Rightarrow & e^0 & = & 1 \ = \ a_0 + 0 + 0 + 0 \ldots \\
\text{so that} & a_0 & = & 1 \\
\text{Furthermore} & \frac{d}{dx}(e^x) & = & \frac{d}{dx}(a_0 + a_1 x + a_2 x^2 + a_3 a^3 + a_4 x^4 + \ldots) \\
 & e^x & = & 0 + a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + \ldots) \\
\Rightarrow & e^0 & = & 1 \ = \ a_1 + 0 + 0 + 0 + \ldots \\
\text{so that} & a_1 & = & 1 \\
\text{Also} & \frac{d}{dx}(e^x) & = & \frac{d}{dx}(a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 a^3 \ldots) \\
 & e^x & = & 0 + 2a_2 + 6a_3 x + 12a_4 x^2 + \ldots \\
\Rightarrow & e^0 & = & 1 \ = \ 2a_2 + 0 + 0 + 0 + \ldots \\
\text{so that} & a_2 & = & 1/2 \ = \ 1/2!
\end{array}$$

In the same way it may be shown by successive differentiation that the other coefficients are $1/3!$, $1/4!$, $1/5!$, $1/6!$, etc. so that

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \ldots \qquad 2.8(4)$$

When $x = 1$, the value of e may be calculated.

$$\begin{aligned} e^1 = e &= 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \frac{1}{5!} + \ldots \qquad 2.8(5) \\ \Rightarrow \qquad e &= 2,71828183\ldots \end{aligned}$$

As π, e is an irrational number which is used as the base for natural or Napierian logarithms with the following notation:

$$\text{If} \qquad b = e^c, \qquad \text{then} \qquad c = \log_e b \quad = \quad \ln b$$

2.8.3 The derivative of a logarithmic function

Consider the expression $y = 2x+6$ by which it is understood that y is a function of x which is described by the given equation. It is possible to rearrange this equation in such a way that x is the subject and is thus a function of y. The rearrangement is as follows:

$$x = \frac{1}{2}y - 3$$

Although it is not always possible to exchange the roles of the dependent and independent variables unconditionally, a pair of functions which originate in this way, have properties which are of special interest in this study.

Exchanging the roles of x and y as dependent and independent variables in the exponential function $y = e^x$, yields the following pair of functions:

$$y = e^x \qquad \text{and} \qquad x = \ln y$$

Consider the function $y = y(x)$ and the function $x = x(y)$ which is obtained if the roles of x and y are exchanged. It is assumed that the change of the subject of the equation is possible and valid. For such a pair of functions, a simple relationship exists between their graphs of which possible examples are shown in Figure 2.8-3.

At position P_1 in sketch (a), the gradient of the function $y = y(x)$ is given by

$$\left.\frac{dy}{dx}\right|_{P_1} = \tan\alpha = \tan(90^\circ - \beta) = \cot\beta = 1/\tan\beta$$

and at the corresponding point P_2 on the graph of $x = x(y)$ in which the roles of x and y are interchanged, the gradient is given by

$$\left.\frac{dx}{dy}\right|_{P_2} = \tan\beta$$

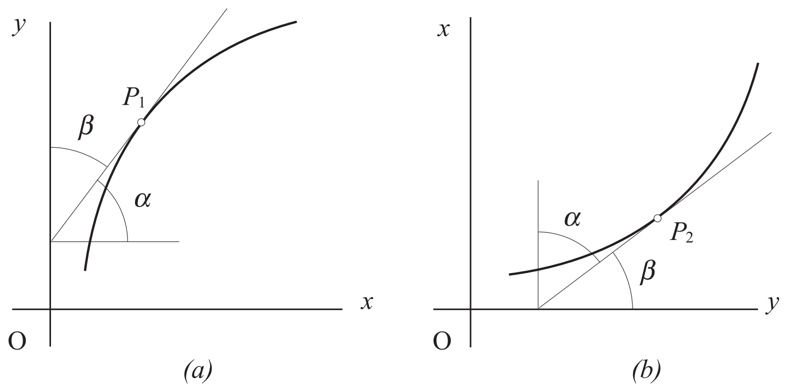


Figure 2.8-3

From this follows an important result which is often used in physics:

$$\frac{dy}{dx} = \left(\frac{dx}{dy}\right)^{-1} \qquad \text{if } \frac{dy}{dx} \neq 0 \qquad 2.8(6)$$

Consider once again the following pair of functions $y = y(x)$ and $x = x(y)$ in which the roles of x and y as dependent and independent variables are interchanged:

$$\begin{aligned} y &= \ln x \quad \text{and} \quad x = e^y \\ \text{From 2.8(3):} \qquad \frac{dx}{dy} &= \frac{d}{dy}(e^y) = e^y = x \\ \text{but} \qquad \frac{dx}{dy} &= \left(\frac{dy}{dx}\right)^{-1} \quad \text{or} \quad \frac{dy}{dx} = \left(\frac{dx}{dy}\right)^{-1} \\ \text{so that} \quad \frac{d}{dx}(y) &= \frac{d}{dx}(\ln x) = \frac{1}{x} \end{aligned}$$

This gives the fifth differentiation formula which is important enough to remember.

$$\frac{d}{dx}(\ln x) = \frac{1}{x} \qquad 2.8(7)$$

This result enables one to differentiate exponential functions with base other than e.

$$\text{If} \qquad y = a^x \quad \text{then} \quad \ln y = x \ln a \quad \text{or} \quad x = \ln y / \ln a$$

$$\begin{aligned} \text{now} \qquad \frac{dx}{dy} &= \frac{d}{dy}\left(\frac{\ln y}{\ln a}\right) = \frac{1}{\ln a}\frac{1}{y} \\ \text{so that} \qquad \frac{dy}{dx} &= (\ln a)y = (\ln a)a^x \end{aligned}$$

This result is the last of the differentiation formulas which are necessary for this course.

$$\frac{d}{dx}(a^x) = (\ln a)a^x \qquad 2.8(8)$$

This result shows that the constant $k(a)$ which occurred in 2.8(2), is given by $k(a) = \ln a$.

Examples:

(a) $y = e^{2x^3}$ Calculate dy/dx.

$$\begin{aligned} \text{Let} \qquad u &= 2x^3 \qquad \Rightarrow du/dx = 6x^2 \\ \text{now} \qquad y &= e^u \qquad \Rightarrow dy/du = e^u = e^{2x^3} \\ \text{and} \qquad \frac{dy}{dx} &= \frac{dy}{du}\frac{du}{dx} = (e^{2x^3}) \times (6x^2) = 6x^2e^{2x^3} \end{aligned}$$

(b) $y = e^{-6x^2}$ Calculate dy/dx.

$$\frac{dy}{dx} = (e^{-6x^2}) \times \frac{d}{dx}(-6x^2) = -12xe^{-6x^2}$$

(c) $y = \ln x^2$ Calculate du/dx.

$$\begin{aligned} \text{Let} \qquad u &= x^2 \qquad \Rightarrow du/dx = 2x \\ \text{now} \qquad y &= \ln u \qquad \Rightarrow dy/du = 1/u = 1/x^2 \\ \text{so that} \qquad \frac{dy}{dx} &= \frac{dy}{du}\frac{du}{dx} = 2x/x^2 = 2x^{-1} \end{aligned}$$

This result may also be calculated in the following way:

$$y = \ln x^2 = 2\ln x \qquad \Rightarrow dy/dx = (d/dx)(2\ln x) = 2x^{-1}$$

(d) $y = \ln[x^3/(x+1)]$ Calculate dy/dx.
In the first step this function is written as the sum of two terms and then the sum rule and the chain rule are used to calculate its derivative. The substitution will not be shown as it was done in the previous problems.

$$y = \ln x^3 - \ln(x+1)$$

$$\begin{aligned} dy/dx &= (x^3)^{-1} \times \frac{d}{dx}(x^3) - (x+1)^{-1} \times \frac{d}{dx}(x+1) \\ &= (x^{-3})(3x^2) - (x+1)^{-1} \times 1 \\ &= 3x^{-1} - (x+1)^{-1} = (2x+3)/(x^2+x) \end{aligned}$$

2.9 Summary of the differentiation formulas and rules

$$\begin{array}{|llllll|} \frac{d}{dx}(ax^n) & = & nax^{n-1} & \frac{d}{dx}(e^{ax}) & = & ae^{ax} \\ \frac{d}{dx}(\sin ax) & = & a\cos ax & \frac{d}{dx}(a^x) & = & (\ln a)a^x \\ \frac{d}{dx}(\cos ax) & = & -a\sin ax & \frac{d}{dx}(\ln x) & = & x^{-1} \end{array}$$

If $u = u(x)$ and $v = v(x)$, then the following apply:

sum rule:	$(d/dx)(u \pm v)$	$=$	$du/dx \pm dv/dx$
product rule:	$(d/dx)(u \times v)$	$=$	$u(dv/dx) + v(du/dx)$
quotient rule:	$(d/dx)(u \div v)$	$=$	$[v(du/dx) - u(dv/dx)] \times v^{-2}$

If $y = y(p)$ and $p = p(x)$, then the following applies:

chain rule: $dy/dx = (dy/dp)(dp/dx)$

On condition that $dx/dy \neq 0$, then

$$dy/dx = (dx/dy)^{-1}$$

It is *imperative* that the contents of this summary are *understood and remembered*. It is *vital* to the study of introductory physics.

2.10 Second and higher-order derivatives

Consider the function $y = x^2 - 4x + 3$. Its graphic representation is shown in Figure 2.10-1(a). The derivative of this function is given by $dy/dx = 2x - 4$, and its graph is shown in Figure 2.10-1(b). From the sketch it should be clear that the function $g = dy/dx$ has a constant gradient of 2. This means that $g = g(x)$ = [the gradient of the function $y = y(x)$] increases by 2 if x increases by 1. The gradient function also has a gradient.

The gradient of the derivative may be calculated by differentiating it to x as follows:

$$\frac{d}{dx}\frac{dy}{dx} = \frac{d}{dx}(2x - 4) = 2$$

This result is called the **second derivative** of y and is indicated by any of the following notations:

$$\frac{d^2y}{dx^2}, \quad \frac{d^2}{dx^2}(y), \quad \frac{d}{dx}\frac{dy}{dx}, \quad \ddot{y}, \quad y'', \quad D^2(y)$$

They are pronounced *d-two-y-d-x-squared, d-two-d-x-squared of y, d-d-x of d-y-d-x, y-double-dot, y-double-prime* and *d-two of y*, respectively.

By means of the derivative of a function the rate of change of the function may be calculated at each position as the independent variable increases. In the same way the derivative of the derivative enables one to calculate the rate at which the gradient changes at a given position when the independent variable increases.

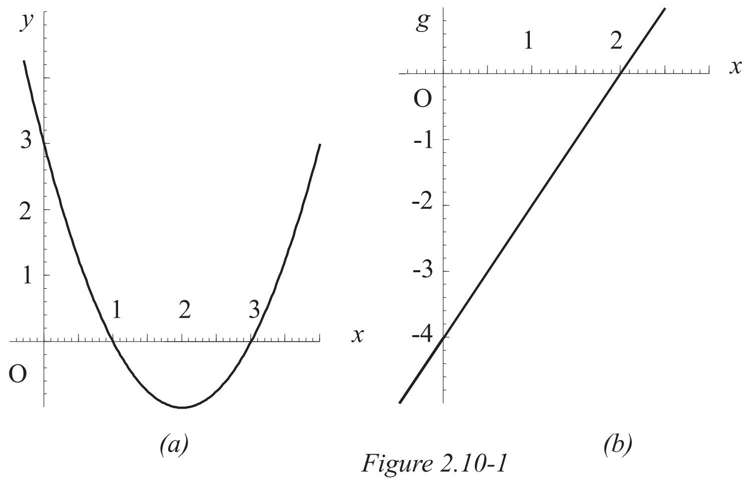


Figure 2.10-1

As it was possible to differentiate the derivative of a function, it will, in general, be possible to differentiate the derivative of the derivative of a function, etc. to obtain the third, fourth and higher-order derivatives. The notation corresponds with that of the second-order derivative namely d^ny/dx^n, in which n is the **order** of the derivative. The meaning of these higher-order derivatives will not be treated in this book but left to the ingenuity and intuition of the reader.

The sign of a second-order derivative has an interesting geometric interpretation as shown in Figure 2.10-2. In sketch (a) a curve is shown of which the gradient constantly increases from left to right, i.e. dy/dx increases with increase in x or d^2y/dx^2 is positive for the portion of the graph which is shown. If the value of d^2y/dx^2 had been larger than that shown in the sketch, the graph would have had a larger **curvature**.

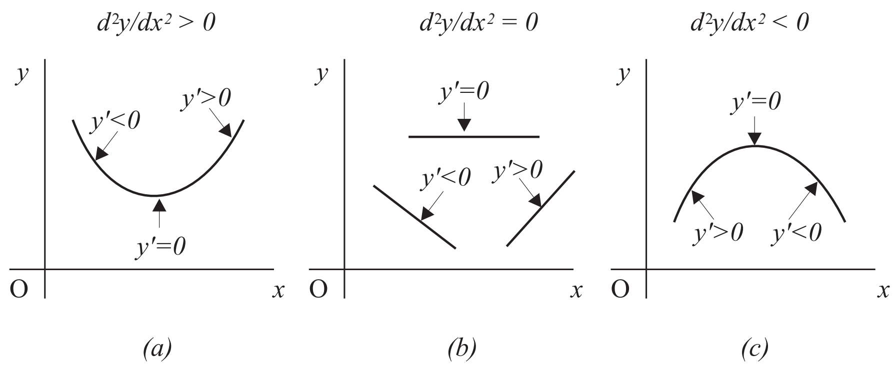


Figure 2.10-2

In sketch (b) three cases are shown in which the gradient remains constant but with a different sign in each case. The graphs are straight with no curvature. The value of the second derivative is zero in each case.

In sketch (c) the gradient decreases with increase in x. The sign of the second derivative is negative over the portion of the graph which is shown and the curve is concave when viewed in the direction in which y increases.

Examples:

(a) $y = 3x^4 - 2x^2 + 3x - 5$. Calculate d^2y/dx^2.

$$\frac{dy}{dx} = 12x^3 - 4x + 3 \quad \Rightarrow \quad \frac{d^2y}{dx^2} = \frac{d}{dx}(12x^3 - 4x + 3) = 36x^2 - 4$$

(b) $y = 3\sin 2x$. Calculate d^3y/dx^3.

$$\frac{dy}{dx} = 6\cos 2x, \qquad \frac{d^2y}{dx^2} = -12\sin 2x, \qquad \frac{d^3y}{dx^3} = -24\cos 2x$$

(c) $y = e^{2x}$. Calculate d^2y/dx^2.

$$\frac{dy}{dx} = 2e^{2x} \qquad \Rightarrow \frac{d^2y}{dx^2} = 4e^{2x}$$

(d) $y = e^{2x^2}$. Calculate d^2y/dx^2.

$$\frac{dy}{dx} = 4xe^{2x^2} \qquad \begin{aligned} \frac{d^2y}{dx^2} &= \frac{d}{dx}(4xe^{2x^2}) \\ &= 16x^2e^{2x^2} + 4e^{2x^2} \quad \text{(product rule)} \\ &= (16x^2+4)e^{2x^2} \end{aligned}$$

2.11 Maxima and minima

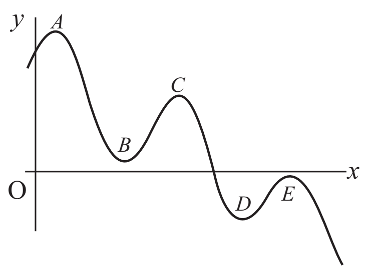


Figure 2.11-1

Consider the graph in Figure 2.11-1. Just to the left of point A, the function increases with increasing x (the gradient is positive) and just to the right of it, it decreases with increasing x (the gradient is negative.) Point A on the graph is called a **stationary point** or a **relative extreme value** of the function which the graph represents.

Just to the left of B, y decreases with increasing x and just to the right of it, it increases. B is also a stationary point or relative extreme value.

The value of the function, y, is larger at position A than at other positions in the immediate vicinity. A is called a **local maximum**. Positions C and E are also local maxima.

At B the value of the function is less than at other positions in its immediate vicinity and it is called a **local minimum**. The function also has a local minimum at position D.

The concepts of *maximum*, *minimum* and *immediate vicinity* may be confusing. Note that the maximum value at E is less than the minimum value at B. The terms *maximum* and *minimum* refer to values of the function at a *relative extreme value* in relation to positions which are very near the position. The confusion disappears if one notes the following:

1. At each relative extreme value (or stationary point), the gradient of the function is zero.
2. An extreme value is a minimum if the gradient increases on both sides of it with increasing x. This condition is met if $d^2y/dx^2 > 0$ at that position. At a relative minimum the graph is concave when viewed from above.
3. An extreme value is a maximum if the gradient decreases on both sides of it with increasing x and the condition is met if $d^2y/dx^2 < 0$ at that position. At a relative maximum, the graph is convex when viewed from above.

Positions of zero gradient may exist on a curve with no extreme value there. It is necessary to have knowledge of the properties of such conditions so that they are not erroneously confused with maxima or minima.

One such condition is when the x-axis or a line parallel to it, is an **asymptote** of the graph of a function. The exponential decay and limited exponential growth functions which are described by Equations 2.1.2(9) and (10) and of which the graphs are shown in Figure 2.1-10, are examples of such behaviour. The rectangular hyperbola which is described in Equation 2.1.2(3) and of which the graph is shown in Figure 2.1-5(b), is another example. The possibility of confusing these states with an extreme value because $dy/dx = 0$, should not happen since the value of d^2y/dx^2 is also equal to zero in each case.

Another possibility for the gradient to be equal to zero without the existence of an extreme value, is at a **point of inflexion**. A point of inflexion is the position where a graph changes from convex to concave or vice versa. Consider the graph in Figure 2.11-2(a). The gradient of the graph is zero at point a. To the left side of it, the graph is convex as viewed from above ($d^2y/dx^2 < 0$, i.e. the gradient is decreasing) and to the right side it is concave as viewed from above ($d^2y/dx^2 > 0$, i.e. the gradient is increasing). At A the value of d^2y/dx^2 is zero. Points of inflexion may also exist at positions where the gradient is not zero. The graph of such a function is shown in Figure 2.11-2(b). The value of d^2y/dx^2 is of course equal to zero at point B.

It may thus be stated that a curve has a point of inflexion at each position where $d^2y/dx^2 = 0$ and the sign of $d^2y/dx^2$is different at a position immediately before and immediately after this position. It is further true that a monotonic continuous function must have a point of inflexion between two consecutive

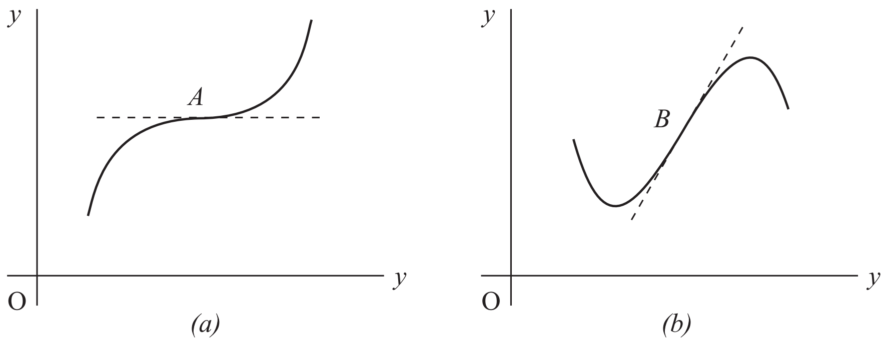


Figure 2.11-2

extreme points. A monotonic function is one which does not have more than one gradient at each point on it, and a continuous function is one which is defined at each value of the independent variable within its domain. For the purposes of this study, points of inflexion are of importance in cases such as that illustrated in Figure 2.11-2(a) where the danger exists that they might be confused with extreme values.

Summary of tests to determine the position and nature of local extreme values of a function $y = y(x)$:

1. At a local extreme value $dy/dx = 0$. At a point of inflexion dy/dx might be equal to zero, but not necessarily.

2. If $d^2y/dx^2 > 0$ at a position where $dy/dx = 0$, the function has a local extreme value at that position and it will be a local minimum.

   If $d^2y/dx^2 < 0$ at a position where $dy/dx = 0$, the function has a local extreme value at that position and it will be a local maximum.

3. If both $dy/dx = 0$ and $d^2y/dx^2 = 0$ at a given position, it necessitates an examination of higher-order derivatives before a conclusion can be reached. The following is stated without proof:

   If $(dy/dx)_{x=a} = 0$ and the smallest positive integer n for which $(d^ny/dx^n)_{x=a} \neq 0$ (i.e. all order derivatives are inspected from the first upwards and the first one which is *not* equal to zero at $x = a$, is considered), then

   (a) for even n and $(d^ny/dx^n)_{x=a} < 0$, $y(a)$ is a local maximum
   > 0, $y(a)$ is a local minimum

(b) for uneven n a point of inflexion exists at $x = a$.

In order to understand the problem of extreme values, it is necessary to study the **Taylor series** expansion of a function about the point $x = a$. This theory is not complicated. The Maclaurin series which was mentioned in section 2.8.1, is a Taylor series about the point $x = 0$.

Problems:

(1) Determine the position and nature of the extreme value(s) of the function $y = x^2 - 4x + 3$.

$$y = x^2 - 4x + 3 \qquad \Rightarrow \; dy/dx = 2x - 4$$

The function *possibly* has an extreme value where $dy/dx = 0$, i.e. where

$$2x - 4 = 0 \quad \text{or} \quad x = 2. \quad \text{There} \quad y = y(2) = (2)^2 - 4(2) + 3 = -1$$

But $d^2y/dx^2 = (d/dx)(2x - 4) = 2$. The second derivative is always positive. The function thus has an extreme value at $x = 2$ which is a local minimum. The minimum value of the function is $y_{min} = y(2) = -1$.

Comment: In the same way it can be shown that the function $y = ax^2+bx+c$ has an extreme value at $x = -b/2a$. The function is symmetrical about $x = -b/2a$. The extreme value is a minimum if $a > 0$ and a maximum if $a < 0$.

(2) Determine the position and nature of each extreme value of the function $y = 2x^3 - 9x^2 + 12x + 4$.

$$y = 2x^3 - 9x^2 + 12x + 4 \qquad \Rightarrow \quad dy/dx = 6x^2 - 18x + 12$$

The function *possibly* has extreme values where $dy/dx = 0$, i.e. where

$$\begin{aligned} 6x^2 - 18x + 12 &= 0 \\ x^2 - 3x + 2 &= 0 \\ (x-2)(x-1) &= 0 \\ \Rightarrow \qquad x = 2 \qquad \text{and} \qquad x &= 1 \\ d^2y/dx^2 = (d/dx)(6x^2 - 18x + 12) &= 12x - 18 \end{aligned}$$

At $x = 1$: $(d^2y/dx^2)_{x=1} = 12(1) - 18 = -6 < 0$
$\Rightarrow$ at $x = 1$ the function has a local maximum.

At $x = 2$: $(d^2y/dx^2)_{x=2} = 12(2) - 18 = +6 > 0$
$\Rightarrow$ at $x = 2$ the function has a local minimum.

$$\begin{aligned} \text{The local maximum is} \quad y(1) &= 2(1)^3 - 9(1)^2 + 12(1) + 4 = 9 \\ \text{The local minimum is} \quad y(2) &= 2(2)^3 - 9(2)^2 + 12(2) + 4 = 8 \end{aligned}$$

(3) An object is projected vertically upwards and its height above the ground, h, is given by $h = 20t - 5t^2$ metre, in which the time, t, is measured in seconds. Calculate the maximum height which it will reach and also when it will be reached.

$$h = 20t - 5t^2 \qquad \Rightarrow \quad dh/dt = 20 - 10t$$

The height, h, *possibly* has an extreme value where $dh/dt = 0$, i.e. where

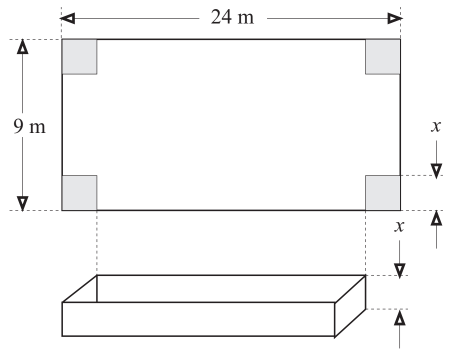
$$20 - 10t = 0 \qquad \text{or} \qquad t = 2 \text{ s}$$

$d^2h/dt^2 = (d/dt)(20 - 10t) = -10 \Rightarrow h$ has an extreme value which can be a maximum only. The maximum is given by

$$h(2) = 20(2) - 5(2)^2 = 20 \text{ m}$$

$\Rightarrow$ At time $t = 2$ s after it is projected upwards, the object reaches its maximum height of 20 m.

(4)



A farmer possesses a rectangular sheet of polyethylene plastic of 24 metres by 9 metres which he wishes to use as a waterproof lining for a rectangular dam with constant depth and vertical walls. To achieve this, equal squares of side length x are cut from the four corners and the edges of the cuts glued to form a hollow rectangular parallelepiped as is shown in the sketch.

First calculate the volume of the dam as a function of x and leave the answer in the form of a polynomial. Calculate the value of x for which the volume will give a maximum value. Calculate the maximum value.

Let the volume of the dam be represented by $V = V(x)$. Then

$$\begin{aligned} V = \text{length} \times \text{breadth} \times \text{height} &= (24 - 2x)(9 - 2x)x \\ &= 216x - 66x^2 + 4x^3 \text{ m}^3 \end{aligned}$$

V *possibly* has an extreme value where $dV/dx = 0$, i.e. where

$$\begin{aligned} dV/dx = (d/dx)(216x - 66x^2 + 4x^3) &= 216 - 132x + 12x^2 = 0 \\ \text{i.e. where} \qquad x^2 - 11x + 18 &= (x - 2)(x - 9) = 0 \\ \text{or} \qquad x &= 2 \quad \text{and} \quad x = 9 \end{aligned}$$

The value $x = 9$ is not physically possible (the width of the sheet is 9 m) and therefore the only possibility left is $x = 2$ m. The sign of the second derivative is calculated at $x = 2$ m to verify the existence of a maximum.

$d^2V/dx^2 = (d/dx)(216 - 132x + 12x^2) = 24x - 132$
and $(d^2V/dx^2)_{x=2} = 24(2) - 132 = -84 < 0$.

For $x = 2$, the volume, V, has a maximum value. The maximum value of the volume is given by

$$V = V(2) = 216(2) - 66(2)^2 + 4(2)^3 = 200 \text{ m}^3$$

2.12 Differentials

For the function $y = y(x)$ the derivative was defined and represented by the notation dy/dx. The symbols dy and dx were not defined and when written separately they have no meaning at this stage. The symbols Δy and Δx were defined and their meanings are well known and they are used quite often.

It is now possible to define dx and dy which are known as **differentials** of x and y respectively, in such a way that they will have a useful meaning. They are defined in such a way that $dy \div dx$ is the same as the gradient of the function at any given position. The meaning of these symbols becomes clear if they are studied by means of the sketch in Figure 2.12-1.

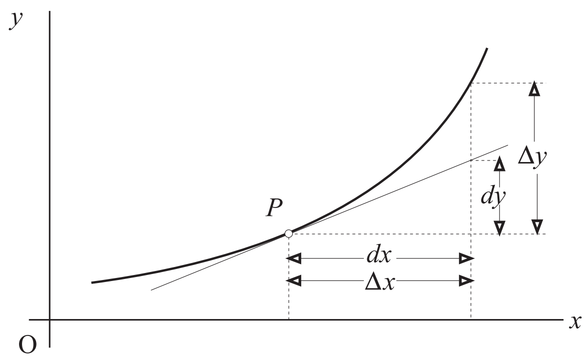


Figure 2.12-1

Consider point $P(x, y)$ on the graph of the function $y = y(x)$ at which a tangent is drawn. If x increases by an amount $\Delta x = dx$, the value of y changes *along* *the curve* by an amount Δy. The corresponding change *along the tangent* is dy. Where Δx and Δy refer to changes along the *curve*, dx $(= \Delta x)$ and dy refer to changes along the *tangent* to the curve.

This allows one to write $$dy = (dy/dx)dx \qquad 2.12(1)$$

In physics the following approximation for the change in a function is often used: If Δx is small, then
$$\Delta y \approx (dy/dx)\Delta x \qquad 2.12(2)$$

The following example will illustrate the use of equation 2.12(1):

Example:

Consider a circle with radius r. Calculate the increase in area if the radius increases by Δr. It is known that Δr is much smaller than r.

(a) First method (rather cumbersome).

The change in the area is given by

$$\begin{aligned}\Delta A &= \pi(r+\Delta r)^2 - \pi r^2 \\ &= \pi r^2 + 2\pi r\Delta r + \pi(\Delta r)^2 - \pi r^2 \\ &= 2\pi r\Delta r + \pi(\Delta r)^2\end{aligned}$$

If Δr is much smaller than r, the term which contains $(\Delta r)^2$ may be disregarded when compared to the other. For this case

$$\Delta A = 2\pi r\Delta r$$

(b) Second method (in which 2.12(2) is used and which is shorter and much more elegant than the first method).

$$A = \pi r^2 \qquad \Rightarrow \quad dA/dr = 2\pi r$$

Then $\Delta A \approx (dA/dr)\Delta r = 2\pi r\Delta r.$

2.13 PROBLEMS: CHAPTER 2

1. Calculate the derivative of the function $y = y(x)$ in each case. Also calculate the gradient of the tangent to its curve at positions $x = -1$ and $x = 2$.

(a) $y = 3x - 7$
(b) $y = 3x^2 - 2x + 14$
(c) $y = 16x^3 - 15x + 18$
(d) $y = 3x^{-3} + 12x^{-2}$
(e) $y = 7x^3 + 7x^{-3}$
(f) $y = 5x^5 - 15x^3 - 8/x$
(g) $y = 5/x^6 + 8/x^3 - (4x)^{-2}$
(h) $y = 6x^{-2} - 4/x + (4/x)^2$

2.(a) $z = 2\sin 0,5\theta$. Calculate $dz/d\theta$ where $\theta = 0$, π and 2π.
(c) $s = 30 + 200t - 5t^2$. Calculate ds/dt where $t = 2$, 20 and 50.
(c) $p = \ln q$. Calculate dp/dq where $q = 0$, 1 and 2.
(d) $y = e^x$. Calculate dy/dx where $x = 0$, -1 and 2.
(e) $g = 3\cos 2\phi$. Calculate $dg/d\phi$ where $\phi = 0$, $\pi/2$ and π.

3.(a) $s = (3t^2 - 5)(2t - 7)$. Calculate ds/dt in two different ways.
(b) $p = (5r^3 - 7)/2r$. Calculate dp/dr in two different ways.
(c) $f = (27t^3 + 8)/(3t + 2)$. Calculate df/dt in two different ways.
(d) $h = \cos^2 p$. Calculate dh/dp in two different ways. First use the product rule and then Equation 1.7(9).
(e) $y = \cot x$. Calculate dy/dx in two different ways. First express $\cot x$ in terms of $\sin x$ and $\cos x$ and apply the quotient rule. Then use the result of example (c) in section 2.7.3 and apply the quotient rule.

4. Calculate the derivatives of the following functions:

(a) $y = \sqrt{2x - 3}$
(b) $y = \sin^3 x$
(c) $y = \cos^4 x$
(d) $y = \sin^2(2x - 5)$
(e) $y = \cos^3(3x^2 - 2)$
(f) $y = \sin\sqrt{3x + 4}$
(g) $y = \ln(\sin x)$
(h) $y = \ln(\sin\sqrt{2x^3 - 3x})$
(i) $y = 2kqa/(a^2 + x^2)^{1/2}$ in which k, q and a are constants.
(j) $y = 3e^{3x^2}$

5. Calculate the first four derivatives of each of the following functions:

(a) $y = \sin x$
(b) $y = \sin 2x$
(c) $y = \cos x$
(d) $y = e^{2x}$
(e) $y = 3x^4 + 2x^3 + 3x^2 + 5x + 6$
(f) $y = 3x^{-4} + 2x^{-3} + 3x^{-2} + 5x^{-1} + 6$

6. Calculate the values of x where the following functions have extreme values. Investigate their nature (maximum or minimum) and the value of the function at each.

(a) $y = 3x^2 - 18x + 12$ (b) $y = -x^2 + 12x + 5$
(c) $y = x^3 - 8x$ (d) $y = 2x^3 - 9x^2 + 12x$
(e) $y = 2x^3 + 3x^2 - 12x + 30$ (f) $y = x^3$

7. In section 2.1.1 an example was considered in which an object was projected from the edge of a precipice. From the data in Table 2.1-1 it was not possible to calculate the maximum height. Use the theory of extreme values and calculate the instant at which the object reaches its maximum height. Use the function which is given in Equation 2.1(2). Calculate the maximum height.

8. A mortar shell is fired over a horizontal field in a direction which makes an angle of θ with the field. The position of the point of impact relative to the position from which it is fired, is represented by x which may be negative or positive as determined by the magnitude of θ. For this problem the domain of θ is given by $0 \leq \theta \leq \pi$. As a function of θ, the position of the point of impact is given by $x = 4000 \sin 2\theta$ metre in which the elevation angle, θ is measured in degrees. Calculate the values of the elevation angle for which x will have extreme values, investigate their nature (maximum or minimum) and then calculate the positions in question. Try to give a physical interpretation of the answers.

9. An existing stone wall on a farm is about 200 m long. The farmer has enough material to erect a fence of length 100 m. He wishes to use a portion of the existing wall as the one side of a rectangular pen for his cattle and the other three sides are to be completed by using the available fencing material. Calculate the dimensions and the area of the pen which is constructed in this way and will have a maximum area.

10. A manufacturer requires cylindrical tin cans in which 500 ml of fruit juice is to be marketed. The curved surface, the lid and the bottom of each tin is made from the same material. The radius of the tin is r cm and the height, h cm. (a) Write h in terms of r. (b) Calculate A, the area of the material required to make such a can, as a function of r. This function must not contain h. (c) Calculate the the value of r for which A will be a minimum. (d) Calculate the ratio r/h for a can of which the area of the material will be a minimum for a given volume.

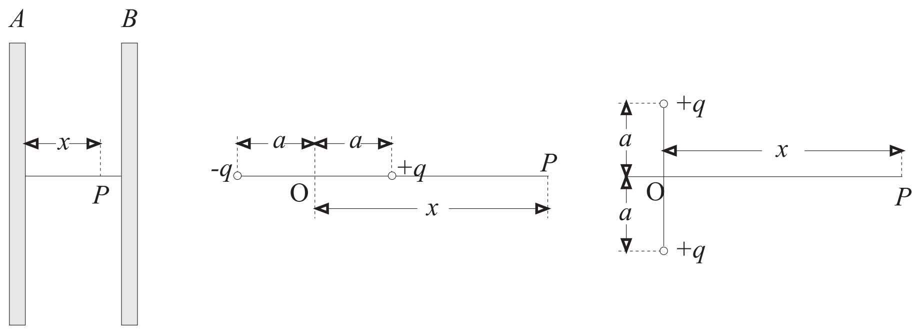
11. Repeat problem 10 for a can which does not have a lid.

12. A point moves along a straight line. A fixed point, O, on this line is used as origin from which the position, x, of the moving point is measured. To the one side of O, x is positive and negative to the other side. If x is constant, the point is at rest. If the point is in motion, x is a function of the time, t. The velocity of a moving point is defined as $v = dx/dt$. If v is not a function of t, the velocity remains constant. If, however, it is a function of t, the point is said to

accelerate and the acceleration is defined as $a = dv/dt = d^2x/dt^2$. Use the two definitions and calculate the velocity and the acceleration of a point of which the position is given by the following functions:

(a) $x = 6t - 5$ (b) $x = 5t^2 - 3t + 5$
(c) $x = 4t - 3t^3 - 2t^2 + 5$ (d) $x = 5\sin 3t$
(e) $x = 5\sin 3t + 5\cos 3t$

13. When the electric potential (the SI units are volts) is constant along a straight line, no electric field exists in the direction of the line. If the electric potential, V, depends upon the position along the line, an electric field exists. The intensity of this field, E, is defined by the equation $E = -dV/dx$ in which x is the position along the line. E is measured in volts per metre ($V\,m^{-1}$). Use the definition to answer the following questions:



(i) *(ii)* *(iii)*

(a) Figure (i) shows two flat parallel metal plates which carry electric charges of equal magnitude but opposite sign. At position P which is at distance x from plate A, the electric potential is given by $V = 2000\,x$ volts. Calculate the intensity of the electric field between the plates.

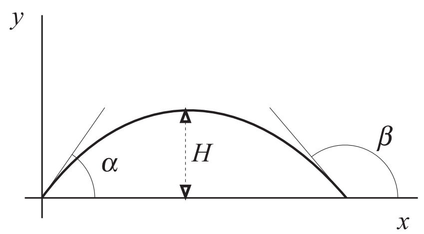
(b) At a distance of x metre from a single point charge, the electric potential is given by $V = 200\,x^{-1}$ volts. Calculate the intensity of the electric field at distance x from the charge.

(c) Figure (ii) shows two electric point charges of equal magnitude but opposite sign at position $x = a$ and $x = -a$ metre relative to point O. Point P is at position x ($x > a$) relative to O and the electric potential there is given by the function $V = 2kqa(x^2 - a^2)^{-1}$ volt. k, q and a are constants. Calculate the intensity of the electric field at P.

(d) Two identical electric point charges, $+q$, are on either side of the x-axis at $y = -a$ and $y = a$ metre respectively. The electric potential at a position P

on the x-axis, is given by $V = 2kq(a^2 + x^2)^{-1/2}$ volt. Calculate the intensity of the electric field at P. Investigate the possibility of local extreme values of E in the region $-5a \leq x \leq 5a$. If such values exist, determine their positions and nature.

14.



An object is projected over a horizontal field at an initial speed of 40 $\mathrm{m\,s^{-1}}$. It follows a trajectory which resembles the curve in the sketch. The horizontal position measured from O is indicated by x and the vertical height by y. The Cartesian equation of the trajectory is

$$y = -x^2/20 + x/\sqrt{3}$$

in which x and y are measured in metre. Use the equation for the trajectory to calculate the following: (a) The value of x for which the object reaches its maximum height. (b) At what angle, α, the object leaves the origin and at what angle, β, it strikes the ground.

15. At instant $t = 0$ s, 10^5 radioactive atomic nuclei exist in a sample of material. Since radioactive nuclei decay, there will be less at any later instant. The number of radioactive nuclei, N, which exist at time t, is given by the function $N = N(t) = 10^5 e^{-t/50}$.
(a) Calculate dN/dt, the rate at which the radioactive nuclei decay. Explain the negative sign of the answer.
(b) Show that dN/dt is directly proportional to N, the number of radioactive nuclei which still exist at a given instant.
(c) Calculate the time when $0,5 \times 10^5$ radioactive nuclei will exist in the sample. Also calculate the time when $0,25 \times 10^5$ will exist.

16. A burette of which the cross-sectional area, A, is constant, contains water to a height $h = h_0$. When the stopcock is opened the water flows out at a rate of $A\,dh/dt$. The height of the water in the tube is given by the function $h = h_0 e^{-\lambda t}$ metre in which the time, t, is measured in seconds and λ is a positive constant.
(a) Calculate the rate (volume per unit time) at which the water flows from the burette and explain the negative sign of the answer.
(b) Show that the flow rate is directly proportional to the height of the water in the burette.

17. One of the many surprising results of Einstein's general theory of relativity, is that the apparent mass of an object depends on its speed. The mass, m, of an object is given by the function $m = m_0(1 - v^2/c^2)^{-1/2}$ in which m_0 is a constant which is known as the rest mass of the object. The speed of the object is v and c is the constant speed of light in free space (vacuum). Calculate dm/dv, the

rate at which the apparent mass of an object changes as its speed increases.

18. A square has a side length of x metre. Write its area, A, as a function of x. Calculate the rate of change of the area if the side length increases. Use this result to calculate the change in area if the side length increases by a small amount Δx.

19. A popular model of rain gauge is conical with base radius R and depth (height) H. The dimensions are in metre. Water is poured into it to a depth of h above the apex and the circular surface has a radius of r. Write the volume, V, of the water in terms of h and r. Use the dimensions of the conical gauge to eliminate r from this expression. Use this result to write the height h of the water in the gauge as a function of its volume, V. Calculate the rate at which the height changes as the volume increases. Use this result to calculate the increase in the height if a small amount of water, ΔV, is added to an existing volume, V.

Chapter 3

INTEGRAL CALCULUS

3.1 The indefinite integral

When a function is known and it is required to calculate its derivative, it is done according to the methods which are explained in chapter 2. The name of the process is *differentiation* and the verb is *differentiate*.

Now the opposite process will be considered. The derivative is known and the function has to be calculated. The name of the process is **integration** and the verb is **integrate**.

If one is skilled in the technique of differentiation, then simple integration can be done forthwith as the following examples illustrate:

(1) $dy/dx = x^2$. Calculate $y = y(x)$.

The answer is obtained by using one's experience in differentiation and by answering the following question: Which function has to be differentiated to give an answer of x^2? The answer to this question is the function which has to be determined.

$$(d/dx)(\text{of which function of } x?) = x^2. \qquad \text{By inspection: } y = \frac{1}{3}x^3$$

The correctness of the answer may, and should always be tested by differentiating it.

$$(d/dx)(\frac{1}{3}x^3) = x^2. \quad \text{The integration is correct.}$$

From experience the reader should know that if *any* constant is added to $\frac{1}{3}x^3$, the derivative will still be x^2, thus

$$(d/dx)(\frac{1}{3}x^3 + k) = x^2$$

At this stage it seems that no better answer can be given than one in which an unknown constant k appears. For this reason it is called an **indefinite integral**.

Although this undetermined constant seems to have nuisance value only, its numerical determination which will be discussed in different sections of this book, plays a most important role in physics. For this reason it should **never** be omitted when giving the answer to an indefinite integral. It is called an **integration constant**.

(2) $dp/dt = t^3 - 3.$ Calculate $p = p(t)$.

(d/dt)(which function of t?) $= t^3 - 3$. By inspection: $p = \frac{1}{4}t^4 - 3t + k$

Test: $(d/dt)(\frac{1}{4}t^4 - 3t + k) = t^3 - 3$. The integration is correct.

(3) $(dv/dt) = \sin 3t.$ Calculate $v = v(t)$.

By inspection : $v = -\frac{1}{3}\cos 3t + c$, in which $c =$ constant.

The test for the correctness of this answer is left to the reader.

The process by which the answers to the preceding three problems were found, is known as **integration**. The term *anti-differentiation* would also have been valid and quite descriptive. A notation exists by means of which calculations of this kind are indicated. Using this notation, example (1) is written as follows:

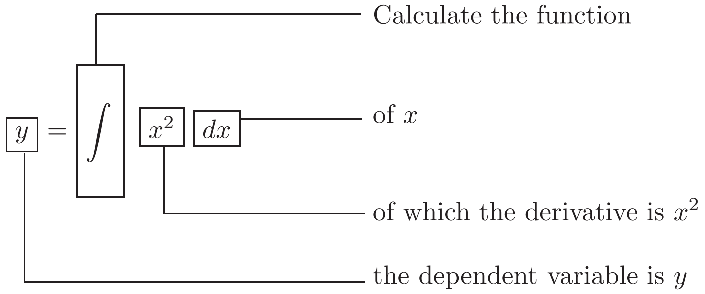
$$y = \int x^2 dx = \frac{1}{3}x^3 + k$$

The origin and actual meaning of this notation will become clear later in this chapter. At this stage it is useful to interpret it as an instruction to perform an integration as follows:

$\int$: This symbol is called an **integral sign** and may be taken to mean an instruction to calculate a function.

dx : The **differential** indicates that a function of x is to be calculated.

x^2 : Whatever is between the integral sign and the differential (in this case x^2), is known as the **integrand** and it is the derivative of the function which is to be calculated.



Examples:

(1) $\int (6x^2 - 8x + 5)\, dx = \frac{6}{3}x^3 - \frac{8}{2}x^2 + 5x + k = 2x^3 - 4x^2 + 5x + k$

(2) $\int (4\cos 2x)\, dx = \frac{4}{2}\sin 2x + c = 2\sin 2x + c$

(3) $\int (5\sin 2x + 3e^{2x})\, dx = -\frac{5}{2}\cos 2x + \frac{3}{2}e^{2x} + k = -2,5\cos 2x + 1,5e^{2x} + k$

(4) $\int (\sin^2 x\, dx) = \int \frac{1}{2}(1 - \cos 2x) dx = 0,5x - 0,25\sin 2x + k$

Comment: Note that when the integrand consists of more than one term, it is always written between brackets.

3.2 Integration formulas

Inspection of the differentiation formulas and rules, leads to the following integration formulas which are sufficient for this course. For constant numbers b,n

and k, the following are valid:

$$
\begin{array}{|lcl|}
\hline
\int bx^n\,dx & = & (b/[n+1])x^{n+1}+k, \quad n \neq -1 \\
\int \frac{1}{x}\,dx \;=\; \int x^{-1}dx & = & \ln|x|+k \\
\int (\sin bx)\,dx & = & -(1/b)\cos bx + k \\
\int (\cos bx)\,dx & = & (1/b)\sin bx + k \\
\int e^{bx}\,dx & = & (1/b)e^{bx}+k \\
\int b^{nx}\,dx & = & (1/n\ln b)b^{nx}+k \\
\int \frac{f'(x)}{f(x)}\,dx & = & \ln|f(x)|+k \\
\int b\,f(x)\,dx & = & b\int f(x)\,dx \\
\int [f(x)\pm g(x)]\,dx & = & \int f(x)\,dx \pm \int g(x)\,dx \\
\hline
\end{array}
$$

3.3 The integration constant

Consider the function $y = 2x+3$. If its gradient is to be calculated, the procedure is simple and gives an unambiguous answer: $dy/dx = 2$. If the problem is to calculate the function of which the gradient is 2, the solution is not quite as straightforward. It is known that the answer is given by $y = \int 2\,dx = 2x + k$

but the presence of the integration constant, k, makes the answer ambiguous.

The equation of a straight line is $y = mx+k$ in which m is its gradient and k the intercept on the ordinate axis. If the function is differentiated, the k disappears and confirms what is already known, namely that the intercept on the ordinate axis has no influence on the gradient. If one has to determine the function of which the gradient is 2, the best answer would be a **family** of straight lines, $y = 2x + k$ which all have a common gradient. If one member of the family is to be identified, supplementary information is needed. The co-ordinates of one position on a line will identify it unambiguously. If the co-ordinates of such a point are known, the value of k may be calculated.

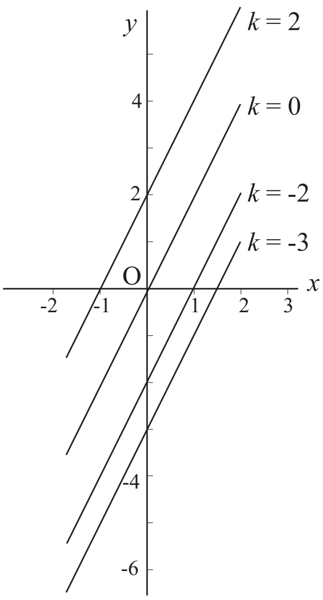


Figure 3.3-1

The graph in Figure 3.3-1 shows a number of the members of the family of straight lines of which the gradient is 2 and which are described by the equation $y = 2x + k$. If the line which contains the point (1, 4) is to be identified, these values are substituted into the equation as follows:

$$\begin{aligned} y &= 2x + k \\ 4 &= 2 + k \\ \text{so that} \quad k &= 2 \end{aligned}$$

The required equation is

$$y = 2x + 2$$

The same procedure may be followed for any other line of the family.

Example:

Calculate the function of which the derivative is $2x - 4$ and which contains the point (2, -1).

$$dy/dx = 2x - 4 \quad \Rightarrow \quad \begin{aligned} y &= \int (2x-4)dx \\ &= x^2 - 4x + k \end{aligned}$$

This answer represents a family of parabolas. It remains to calculate the intercept on the ordinate axis of the one which contains the given point.

$$y = x^2 - 4x + k$$

$$\begin{array}{rrcl} & -1 & = & 2^2 - 4(2) + k \\ \Rightarrow & k & = & 3 \\ \text{so that} & y & = & x^2 - 4x + 3 \end{array}$$

The graph of this function is shown in Figure 2.10-1.

3.4 The definite integral

In this section the reader is introduced to the concept of a definite integral. It will be defined and then a number of examples of calculations will be shown. The actual meaning and use of the concept will be treated in the following sections.

If the derivative of the function $y = y(x)$ is given by $dy/dx = g(x)$, the following is true:

$$\int g(x)\,dx = y(x) + k, \quad \text{in which } k \text{ is constant}$$

The **definite integral** of $g(x)$ between the **limits** $x = a$ and $x = b$ (in this order) is defined by the following:

$$\int_a^b g(x)\,dx = [y(x)]_a^b = y(x)|_a^b = y(b) - y(a) \qquad 3.4(1)$$

In this relationship, a is called the **lower limit of the integral** and b the **upper limit**. Their order is of prime importance. The answer cannot contain the independent variable since it is replaced by a and b when the answer is calculated in accordance with the definition. In the first example the integration constant will be introduced to show that it plays no role in the final answer and may thus always be omitted when a definite integral is calculated.

Examples:

(a)

$$\begin{aligned} \int_1^3 (4x-3)dx &= [2x^2 - 3x + k]_1^3 \\ &= [2(3)^2 - 3(3) + k] - [2(1)^2 - 3(1) + k] \\ &= [9 + k] - [-1 + k] \ = \ 10 \end{aligned}$$

This result shows very clearly that it is superfluous to make use of an integration constant when a definite integral is calculated.

(b)

$$\int_3^1 (4x-3)dx = [2x^2 - 3x]_3^1 = [-1] - [9] = -10$$

Examples (a) and (b) illustrate a theorem which will not be proved but of which the reader should be thoroughly aware. *If the upper and lower limits of a definite integral are interchanged, the sign of the answer changes.*

(c) $\displaystyle\int_0^{\pi/2} (\cos 2x)dx = [0,5\sin 2x]_0^{\pi/2} = [0,5\sin\pi] - [0,5\sin 0] = 0$

$$
\begin{aligned}
\text{(d)} \int_0^{\pi/2} (\sin x - 3\cos 3x)dx &= [-\cos x - \sin 3x]_0^{\pi/2} \\
&= [-\cos(\pi/2) - \sin(3\pi/2)] - [-\cos 0 - \sin 0] \\
&= [-0 - (-1)] - [-1 - 0] \\
&= 2
\end{aligned}
$$

3.5 The definite integral represented by an area

Consider the function $y = y(x)$ of which the derivative is given by $dy/dx = g(x)$ so that

$$\int g(x)\,dx = y(x) + k \qquad 3.5(1)$$

Figure 3.5-1 shows a graphic representation of a portion of the function $g = g(x)$. At position C the value of x is a, and at F, $x = b$. Consider an arbitrary value of x at D with corresponding value of the function of g. At E the x-value is $x + \Delta x$ and the corresponding value of the function, $g + \Delta g$.

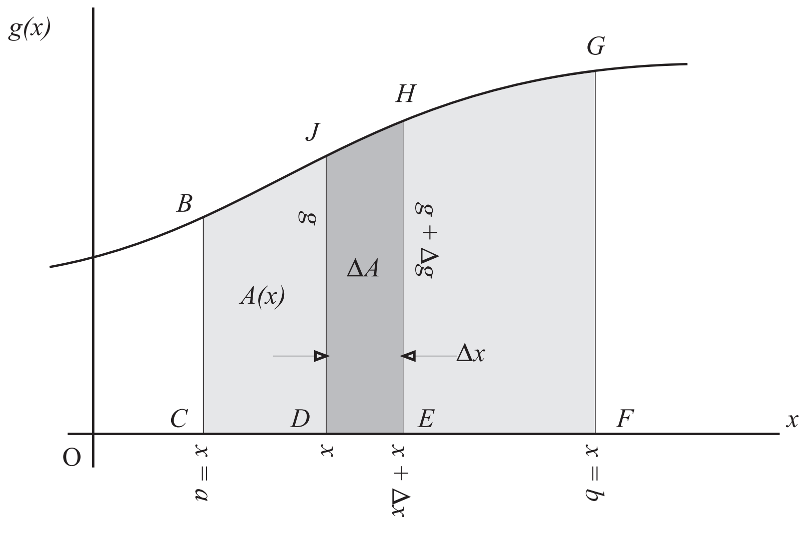


Figure 3.5-1

Let A be the area between the graph $g = g(x)$ and the x-axis from the fixed abscissa $x = a$ to the arbitrary abscissa $x = x$. In the sketch, A is the area of figure $BCDJ$. A is thus a function of x and when x increases by an amount Δx, A changes by an amount ΔA and it may be written that

$$\begin{aligned} A(x+\Delta x) &= A(x) + \Delta A \\ \text{and} \qquad \Delta A &= A(x+\Delta x) - A(x) \\ &= \text{area of } JDEH \end{aligned}$$

If Δx is small, figure $JDEH$ may be approximated by a trapesium of which the area is given by

$$\begin{aligned} \Delta A &= \frac{JD + HE}{2} \times DE = \frac{1}{2}(g + [g + \Delta g]) \times \Delta x \\ \text{so that} \quad \frac{\Delta A}{\Delta x} &= \frac{1}{2}(2g + \Delta g) \end{aligned}$$

If $\Delta x \to 0$, then $\Delta A \to 0$ and $\Delta g \to 0$ and

$$\lim_{\Delta x \to 0} \frac{\Delta A}{\Delta x} = \frac{dA}{dx} = g(x) \qquad 3.5(2)$$

From this it follows directly that

$$A = \int g(x)\,dx = y(x) + k \qquad \text{[from 3.5(1)]} \qquad 3.5(3)$$

in which k is a constant which may be calculated if the value of A is known for a given value of x. According to the way in which A was defined, $A = 0$ where $x = a$ (i.e. the area between $x = a$ and $x = a$). From Equation 3.5(3) it follows that

$$\begin{aligned} 0 &= y(a) + k \quad \Rightarrow \quad k = -y(a) \\ \text{and} \quad A &= y(x) - y(a) \end{aligned}$$

At $x = b$:

$$A = y(b) - y(a) = \int_a^b g(x)\,dx \qquad 3.5(4)$$

which is the area between the graph of $g = g(x)$ and the x-axis and which lies between $x = a$ and $x = b$, i.e. the area $BCFG$ in Figure 3.5-1.

In the study of physics this result is of great importance since the area under a given curve, often represents a physical quantity. The units of the area are determined by the units along the two axes. If, for example, g is measured in metres per second and x in seconds, the area represents metres. If g is measured in ampère and x in seconds, the area represents coulombs.

In the deduction of this result, the abscissa was represented by x and the ordinate by g. It should be noted that any symbols may be used. Similarly

$$\int_p^q y(t)\,dt$$

would represent the area between the graph of the function $y = y(t)$ and the t-axis from $t = p$ to $t = q$.

In Figure 3.5-1 a portion of the graph which lies above the abscissa-axis was considered. In the examples which follow, cases in which the graph lies below the axis will also be considered and an important extended interpretation of the result will have to be treated.

Examples:

(a) Calculate the area between the curve $y = t^2 - 4t + 5$ and the t-axis which lies between $t = 1$ and $t = 4$. y is the gas flow in a pipe in litres per second and t the time in seconds. The area is shown in Figure 3.5-2(a).

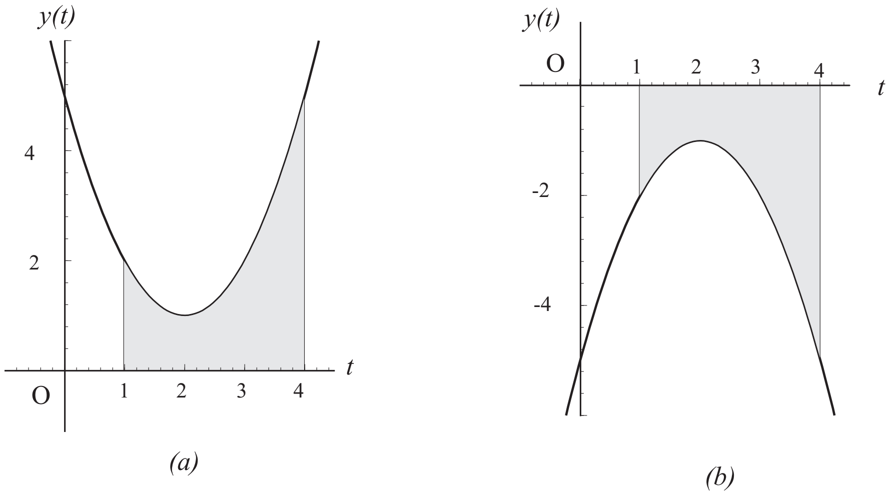


Figure 3.5-2

$$\begin{aligned}
A &= \int_1^4 (t^2 - 4t + 5)dt = \frac{1}{3}t^3 - 2t^2 + 5t|_1^4 \\
&= [\frac{1}{3}(4)^3 - 2(4)^2 + 5(4)] - [\frac{1}{3}(1)^3 - 2(1)^2 + 5(1)] \\
&= 9\frac{1}{3} - 3\frac{1}{3} \\
&= 6 \text{ litres}
\end{aligned}$$

(b) Calculate the area between the graph of the function $y = -t^2 + 4t - 5$ and the t-axis between $t = 1$ and $t = 4$. The graph is the mirror image about the t-axis of that in example (a). As in the previous example, y is the gas flow in a pipe in litres per second and t the time in seconds. The area is shown in Figure 3.5-2(b).

$$\begin{aligned} A &= \int_1^4 (-t^2 + 4t - 5)dt = -\frac{1}{3}t^3 + 2t^2 - 5t|_1^4 \\ &= -9\frac{1}{3} + 3\frac{1}{3} \\ &= -6 \text{ litres} \end{aligned}$$

The minus sign indicates that the area which was calculated, lies under the t-axis. If the actual *magnitude* of the area was to be calculated, the sign would have to be disregarded and the answer given as 6 litres. Negative area (as area) has no meaning. The physical interpretation of the negative answer of example (b) is that the flow was in the opposite sense.

(c) Calculate the area between the graph of $y = t^2 - 4t + 3$ and the t-axis between (i) $t = 0$ and $t = 1$, (ii) $t = 1$ and $t = 3$, (iii) $t = 0$ and $t = 3$. As in the previous questions, y is the gas flow in a pipe in litres per second and t the time in seconds. These areas are shown in Figure 3.5-3.

(i) $$\begin{aligned} A &= \int_0^1 (t^2 - 4t + 3)dt \\ &= \frac{1}{3}t^3 - 2t^2 + 3t|_0^1 \\ &= 1\frac{1}{3} \text{ litres} \end{aligned}$$

(ii) $$\begin{aligned} B &= \int_1^3 (t^2 - 4t + 3)dt \\ &= \frac{1}{3}t^3 - 2t^2 + 3t|_1^3 \\ &= -1\frac{1}{3} \text{ litres} \end{aligned}$$

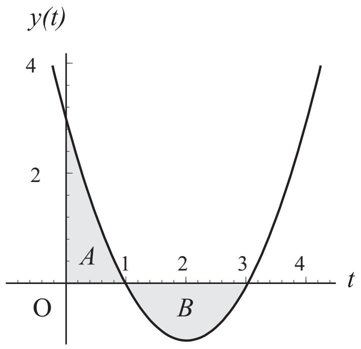


Figure 3.5-3

This answer is negative and indicates that the flow is in the opposite sense to that in the previous answer. The area representing the negative amount is under the t-axis.

(iii) The magnitude of the area between $y = y(t)$, the t-axis from $t = 0$ and $t = 3$ is $|A| + |B| = 1\frac{1}{3} + 1\frac{1}{3} = 2\frac{2}{3}$ square units. In this case it indicates that $2\frac{2}{3}$ litres of gas have flowed and the answer disregards the sense of the flow. If the integral is calculated between the given limits, it gives the following answer:

$$\int_0^3 (t^2 - 4t + 3)dt = \frac{1}{3}t^3 - 2t^2 + 3t|_0^3 = 0$$

The zero value indicates that the magnitudes of the areas above and below the t-axis are equal. The gas flow was the same in both ways, i.e. no net flow.

If the intersection points of a graph and the abscissa axis are not known and a definite integral which is calculated between given limits is positive, the answer should be interpreted as net area above the abscissa axis and when it is negative, as net area below the axis. A zero answer indicates equal areas above and below the axis.

3.6 The definite integral as the limit of a sum

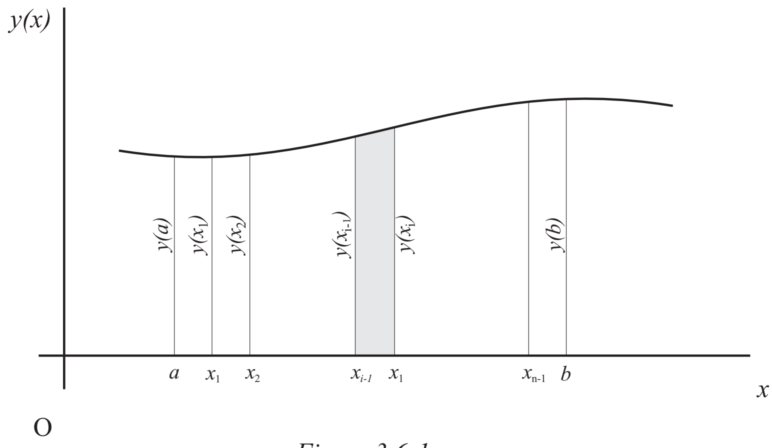


Figure 3.6-1

Figure 3.6-1 shows a portion of the graph of the function $y = y(x)$ between the limits $x = a$ and $x = b$. From 3.5(4) it is known that the area between the graph of the function and the x-axis from $x = a$ to $x = b$ is given by

$$A = \int_a^b y(x)\,dx \qquad 3.6(1)$$

The area may be calculated in a different way. The interval between $x = a$ and $x = b$ along the x-axis, is divided into n equal intervals, each with magnitude $\Delta x = (b - a)/n$. The lower limits of these intervals are the x-values a, x_1,

$x_2, \ldots x_{i-1}, x_i \ldots x_{n-1}$. According to the notation which is used, $x_0 = a$ and $x_n = b$. The total area is divided into n strips, which, if Δx is small enough (or n large enough), may each be approximated by a trapezium. Consider strip number i (which is limited by x_{i-1} and x_i). Its area is given by

$$\Delta A_i \approx \frac{y(x_{i-1}) + y(x_i)}{2} \times \Delta x \qquad 3.6(2)$$

The entire area between the limits $x = a$ and $x = b$ is given by

$$\begin{aligned} A &\approx \Delta A_1 + \Delta A_2 + \Delta A_3 \ldots \Delta A_{n-1} + \Delta A_n \\ &= \sum_{i=1}^{n} \Delta A_i \end{aligned} \qquad 3.6(3)$$

In this notation, the Greek letter Σ (sigma) represents the instruction to calculate the sum of the terms indicated by the index i which in consecutive terms assumes the values form 1 to n. From Equations 3.6(2) and 3.6(3) it follows that

$$A \approx \sum_{i=1}^{n} \frac{y(x_{i-1}) + y(x_i)}{2} \times \Delta x$$

This approximation may be replaced by an equality if $\Delta x \to 0$ (i.e. $n \to \infty$). If $\Delta x \to 0$, then $y(x_{i-1}) \to y(x_i)$ so that

$$\begin{aligned} A &= \lim_{\Delta x \to 0} \sum_{i=1}^{n} \frac{y(x_{i-1}) + y(x_i)}{2} \times \Delta x \\ &= \lim_{\Delta x \to 0} \sum_{i=1}^{n} y(x_i) \Delta x = \lim_{n \to \infty} \sum_{i=1}^{n} y(x_i) \Delta x \end{aligned}$$

To say that the index i assumes all the numerical values from 1 to n, is equivalent to the statement that x assumes all the values from a to b with intervals of Δx so that

$$A = \lim_{\Delta x \to 0} \sum_{x=a}^{b} y(x) \Delta x \qquad 3.6(4)$$

This is an exact answer for the calculation of the area under consideration. The fact that an infinite number of intervals of which the magnitude tends to zero are used, makes the practical calculation of the area by this method impossible. Another exact way of calculating the area by means of an integral is known. By equating the results of Equations 3.6(1) and 3.6(4), we have

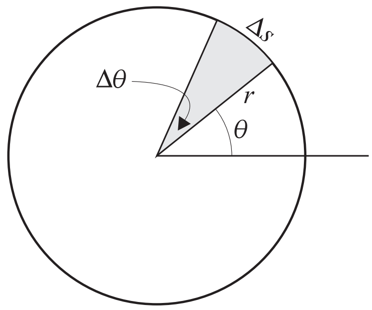
$$\boxed{\lim_{\Delta x \to 0} \sum_{x=a}^{b} y(x)\Delta x = \int_a^b y(x)\,dx} \qquad 3.6(5)$$

This result shows that a definite integral is the limit of a sum. This interpretation explains the origins of the word *integrate* and the integral sign which is a deformed letter S which was introduced by Wilhelm Leibniz. The interpretation also explains the necessity for writing the differential dx after the integrand.

Few techniques are used more often by physicists than the definite integral. It is a most powerful tool without which many results would be impossible, or at the very best, extremely difficult to calculate. It requires much practice before one is skilled in its use and the reader should constantly keep in mind that it will be one of his most important mathematical tools and that time spent to become familiar with it, is a good investment. A number of examples will illustrate its use.

Examples:

(a)



In this example an integral is used to derive the formula for the area of a circle of which the radius is r metre.

Consider a small sector formed by two radii which make an angle $\Delta\theta$. The angle subtends an arc Δs as shown in the sketch. From the definition of radian measure it follows that

$$\Delta s = r \times \Delta\theta$$

Since the angle is very small, the sector may be approximated by a triangle with base Δs and height r.

When approximated by a triangle, the area of the sector is given by

$$\Delta A \approx \frac{1}{2}(\text{base}) \times (\text{height}) = \frac{1}{2}\Delta s \times r = \frac{1}{2}(r\,\Delta\theta) \times (r) = \frac{1}{2}r^2\Delta\theta$$

The area of the circle is given by

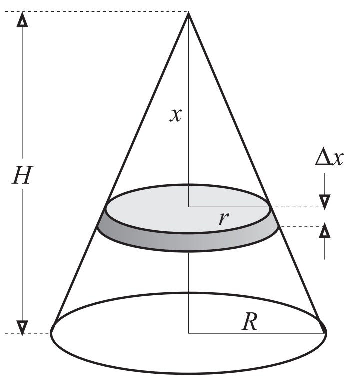
$$A = \lim_{\Delta\theta \to 0} \sum_{\theta=0}^{2\pi} \frac{1}{2}r^2\Delta\theta \quad = \quad \int_0^{2\pi} \frac{1}{2}r^2\,d\theta$$

$$= \frac{1}{2}r^2\theta|_0^{2\pi} = \frac{1}{2}r^2(2\pi) - 0 = \pi r^2$$

The way in which this result was calculated is rather clumsy. Furthermore, the "approximation" was not necessary since the result for the area of the small sector is exactly correct. A person who is skilled in the use of definite integrals would do the calculation without referring to a sum since he always thinks of a definite integral as a sum. The calculation will be repeated by making direct use of differentials.

Consider a sector which is formed by two radii of length r which make an angle $d\theta$. The length of the arc which subtends the angle is given by $ds = r\,d\theta$, and the area of the sector by $dA = \frac{1}{2}r \times r\,d\theta = \frac{1}{2}r^2 d\theta$. The integral remains to be calculated and the answer is the same as that of the previous calculation.

(b).



A definite integral will be used to calculate the formula for the volume of a right circular cone with base radius R and height H.

Consider a circular slice with thickness Δx at distance x from the apex of the cone and parallel to its base. From the geometry of a cone it follows that

$$r = (R/H)x$$

If Δx is small, the volume of the slice may be approximated by that of a cylinder as follows:

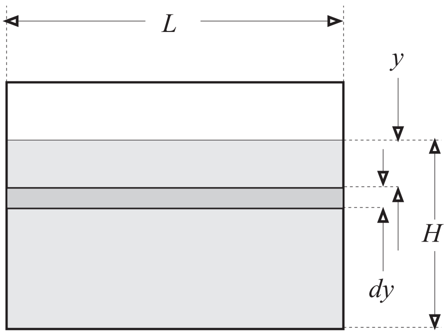
$$\Delta V \approx \pi r^2 \Delta x = \pi(R^2/H^2)x^2 \Delta x$$

The volume of the cone is given by

$$\begin{aligned} V = \lim_{\Delta x \to 0} \sum_{x=0}^{H} \pi(R^2/H^2)x^2 \Delta x &= \int_0^H \pi(R^2/H^2)x^2 dx \\ &= \frac{1}{3}\pi(R^2/H^2)x^3|_0^H \\ &= \frac{1}{3}\pi R^2 H \end{aligned}$$

As in the previous example, the calculation may be done by avoiding reference to a sum. In this problem the approximation was necessary since the slice is not a true cylinder. In the limit its difference from a cylinder is of no consequence whatsoever. Consider a circular slice of thickness dx at distance x from the apex of the cone and parallel to its base. Its volume is given by $dV = \pi r^2 dx = \pi(R^2/H^2)x^2 dx$. The volume of the cone follows by integration between the limits $x = 0$ and $x = H$.

(c)



The sketch shows a rectangular dam wall with horizontal width L which holds water of density ρ kg m^{-3} to a depth of H metres. The hydrostatic pressure of the water at depth y below the surface, is given by

$$p = \rho g y \ \text{Pa}\ (\text{N m}^{-2})$$

in which g = magnitude of gravitational acceleration. Calculate the total force exerted by the water on the dam wall as a result of the hydrostatic pressure.

Since the pressure varies with depth, it is not possible to calculate the force by multiplying the pressure by the area. If a narrow horizontal strip of area which spans the entire width of the wall is considered, this method of calculation would be in order since the variation of pressure over the height of the strip, may be neglected. The total force is equal to the sum (integral) of the forces on all the strips on which the water exerts pressure. The calculation is made by direct reference to differentials.

Consider a horizontal element of area of which the width is dy and which is at depth y below the surface. The area of the strip is $dA = L\,dy$, and the force on it is given by

$$(\text{pressure}) \times (\text{area}) = p\,dA = (\rho g y)(L\,dy)$$

The total force against the dam wall is given by the following definite integral:

$$\begin{aligned} F = \int_0^H \rho g L\, y\, dy &= \rho g L \int_0^H y\, dy \\ &= \rho g L(\frac{1}{2}y^2)\Big|_0^H = \frac{1}{2}\rho g L H^2 \end{aligned}$$

3.7 The calculation of integration constants

In section 3.3 it was shown that a known pair of co-ordinates is required to calculate an integration constant. This may be done in a more elegant way by using a definite integral as will be illustrated in the following examples. In each example the method used in 3.3 will be done first and then a definite integral will be used to perform the same task.

Examples:

(a) $dy/dx = 6x^2 - 4x + 5$. It is known that $y(1) = 2$. Calculate $y = y(x)$.

1. The method as used in section 3.3:

$$y = \int (6x^2 - 4x + 5)dx = 2x^3 - 2x^2 + 5x + k$$

   It is known that $y = 2$ if $x = 1$. Substitute these values in the above equation:

$$\begin{aligned} 2 &= 2(1)^3 - 2(1)^2 + 5(1) + k = 5 + k \quad \Rightarrow \quad k = -3 \\ \text{so that } \; y &= 2x^3 - 2x^2 + 5x - 3 \end{aligned}$$

2. By means of a definite integral:

$$\text{Since } \; dy/dx = 6x^2 - 4x + 5 \quad \Rightarrow \quad dy = (6x^2 - 4x + 5)dx$$

   Both sides are integrated and the following pairs of co-ordinates are used as limits: *Lower limits:* On the left side 2 (the value of y) and on the right side, 1 (the corresponding value of x). *Upper limits:* On the left side y and on the right side x.

$$\begin{aligned} \int_2^y dy &= \int_1^x (6x^2 - 4x + 5)dx \\ y|_2^y &= 2x^3 - 2x^2 + 5x|_1^x \\ y - 2 &= (2x^3 - 2x^2 + 5x) - (2 - 2 + 5) \\ \Rightarrow \quad y &= 2x^3 - 2x^2 + 5x - 3 \end{aligned}$$

Question: Would it be admissible to exchange the upper and lower limits on both sides? If the answer is not clear, test and see what happens.

(b) A body is limited to motion along a straight line. Its position relative to a fixed point on this line which is called the origin, is represented by x. x is positive for positions to the one side of the origin and negative for positions to the opposite side. The velocity of the body is defined as $v = dx/dt$. For the body under consideration the velocity is a function of time and is given by $v = dx/dt = -6 \sin 2t \text{ m s}^{-1}$ in which the time, t, is measured in seconds. If $t = 0$, $x = 0$ m (or $x(0) = 0$ m). Calculate the position of the body as a function of time, i.e. $x = x(t)$.

1. By the method used in section 3.3:

$$x = \int (-6\sin 2t)dt = 3\cos 2t + k$$

It is given that $x = 0$ when $t = 0$. Substitute these values in the above equation:

$$\begin{array}{rrcl} & 0 & = & 3\cos 0 + k = 3 + k \quad \Rightarrow \quad k = -3 \\ \text{so that} & x & = & 3\cos 2t - 3 \ \text{ metres} \end{array}$$

2. By means of a definite integral:

$$\text{Since} \quad dx/dt = -6\sin 2t \quad \Rightarrow \quad dx = (-6\sin 2t)dt$$

Integrate on both sides, using the following pairs of co-ordinates as limits: *Lower limits:* On the left side 0 (the value of x) and on the right side 0 (the corresponding value of t). *Upper limits:* On the left side x and on the right side, t.

$$\begin{aligned} \int_0^x dx &= \int_0^t (-6\sin 2t)dt \\ x|_0^x &= 3\cos 2t|_0^t \\ x - 0 &= 3\cos 2t - 3\cos 0 = 3\cos 2t - 3 \\ \Rightarrow \quad x &= 3\cos 2t - 3 \ \text{ metres} \end{aligned}$$

3.8 PROBLEMS: CHAPTER 3

1. Calculate $y = y(x)$ in each case if it is known that dy/dx is given by the following:

(a) $3x^2 - 6x + 5$
(b) $3x^4 + 2x^3 - 5x^2 + 7x$
(c) $7x^3 + 7x^{-3}$
(d) $4x^2 + 6x^{-2}$
(e) $16x^7 + 29x^{-9}$
(f) $4x^3 + 3x^2 + 2x + 1 - x^{-1} - x^{-2} - x^{-3}$
(g) $6e^{3x}$
(h) $4 \sin 2x$
(i) $6 \cos(3x)$
(j) $(12x^3 + 5)/(3x^4 + 5x)$

2. Calculate the following indefinite integrals:

(a) $\int (4x^2+2x-3)dx$
(b) $\int (8x^6-12x^3-5)dx$
(c) $\int (8s^3-3-4s^{-3})ds$
(d) $\int (20-10t)dt$
(e) $\int -18x\,dx$
(f) $\int -18\,dx$
(g) $\int (-18x^{-1})dx$
(h) $\int (v-gt)dt$ v and g are constant
(i) $\int \left(\int -10t\,dt\right) dt$
(j) $\int \left(\int -g\,dt\right) dt$ g is constant
(k) $\int 5\cos 4x dx$
(l) $\int 3\sin 2x dx$

2. Calculate the following definite integrals:

(a) $\int_2^5 3x\,dx$
(b) $\int_{0^\circ}^{90^\circ} \sin x dx$
(c) $\int_0^{\pi/2} \sin 2x\,dx$
(d) $\int_{-1}^{0} (1+3x-2x^2)dx$

(e) $\int_0^{\pi/6} \cos 3x dx$

(f) $\int_0^{\pi/2} (\cos\theta - \sin 2\theta) d\theta$

(g) $\int_1^4 \sqrt{x} dx$

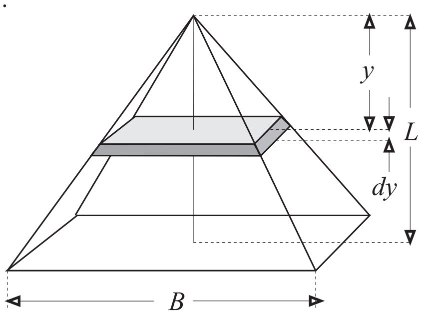
(h) $\int_0^{10} x^{-0,6} dx$

4. Calculate the area bounded by the t-axis and the graph of the function $v = 3 + 4t$, between $t = 1$ and $t = 5$. The units of v are metres per second. What are the units of the quantity which the area represents?

5. Make a rough sketch of the graph of the function $y = 4x^3 - 12x^2 + 8x$. Calculate the area between the graph of the function and the x-axis (i) from $x = 0$ to $x = 1$, (ii) $x = 1$ to $x = 2$. Use the two quantities to calculate the definite integral of the function between the limits $x = 0$ and $x = 2$.

6. Sketch the graphs of the functions $y = x^2$ and $y = 2x$ on one and the same set of axes. First calculate the co-ordinates of their points of intersection and then the area between the graphs.

7.

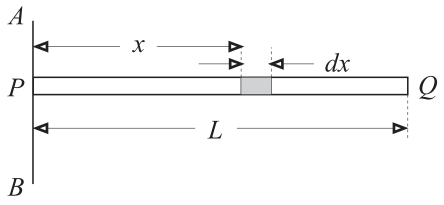


The sketch shows a right pyramid on a square base with side length B metres. The height of the pyramid is L metres. Consider a section parallel to the base at distance y from the apex and with infinitesimal thickness dy. First write down the volume, dV, of the section in terms of y, L, and B, only. (*Hint:* Let the side length of the square section be represented by b. First write dV in terms of b and dy. Then eliminate b by writing it in terms of y, L and B.) Use the expression for dV to calculate the volume of the pyramid.

8. When a gas of volume V cubic metres is at a pressure of p pascal (newton per square metre), work is done when the volume changes. The work done when the volume changes by an infinitesimal amount dV, is given by $dA = p\,dV$. A change in volume generally changes the pressure and the way in which it is changed depends on the way in which the volume is changed. An ideal gas of which the volume changes **isothermally** (the temperature remains constant), obeys Boyle's law: $pV = k$ in which k is constant. If the volume changes **adiabatically** (no heat crosses the border of the system), it obeys the law

$pV^\gamma = K$ in which K and γ are constant quantities. Calculate the work done when an ideal gas changes from the initial state p_1, V_1 to the final state p_2, V_2 during an (a) isothermal, (b) adiabatic change.

9.



The sketch shows a thin straight homogeneous rod with length L metres and mass M kilograms. The mass per unit length is given by $\mu = M/L\ \mathrm{kg\,m^{-1}}$. AB is an axis perpendicular to the rod, PQ, and dx is an infinitesimal element of length at position x from AB. The mass of this element is given by $dm = \mu dx = (M/L)dx$. The quantity $dI = x^2 dm$ is known as the **moment of inertia** of the element about the axis AB. (a) Calculate the moment of inertia of the entire rod about axis AB. (b) Calculate the the moment of inertia of the rod about an axis parallel to AB and through the centre of the rod.

Chapter 4

VECTOR ALGEBRA

4.1 Introductory discussion and definitions

Many quantities which are encountered in the study of physics are relatively simple in the sense that they can be described by a single number. Calculations involving such numbers are simple since they require the rules of ordinary arithmetic. Other quantities are more involved since they require the use of more than one number each. A physical quantity in which direction is involved, is an example of such a quantity. In this chapter an elegant way to describe such quantities will be developed and the rules which are required to use them in calculations, will be investigated. These rules are known as **vector algebra**. In this introductory discussion the necessary definitions will be given.

1. A **scalar** is a physical quantity which has magnitude but no direction. Examples: time, length, mass, temperature, etc. A scalar is usually represented by a Roman or Greek character as follows: x, y, a, A, D, α, β, etc. (never boldface). A scalar is specified by a single number.

2. Provisionally a vector may be defined as a physical quantity which has magnitude and direction. A more stringent definition which is required for advanced work, and which depends on its so-called **transformation properties**, falls outside the scope of this book. Examples of vector quantities are: force, displacement, acceleration, electric field intensity, etc. Unfortunately many different notations are used to indicate vectors. The following are found in textbooks: $\bar{H}$, $\vec{H}$, $\mathcal{H}$, $\underline{H}$, $\mathbf{H}$, $\vec{H}$. Greek letters are also used: $\bar{\mu}$, $\vec{\mu}$, etc. In most modern physics textbooks, boldface characters are used to indicate vectors. Such a notation is not suitable for

handwriting and typewriters and for this reason the first notation ($\bar{H}$, $\bar{a}$, etc.) will be used in this book.

3. A vector may be specified by its magnitude and direction relative to a suitable frame of reference. A vector can never be specified by a single number. In the following *two-dimensional* example, *two* numbers are necessary: A velocity of 20 $\mathrm{m\,s^{-1}}$ in the direction N 30°E. Although this way of specifying a vector will be used initially in this discussion, it is not always the best and another will be developed as the study progresses.

4. The **magnitude, modulus** or **norm** of vector $\bar{A}$, is indicated by $|\bar{A}|$. Often, when no possibility of confusion exists, the following notation is used for the modulus of a vector: $A = |\bar{A}|$. It is defined as a **positive number**.

5. A vector may be represented graphically by means of a directed line segment of which the length represents the magnitude of the vector according to a suitable scale. The end which terminates in an arrowhead is called the **terminal point** of the vector and the other point, the **initial point**.

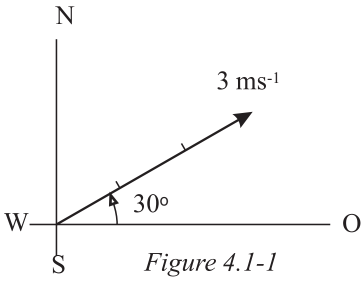


Figure 4.1-1

Figure 4.1-1 shows a graphical representation of a velocity of 3 $\mathrm{m\,s^{-1}}$ in the direction 30° north of east. The direction may also be indicated as 60° east of north, E 30°N or N 60°E. A navigator would specify the direction simply as 60°. According to this notation, north is taken as 0° (or 360°), and the angle is measured clockwise.

6. A **unit vector** is one of which the magnitude is unity. Unit vectors which do not have units are used to specify directions. Since they indicate direction only, they are dimensionless. The following notation will be used to indicate unit vectors: $\hat{A}$, $\hat{a}$, $\hat{\mu}$, $\hat{x}$, $\hat{y}$, etc. If a given vector is $\bar{A}$, a unit vector may be constructed which has the same direction as $\bar{A}$.

$$\hat{A} = \bar{A}/|\bar{A}| = \bar{A}/A \qquad 4.1(1)$$

Example:
$$\bar{A} = 20\ \mathrm{m\,s^{-1}},\ \text{north} \quad \Rightarrow \quad |\bar{A}| = 20\ \mathrm{m\,s^{-1}}$$
$$\text{and } \hat{A} = (20\ \mathrm{m\,s^{-1}},\ \text{north})/(20\ \mathrm{m\,s^{-1}}) = 1\ \text{north}$$

The way in which this unit vector is specified is rather awkward (in fact, fairly useless) and a more elegant way will be developed later in the chapter.

7. The **negative vector**, $-\bar{A}$ has the same magnitude as $\bar{A}$, but it is in the opposite direction. It is said that they are **anti-parallel**.

8. A vector $\bar{A}$ multiplied by a scalar quantity k, gives a new vector $k\bar{A}$ of which the magnitude is $|k|$ times the magnitude of $\bar{A}$ and with direction parallel or antiparallel to $\bar{A}$, depending on whether k is positive or negative. If any vector with finite magnitude is multiplied by zero, the answer is the **zero vector** which is written as $\bar{0}$. The zero vector does not have direction.

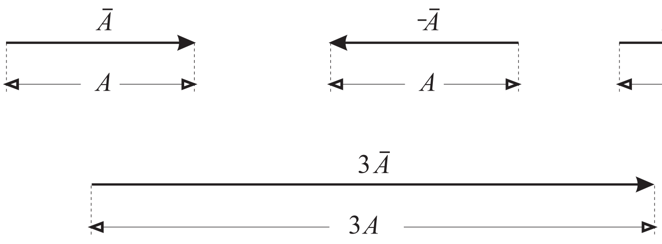


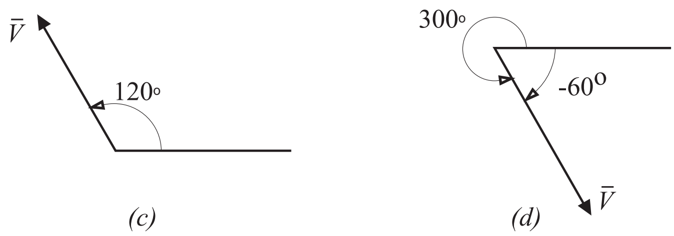
Figure 4.1-2

9. The **component of a vector in a given direction** is defined as the magnitude of the vector multiplied by the cosine of the angle between the direction of the vector and the given direction. For this purpose the smallest angle θ is used $(0 \leq \theta \leq \pi)$ and it is measured as follows: Measure the smallest angle from the given direction to the direction of the vector. If it is measured anticlockwise, it is taken to be positive and when it is measured clockwise, it is negative.[1] Since $\cos\theta = \cos(-\theta)$, the effect would be the same if positive angles only between 0 and 2π are used. It is useful to remember that $\cos\theta = -\cos(\pi - \theta)$.

   Other terms used for the component of a vector in a given direction, are the **projection** of the vector on the given direction and the **mapping** of the vector on the given direction.

Figure 4.1-3

[1]This method of measurement is introduced since it is used in calculators which are programmed for conversions from Cartesian co-ordinates to polar co-ordinates.



(c) *(d)*

Figure 4.1-3

The components of $\bar{V}$ in the given direction OX (in this order) for the four examples illustrated in Figure 4.1-3 are given by the following:

$$\begin{array}{lllll}
\text{(a)} & V\cos 60^\circ & = & 0,5V & \\
\text{(b)} & V\cos 90^\circ & = & 0 & \\
\text{(c)} & V\cos 120^\circ & = & -V\cos 60^\circ & = -0,5V \\
\text{(d)} & V\cos(-60^\circ) & = & V\cos 300^\circ & = \ \ 0,5V
\end{array}$$

The advantages of measuring the angles in the way explained above, become evident when the components of a vector are also to be calculated in a second direction, perpendicular to the first.

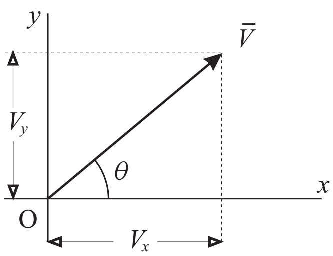


Figure 4.1-4

Consider two such directions which form a two-dimensional orthogonal frame of reference. The two straight lines which form the axes, are distinguished by naming them the x-axis and y-axis respectively. The component of V along the x-axis is called its x-component (indicated by V_x) and that along the y-axis, its y-component (indicated by V_y). The two components are calculated from V, the magnitude of the vector, and θ, the angle between $\bar{V}$ and the x-axis.

$$\begin{array}{lrcll}
\text{Now} & V_x & = & V\cos\theta & \\
\text{and} & V_y & = & V\cos(\pi/2-\theta) = V\sin\theta & 4.1(2) \\
\text{note that} & V_x^2+V_y^2 & = & V^2\cos^2\theta + V^2\sin^2\theta & \\
 & & = & V^2(\cos^2\theta+\sin^2\theta) = V^2 & \\
\text{so that} & V & = & |\bar{V}| = (V_x^2+V_y^2)^{\frac{1}{2}} & 4.1(3)
\end{array}$$

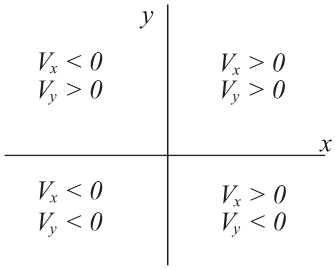


Figure 4.1-5

If the vector lies in the second quadrant, $\pi/2 \leq \theta \leq \pi$ and then $\cos\theta < 0$ and $\sin\theta > 0$ so that V_x is negative and V_y, positive.

For a vector in the third quadrant, $-\pi \leq \theta \leq -\pi/2$ so that $\cos\theta < 0$ and $\sin\theta < 0$ and both V_x and V_y are negative.

In the fourth quadrant, $-\pi/2 \leq \theta \leq 0$ so that $\cos\theta > 0$ and $\sin\theta < 0$. In this case $V_x > 0$ and $V_y < 0$. These results are summarised in Figure 4.1-5.

All calculators which are known as *scientific models* provide a function by means of which the x and y-components are obtained directly if the magnitude of the vector and the angle between the vector and the x-axis (known as the **azimuth angle**) are keyed in, in the prescribed manner. The inverse function is also available. It is of prime importance that the reader becomes skilled in the use of these functions. On some models the function is indicated by the notation $(r, \theta) \rightarrow (x, y)$ with inverse $(x, y) \rightarrow (r, \theta)$. Other models use the notations $P \rightarrow R$ and the inverse is $R \rightarrow P$. In the literature supplied with the calculator, (r, θ) and P are referred to as **polar notation** and (x, y) or R as the **rectangular notation**. If such functions are not available on a calculator, the calculations have to be made as illustrated in the example which follows.

Example: Consider a frame of reference in which the x-axis is to the east and the y-axis, north. At a given instant an aircraft has a speed of 200 km h^{-1} while flying horizontally. Calculate the x and y-components if the direction in which it flies is (a) N 40°E, (b) N 30°W, (c) W 15°S, (d) south, (e) S 30°E.

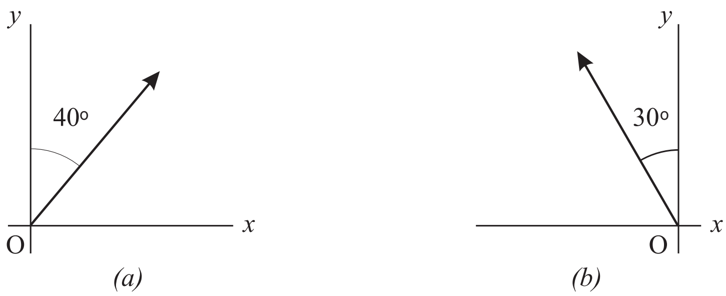


Figure 4.1-6

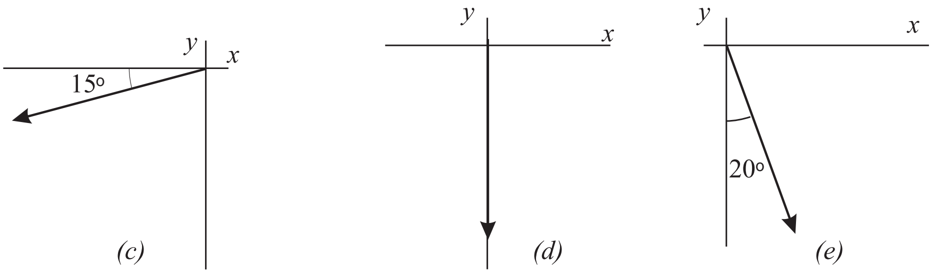


Figure 4.1-6

$$\begin{array}{lllllll}
\text{(a)} & V_x & = & 200\cos 50^\circ & = & 200(0{,}6428) & = 128{,}6\ \mathrm{m\,s^{-1}} \\
 & V_y & = & 200\sin 50^\circ & = & 200(0{,}7660) & = 153{,}2\ \mathrm{m\,s^{-1}} \\
\text{(b)} & V_x & = & 200\cos 120^\circ & = & 200(\text{-}0{,}5000) & = \text{-}100{,}0\ \mathrm{m\,s^{-1}} \\
 & V_y & = & 200\sin 120^\circ & = & 200(0{,}8660) & = 173{,}2\ \mathrm{m\,s^{-1}} \\
\text{(c)} & V_x & = & 200\cos(-165^\circ) & = & 200(\text{-}0{,}9659) & = \text{-}193{,}2\ \mathrm{m\,s^{-1}} \\
 & V_y & = & 200\sin(-165^\circ) & = & 200(\text{-}0{,}2588) & = \text{-}51{,}8\ \mathrm{m\,s^{-1}} \\
\text{(d)} & V_x & = & 200\cos(-90^\circ) & = & 200(0{,}0000) & = 0{,}0\ \mathrm{m\,s^{-1}} \\
 & V_y & = & 200\sin(-90^\circ) & = & 200(\text{-}1{,}0000) & = \text{-}200{,}0\ \mathrm{m\,s^{-1}} \\
\text{(e)} & V_x & = & 200\cos(-70^\circ) & = & 200(0{,}3420) & = 68{,}4\ \mathrm{m\,s^{-1}} \\
 & V_y & = & 200\sin(-70^\circ) & = & 200(\text{-}0{,}9396) & = \text{-}187{,}9\ \mathrm{m\,s^{-1}}
\end{array}$$

4.2 Vector addition

4.2.1 The vector parallelogram

The addition of two vectors is quite a different process from the addition of two scalar quantities and it involves a rather cumbersome process which is known as the vector parallelogram. The answer to the "addition" of two vectors is called their **resultant** and the following notation is used:

$$\bar{A} + \bar{B} = \bar{R} \qquad 4.2(1)$$

It should be noted that even if the equals sign is used, it does not necessarily mean that the two vectors $\bar{A}$ and $\bar{B}$ when representing physical quantities, will have the same effect on a system as the resultant $\bar{R}$.

Consider the two vectors $\bar{A}$ and $\bar{B}$ in Figure 4.2-1 of which the resultant $\bar{R}$ is

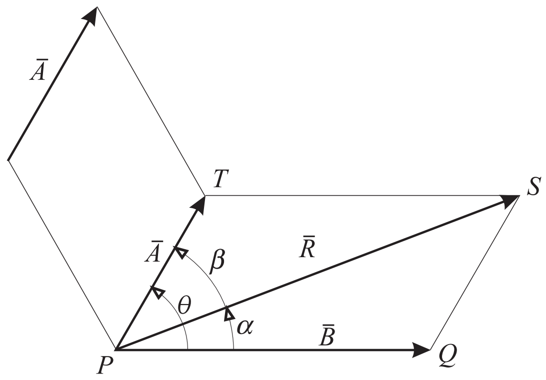


Figure 4.2-1

to be calculated. The process is as follows: One of the two vectors is given a parallel displacement so that their initial points coincide. With these two vectors as adjacent sides, a parallelogram is completed. The resultant, $\bar{R}$, is the vector represented by the diagonal from the common initial point of $\bar{A}$ and $\bar{B}$ to the opposite vertex of the parallelogram as shown in the sketch. The magnitude of $\bar{R}$ may be determined by accurate construction and measurement or it may by calculated by using the cosine rule and the sine rule as follows:

In $\Delta PQS (QS = PT = A)$ in Figure 4.2-1 we have:

$$\begin{aligned} R^2 &= A^2 + B^2 - 2AB\cos(\pi - \theta) = A^2 + B^2 - 2AB(-\cos\theta) \\ &= A^2 + B^2 + 2AB\cos\theta \\ \text{so that}\quad R &= (A^2 + B^2 + 2AB\cos\theta)^{\frac{1}{2}} \end{aligned} \qquad 4.2(2)$$

The direction of $\bar{R}$ can be determined by calculating either angle α or angle β. In ΔPQS

$$(\sin\alpha)/QS = (\sin[\pi - \theta])/PS \quad \Rightarrow \quad \sin\alpha = (QS\sin[\pi - \theta])/PS = (A\sin\theta)/R$$

$$\text{so that} \qquad \alpha = \arcsin([A\sin\theta]/R) \qquad 4.2(3)$$

From the preceding calculation it should be clear that the construction of the parallelogram is superfluous since ΔPQS could have been constructed directly by placing the initial point of $\bar{A}$ at the terminal point of $\bar{B}$. The vector represented by the side which completes the triangle, is the resultant and its direction is from the open initial point to the open terminal point. Some prefer to speak of the **triangle of vectors** instead of the parallelogram.

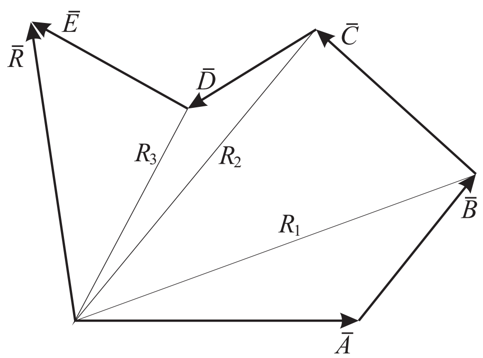


Figure 4.2-2

For the calculation of the resultant of more than two coplanar vectors, the resultant of any two may be added to a third and the new resultant to a fourth, etc. The method leads to a so-called **vector polygon** in which the vectors may be displaced in any order, parallel to their original directions to form a polygon. The terminal point of each vector must coincide with the initial point of the next. The side which closes the polygon, represents the resultant and its direction is the same as that for the triangle of vectors. This process is illustrated in Figure 4.2-2. The magnitude and direction may also be determined by construction and measurement. The use of the polygon of vectors cannot be recommended since it leads to cumbersome calculations in which calculation errors may easily occur.

For the definition of the difference between two vectors, the concept of a negative vector is used.

$$\bar{A} - \bar{B} = \bar{A} + (-\bar{B}) \qquad 4.2(4)$$

4.2.2 The method of mutually perpendicular components for the addition of coplanar vectors

This method is in agreement with the parallelogram method since it is derived from it. Although the proof will not be given in this book, the method will be described in detail.

The first step in this method is the parallel displacement of all the vectors so that their initial points coincide. The common initial point is used as the origin of a two-dimensional frame of reference. Provided they are mutually perpendicular, any coplanar orientation of the axes is admissible. For this calculation the axes will be called the x-axis and y-axis respectively. The method is illustrated by the example in Figure 4.2-3.

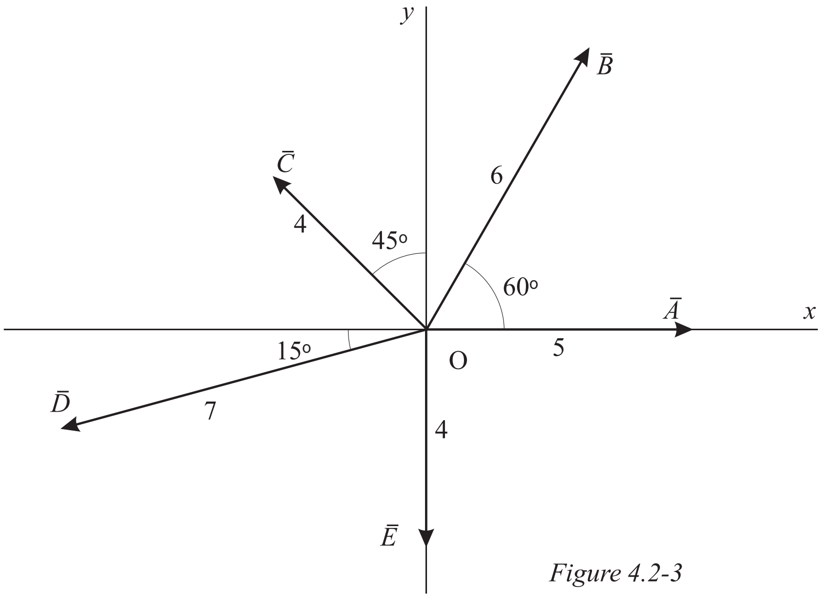


Figure 4.2-3

First the sum of all the components along the x-axis is calculated. This sum is represented by X. Then the sum of the components along the y-axis is calculated and it is denoted by Y. In the calculation which is shown below, we start with vector $\bar{A}$ and then proceed to the others in a counter-clockwise sense.

$$\begin{aligned}
X &= 5\cos 0^\circ \quad + 6\cos 60^\circ \quad + 4\cos 135^\circ \quad + 7\cos(-165^\circ) + 4\cos(-90^\circ)\\
&= 5(1{,}0000) + 6(0{,}5000) + 4(\text{-}0{,}7071) + 7(\text{-}0{,}9659) \quad + 4(0{,}0000)\\
&= 5{,}0000 \quad + 3{,}0000 \quad - 2{,}8284 \quad - 6{,}7614 \quad + 0{,}0000\\
&= -1{,}5898
\end{aligned}$$

$$\begin{aligned}
Y &= 5\sin 0^\circ \quad + 6\sin 60^\circ \quad + 4\sin 135^\circ \quad + 7\sin(-165^\circ) + 4\sin(-90^\circ)\\
&= 5(0{,}0000) + 6(0{,}8660) + 4(0{,}7071) + 7(\text{-}0{,}2588) \quad + 4(\text{-}1{,}0000)\\
&= 0{,}0000 \quad + 5{,}1961 \quad + 2{,}8284 \quad - 1{,}8117 \quad - 4{,}0000\\
&= 2{,}2128
\end{aligned}$$

The resultant of the vectors which are shown in Figure 4.2-4, is the vector $\bar{R}$ of which the x-component is $X = -1,5898$ and the y-component, $Y = 2,2128$. This result is shown in Figure 4.2-4.

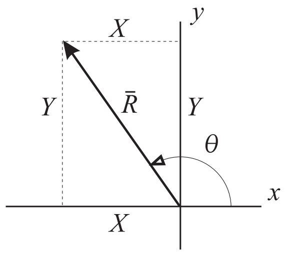


Figure 4.2-4

From this follows:

$$\begin{aligned} R &= (X^2 + Y^2)^{\frac{1}{2}} \\ &= ([-1,5898]^2 + [2,2128]^2)^{\frac{1}{2}} \\ &= (2,5274 + 4,8964)^{\frac{1}{2}} \\ &= (7,4238)^{\frac{1}{2}} \\ &= 2,7246 \end{aligned}$$

$$\begin{array}{lrcl} \text{Further} & \tan\theta & = & Y/X = (2,2128)/(-1,5898) = -1,3918 \\ \text{so that} & \theta & = & \arctan(-1,3918) \\ & & = & 125,7^\circ \end{array}$$

Now all the information about the resultant is known.

If a scientific calculator is available which provides the function for the transformation between polar and rectangular notations and also two memory positions in which the sums of the x and y components can be accumulated, the above calculation may be performed without writing down any intermediate steps. The procedure is fairly simple and it is a good investment to explore these facilities on a calculator.

4.2.3 The specification of a vector in terms of its components along two mutually perpendicular axes in two dimensions

The addition of vectors as illustrated in sections 4.2.1 and 4.2.2 requires much calculation. It is surely not as simple as the addition of scalar quantities. One of the reasons for this is that a vector is a more complicated concept than a scalar. A second, and probably more important reason, is that people often prefer to specify vectors in terms of their magnitudes and directions relative to a suitable frame of reference.

From section 4.2.2 it should be clear that a vector is *fully* specified if its components along two mutually perpendicular axes are known. Vector $\bar{B}$ in Figure 4.2-3 has a magnitude of 6 units and its azimuth angle is 60°. This information inevitably means that $B_x = 3,0000$ and $B_y = 5,1961$. Similarly in Figure 4.2-4, the information that $R_x = -1,5898$ and $R_y = 2,2128$ unambiguously specifies a vector $\bar{R}$ of which the magnitude is 2,7246 units and of which the azimuth angle is 125,7°.

If the problem shown in Figure 4.2-3 could be repeated but with each vector specified in terms of its components, it becomes very simple.

$$\begin{array}{llll}
\bar{A}: & A_x = 5,0000 & A_y = 0,0000 \\
\bar{B}: & B_x = 3,0000 & B_y = 5,1961 \\
\bar{C}: & C_x = -2,8284 & C_y = 2,8284 \\
\bar{D}: & D_x = -6,7614 & D_y = -1,8117 \\
\bar{E}: & E_x = 0,0000 & E_y = -4,0000
\end{array}$$

R_x is simply the sum of all the x-components and R_y the sum of all the y-components.

$$\bar{R}: \quad R_x = -1,5898 \quad R_y = 2,2128$$

which specifies $\bar{R}$ in full and is thus also a satisfactory answer to the problem.

By introducing this way of specifying a vector, a very cumbersome calculation is reduced to two simple additions. Many ways exist for the specification of a vector in terms of its components. Although it is seldom used, one way is that which is used above.

$$\bar{B}: \quad B_x = 3,0000 \quad B_y = 5,1961$$

In the so-called **matrix notation**, the vector is written as an ordered pair of numbers with the x-component first.

$$\bar{B}: \quad (3,0000; 5,1961)$$

Matrix notation has many advantages and is used extensively in the study of physics but it does have the problem that the frame of reference which is used, cannot be indicated. In introductory physics it is preferable to make use of **component vectors** along mutually perpendicular axes.

Consider a rectangular frame of reference. Let $\hat{x}$ be a unit vector along the x-axis and $\hat{y}$ a unit vector along the y-axis. These unit vectors are referred to as the **base vectors** of the frame of reference. For vector $\bar{V}$ in Figure 4.2-5, the two components along the axes are as follows:

$$V_x = V\cos\alpha \quad \text{and} \quad V_y = V\sin\alpha = V\cos\beta$$

The **component vectors** of vector $\bar{V}$ along the axes of the system are defined as follows:

$$V_x\hat{x} = (V\cos\alpha)\hat{x} \quad \text{and} \quad V_y\hat{y} = (V\cos\beta)\hat{y}$$

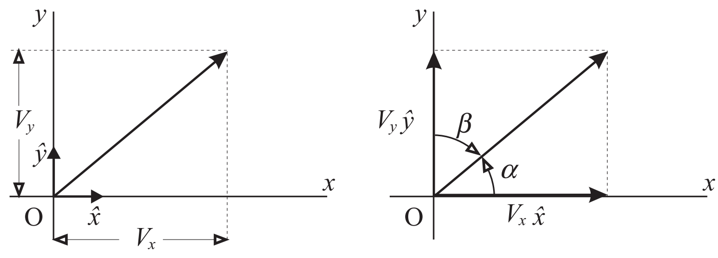


Figure 4.2-5

The component vector $V_x\hat{x}$ is in the direction of the positive x-axis (i.e. $\hat{x}$) with a magnitude of $V\cos\alpha$ and $V_y\hat{y}$ is in the direction $\hat{y}$ with magnitude $V\cos\beta$. According to the definition of vector addition, it may be written that

$$\bar{V} = V_x\hat{x} + V_y\hat{y} \qquad 4.2(5)$$

which is the notation to which preference is given in the study of introductory physics. The magnitude (modulus or norm) of $\bar{V}$ may be calculated from this as follows:

$$V = |\bar{V}| = (V_x^2 + V_y^2)^{\frac{1}{2}} \qquad 4.2(6)$$

The angles which it makes with the co-ordinate axes are

$$\begin{aligned} \alpha &= \arccos(V_x/V) \qquad (0 \le |\alpha| \le \pi) \\ \beta &= \arccos(V_y/V) \qquad (0 \le |\beta| \le \pi) \qquad 4.2(7) \end{aligned}$$

Comment: A component vector is a vector and should not be confused with the component of a vector in a given direction which is a scalar quantity.

4.2.4 The specification of a vector in terms of its component vectors along mutually perpendicular axes in three dimensions

The space in which we live requires three independent numbers to specify the position of a point unambiguously. For this reason we speak of three-dimensional space. It is impossible to solve some problems in physics in two-dimensional frames of reference such as those which we have used up to know.

For three-dimensional problems we make use of a frame of reference with three mutually perpendicular axes with base vectors $\hat{x}$, $\hat{y}$ and $\hat{z}$. Such a frame of reference is called a **three-dimensional Cartesian frame of reference** (*Renè*

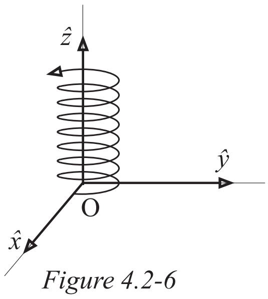


Figure 4.2-6

Descartes, 1596 – 1650, French mathematician, physicist and philosopher). In this book only **right-handed systems** will be treated. It is said that $\hat{x}$, $\hat{y}$ and $\hat{z}$ (in this order) form a right-handed system if the handle of a conventional corkscrew (right-handed screw) which is along the x-axis is rotated through 90° to coincide with $\hat{y}$ and the screw advances in the direction of $\hat{z}$. See Figure 4.2-6.

Consider vector $\bar{V}$ in such a frame of reference with base vectors $\hat{x}$, $\hat{y}$ and $\hat{z}$ along the axes Ox, Oy and Oz respectively as shown in Figure 4.2-7.

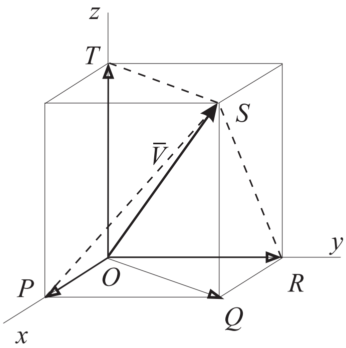


Figure 4.2-7

From the terminal point, S, of the vector, perpendiculars are dropped to the axes. The projections on the axes are the components of $\bar{V}$ in the directions of the base vectors. $V_x = OP$, $V_y = OR$ and $V_z = OT$. By multiplying each component by its corresponding base vector, the component vectors, $V_x\hat{x}$, $V_y\hat{y}$ and $V_z\hat{z}$ are obtained.

In rectangle $OPQR$ we have

$$V_x\hat{x} + V_y\hat{y} = \overline{OQ}$$

and in rectangle $OQST$,

$$\overline{OQ} + V_z\hat{z} = \bar{V}$$

$$\Rightarrow \quad \bar{V} = V_x\hat{x} + V_y\hat{y} + V_z\hat{z} \qquad 4.2(8)$$

which is in agreement with the notation by which a vector was represented in two dimensions as in Equation 4.2(5). To calculate the magnitude (modulus or norm) of this vector, first consider ΔORQ. Here angle $ORQ = 90°$, $RQ = OP = V_x$ and $OR = V_y$ so that

$$OQ^2 = V_x^2 + V_y^2$$

In ΔOQS, angle $OQS = 90°$ and $QS = OT = V_z$ so that

$$\begin{aligned} V^2 &= OQ^2 + V_z^2 = V_x^2 + V_y^2 + V_z^2 \\ \text{and} \quad V &= |\bar{V}| = (V_x^2 + V_y^2 + V_z^2)^{\frac{1}{2}} \end{aligned} \qquad 4.2(9)$$

Let α = angle POS = the angle between $\bar{V}$ and $\hat{x}$, then we have in ΔPOS (angle $OPS = 90°$)

$$\cos\alpha = OP/OS = V_x/V$$

In the same manner we have for β = angle ROS and γ = angle TOS

$$\begin{aligned} \cos\beta &= V_y/V \qquad \text{and} \qquad \cos\gamma = V_z/V \\ \text{so that} \qquad \alpha &= \arccos(V_x/V) \\ \beta &= \arccos(V_y/V) \\ \gamma &= \arccos(V_z/V) \end{aligned} \qquad 4.2(10)$$

The following three numbers

$$\cos\alpha = V_x/V, \quad \cos\beta = V_y/V, \quad \cos\gamma = V_z/V \qquad 4.2(11)$$

are called the **direction cosines** of vector $\bar{V}$. The direction cosines of a vector have the following interesting property:

$$\begin{aligned} (V_x/V)^2 + (V_y/V)^2 + (V_z/V)^2 &= (V_x^2 + V_y^2 + V_z^2)/V^2 = 1 \\ \text{so that} \quad \cos^2\alpha + \cos^2\beta + \cos^2\gamma &= 1 \end{aligned} \qquad 4.2(12)$$

If a vector $\bar{V} = V_x\hat{x} + V_y\hat{y} + V_z\hat{z}$ is given, it is simple to calculate a unit vector $\hat{V}$ which has the same direction as $\bar{V}$. From Equations 4.1(1) and 4.2(9) follows

$$\begin{aligned} \hat{V} &= \bar{V}/V = (V_x/V)\hat{x} + (V_y/V)\hat{y} + (V_z/V)\hat{z} && 4.2(13) \\ &= (\cos\alpha)\hat{x} + (\cos\beta)\hat{y} + (\cos\gamma)\hat{z} && 4.2(14) \end{aligned}$$

in which $\cos\alpha$, $\cos\beta$ and $\cos\gamma$ are the direction cosines of vector $\bar{V}$ in the given frame of reference.

4.2.5 The position vector

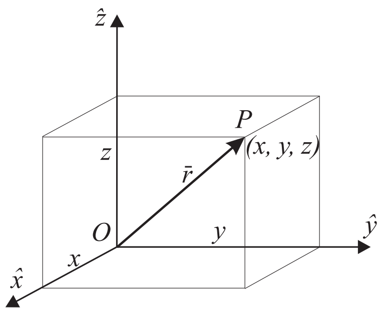


Figure 4.2-8

One of the most important and surely the most fundamental vector in the study of physics, is the **position vector** which specifies the position of a point relative to a given frame of reference. Consider a point P with coordinates (x, y, z). If the initial point of the vector $\bar{r} = x\hat{x} + y\hat{y} + z\hat{z}$ is placed at the origin of the frame of reference, its terminal point will be at the position of P.

This vector is the position vector of P and its magnitude is given by $r = (x^2 + y^2 + z^2)^{\frac{1}{2}}$. This is the distance from the origin to point P.

The unit vector in the direction of $\bar{r}$, is given by

$$\hat{r} = \bar{r}/r = (x\hat{x} + y\hat{y} + z\hat{z})/(x^2 + y^2 + z^2)^{\frac{1}{2}}$$

Examples:

(a)

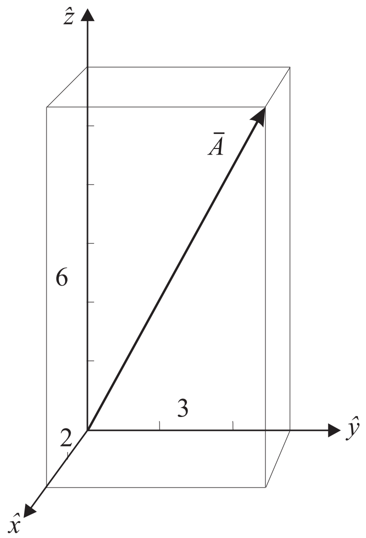


Figure 4.2-9

Consider the vector $\bar{A} = 2\hat{x} + 3\hat{y} + 6\hat{z}$. It is shown in Figure 4.2-9 in which its components are clearly visible. The following are to be calculated: The magnitude of $\bar{A}$, the unit vector $\hat{A}$, the direction cosines of $\bar{A}$ and the angles which it makes with the axes.

$$\begin{aligned} A &= (2^2 + 3^2 + 6^2)^{\frac{1}{2}} = (49)^{\frac{1}{2}} = 7 \\ \hat{A} &= \bar{A}/A \\ &= (2/7)\hat{x} + (3/7)\hat{y} + (6/7)\hat{z} \end{aligned}$$

The direction cosines are as follows:

$$\begin{aligned} \cos\alpha &= A_x/A = 2/7 \\ \cos\beta &= A_y/A = 3/7 \\ \cos\gamma &= A_z/A = 6/7 \end{aligned}$$

The angles which $\bar{A}$ makes with the base vectors are given by

$$\alpha = \arccos(2/7) = 73°24' \qquad \beta = \arccos(3/7) = 64°37' \qquad \gamma = \arccos(6/7) = 31°0'$$

(b) The calculations of the previous example are repeated for vector $\bar{B} = 4\hat{x} - 4\hat{y} - 7\hat{z}$. It is left to the reader as an exercise to sketch the vector in the same manner as that in example (a).

$$\begin{aligned} B &= (16 + 16 + 49)^{\frac{1}{2}} = (81)^{\frac{1}{2}} = 9 \\ \hat{B} &= \bar{B}/B \\ &= (4/9)\hat{x} - (4/9)\hat{y} - (7/9)\hat{z} \end{aligned}$$

The direction cosines are as follows:

$$\begin{aligned}\cos\alpha &= B_x/B = \ \ 4/9\\ \cos\beta &= B_y/B = -4/9\\ \cos\gamma &= B_z/B = -7/9\end{aligned}$$

The angles which it makes with the base vectors are as follows:

$$\begin{aligned}\alpha &= \arccos(4/9) \ \ = 63°37'\\ \beta &= \arccos(-4/9) = 116°23'\\ \gamma &= \arccos(-7/9) = 141°3'\end{aligned}$$

(c) The vectors $\bar{A}$ and $\bar{B}$ are given by $\bar{A} = 3\hat{x} - 2\hat{y} + 4\hat{z}$ and $\bar{B} = -2\hat{x} + 3\hat{y} - 5\hat{z}$ respectively. Calculate (i) $\bar{A} + \bar{B}$, (ii) $\bar{A} - \bar{B}$, (iii) $\bar{B} - \bar{A}$, (iv) $3\bar{A} + 2\bar{B}$, (v) $2\bar{A} - 3\bar{B}$.

$$\begin{aligned}\text{(i)}\quad \bar{A} + \bar{B} &= (3\hat{x} - 2\hat{y} + 4\hat{z}) + (-2\hat{x} + 3\hat{y} - 5\hat{z})\\ &= (3-2)\hat{x} + (-2+3)\hat{y} + (4-5)\hat{z}\\ &= \hat{x} + \hat{y} - \hat{z}\end{aligned}$$

$$\begin{aligned}\text{(ii)}\quad \bar{A} - \bar{B} &= (3\hat{x} - 2\hat{y} + 4\hat{z}) - (-2\hat{x} + 3\hat{y} - 5\hat{z})\\ &= 5\hat{x} - 5\hat{y} + 9\hat{z}\end{aligned}$$

$$\begin{aligned}\text{(iii)}\quad \bar{B} - \bar{A} &= (-2\hat{x} + 3\hat{y} - 5\hat{z}) - (3\hat{x} - 2\hat{y} + 4\hat{z})\\ &= -5\hat{x} + 5\hat{y} - 9\hat{z}\end{aligned}$$

$$\begin{aligned}\text{(iv)}\quad 3\bar{A} + 2\bar{B} &= 3(3\hat{x} - 2\hat{y} + 4\hat{z}) + 2(-2\hat{x} + 3\hat{y} - 5\hat{z})\\ &= (9\hat{x} - 6\hat{y} + 12\hat{z}) + (-4\hat{x} + 6\hat{y} - 10\hat{z})\\ &= 5\hat{x} + 2\hat{z}\end{aligned}$$

$$\begin{aligned}\text{(v)}\quad 2\bar{A} - 3\bar{B} &= 2(3\hat{x} - 2\hat{y} + 4\hat{z}) - 3(-2\hat{x} + 3\hat{y} - 5\hat{z})\\ &= (6\hat{x} - 4\hat{y} + 8\hat{z}) + (6\hat{x} - 9\hat{y} + 15\hat{z})\\ &= 12\hat{x} - 13\hat{y} + 23\hat{z}\end{aligned}$$

(d) Calculate the resultant of the following vectors and also calculate the magnitude of the resultant: $\bar{A} = 4\hat{x} - 3\hat{y} - 2\hat{z}$, $\bar{B} = \hat{x} - 2\hat{y} + 4\hat{z}$, $\bar{C} = -5\hat{x} + \hat{y} + 3\hat{z}$ and $\bar{D} = 2\hat{x} + 3\hat{y} - 5\hat{z}$.

$$\begin{aligned}\bar{R} &= \bar{A} + \bar{B} + \bar{C} + \bar{D} = 2\hat{x} - \hat{y}\\ R &= (4+1)^{\frac{1}{2}} \ = \ 2,236\end{aligned}$$

(e) The co-ordinates of point A are (2, 2, 2) and those of point B, (4, 5, 6). Calculate the distance AB by using the position vectors of the points.

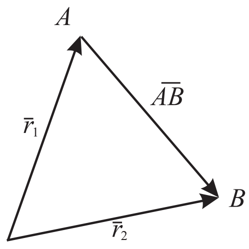


Figure 4.2-10

The position vector of point A is given by $\bar{r}_1 = 2\hat{x}+2\hat{y}+2\hat{z}$ and that of point B by $\bar{r}_2 = 4\hat{x}+5\hat{y}+6\hat{z}$. Both are measured in metres.

To calculate the required distance, the vector $\overline{AB}$ is used.

$$\begin{aligned}
\overline{AB} &= \bar{r}_2 - \bar{r}_1 \\
&= 2\hat{x} + 3\hat{y} + 4\hat{z} \text{ metres} \\
AB &= |\overline{AB}| \\
&= (4 + 9 + 16)^{\frac{1}{2}} \\
&= (29)^{\frac{1}{2}} = 5,385 \text{ metres}
\end{aligned}$$

4.3 The scalar product of two vectors

4.3.1 Definition and derivation

The notation by which the **scalar product** (also known as **dot product** or **inner product**) of vectors $\bar{A}$ and $\bar{B}$ is indicated is $\bar{A}\bullet\bar{B}$. It is read as A *dot* B. If the dot is omitted the result will not be a scalar product and will be devoid of meaning in this course. The scalar product is defined as follows:

$$\bar{A}\bullet\bar{B} = AB\cos\theta \qquad (0 \leq \theta \leq \pi) \qquad 4.3(1)$$

in which A and B are the magnitudes of the two vectors and θ the smallest angle between them. As the definition indicates, $\bar{A}\bullet\bar{B}$ is a scalar quantity.

$\bullet$	$\hat{x}$	$\hat{y}$	$\hat{z}$
$\hat{x}$	1	0	0
$\hat{y}$	0	1	0
$\hat{z}$	0	0	1

Table 4.3-1

From the definition follows:

$$\begin{aligned}
\hat{x}\bullet\hat{x} &= 1 \times 1 \times \cos 0^\circ = 1 \\
\hat{y}\bullet\hat{x} &= 1 \times 1 \times \cos 90^\circ = 0 \\
\hat{z}\bullet\hat{y} &= 1 \times 1 \times \cos 90^\circ = 0
\end{aligned}$$

In the same way the reader may explore all the possible combinations for the formation of scalar products between the base vectors $\hat{x}$, $\hat{y}$ and $\hat{z}$. The results are given in Table 4.3-1.

If the magnitudes of two vectors and the angle between them are known, Equation 4.3(1) may be used directly for the calculation of their scalar product. If, however, the two vectors are specified in terms of their components along the axes of a Cartesian frame of reference, that will not be possible. In such a case the information in Table 4.3-1 can be used to calculate the scalar product.

If $\bar{A} = A_x\hat{x} + A_y\hat{y} + A_z\hat{z}$ and $\bar{B} = B_x\hat{x} + B_y\hat{y} + B_z\hat{z}$, then

$$\begin{aligned}
\bar{A}\bullet\bar{B} &= (A_x\hat{x} + A_y\hat{y} + A_z\hat{z})\bullet(B_x\hat{x} + B_y\hat{y} + B_z\hat{z}) \\
&= A_xB_x\hat{x}\bullet\hat{x} + A_xB_y\hat{x}\bullet\hat{y} + A_xB_z\hat{x}\bullet\hat{z} \\
&\quad + A_yB_x\hat{y}\bullet\hat{x} + A_yB_y\hat{y}\bullet\hat{y} + A_yB_z\hat{y}\bullet\hat{z} \\
&\quad + A_zB_x\hat{z}\bullet\hat{x} + A_zB_y\hat{z}\bullet\hat{y} + A_zB_z\hat{z}\bullet\hat{z} \\
&= A_xB_x + 0 + 0 \\
&\quad + 0 + A_yB_y + 0 \\
&\quad + 0 + 0 + A_zB_z
\end{aligned}$$

This result supplies an algorithm (recipe) by which the scalar product of two vectors may be written down directly if they are known in terms of their component vectors.

$$\bar{A}\bullet\bar{B} = A_xB_x + A_yB_y + A_zB_z \qquad 4.3(2)$$

Example: Calculate the scalar product of $\bar{P} = -2\hat{x} - 3\hat{y} - 7\hat{z}$ and $\bar{Q} = 5\hat{x} - 4\hat{y} + 2\hat{z}$.

$$\begin{aligned}
\bar{P}\bullet\bar{Q} &= (-2\hat{x} - 3\hat{y} - 7\hat{z})\bullet(5\hat{x} - 4\hat{y} + 2\hat{z}) \\
&= -10 + 12 - 14 \\
&= -12
\end{aligned}$$

4.3.2 A number of useful applications of scalar products

If vectors $\bar{A}$ and $\bar{B}$ are known in terms of their component vectors, their scalar product may be used to calculate the angle between them.

$$\begin{aligned}
\text{Since} \quad \bar{A}\bullet\bar{B} &= AB\cos\theta \\
\Rightarrow \quad \cos\theta &= \bar{A}\bullet\bar{B}/AB && 4.3(3) \\
\text{and} \quad \theta &= \arccos(\bar{A}\bullet\bar{B}/AB) && 4.3(4)
\end{aligned}$$

From the scalar product the component of $\bar{A}$ in the direction of $\bar{B}$, i.e. $A\cos\theta$ (which is also known as the projection of $\bar{A}$ on $\bar{B}$ or the mapping of $\bar{A}$ on $\bar{B}$) may also be calculated.

$$\begin{array}{llcll} \text{Since} & \bar{A}\bullet\bar{B} & = & AB\cos\theta & \\ \Rightarrow & A\cos\theta & = & \bar{A}\bullet\bar{B}/B & 4.3(5) \\ & & = & \bar{A}\bullet\hat{B} & 4.3(6) \end{array}$$

In the same way the component of $\bar{B}$ in the direction of $\bar{A}$ may be calculated.

$$B\cos\theta = \bar{A}\bullet\bar{B}/A = \hat{A}\bullet\bar{B}$$

Examples: The following two vectors are known: $\bar{A} = 7\hat{x} - 4\hat{y} + 4\hat{z}$ and $\bar{B} = -3\hat{x} + 2\hat{y} + 6\hat{z}$.

(a) Calculate the angle between them.

$$\begin{array}{llcl} & \bar{A}\bullet\bar{B} & = & AB\cos\theta \\ \Rightarrow & \cos\theta & = & \dfrac{\bar{A}\bullet\bar{B}}{AB} = \dfrac{(7\hat{x} - 4\hat{y} + 4\hat{z})\bullet(-3\hat{x} + 2\hat{y} + 6\hat{z})}{(49 + 16 + 16)^{\frac{1}{2}}(9 + 4 + 36)^{\frac{1}{2}}} \\ & & = & \dfrac{-21 - 8 + 24}{(81)^{\frac{1}{2}}(49)^{\frac{1}{2}}} = \dfrac{-5}{63} \\ \text{and} & \theta & = & \arccos(-5/63) = 94°33' \end{array}$$

(b) Calculate the component of $\bar{A}$ in the direction of $\bar{B}$.

$$A\cos\theta = \bar{A}\bullet\bar{B}/B = -5/7 = -0,7143$$

(c) Calculate the projection of $\bar{B}$ on $\bar{A}$.

$$B\cos\theta = \bar{A}\bullet\bar{B}/A = -5/9 = -0,5556$$

That the last two answers are negative stems from the fact that the angle between the vectors is larger than 90°. In each case the projection has to be made on the extension of the vector in question.

The scalar product of two vectors provides a test to determine whether they are perpendicular to each other (**orthogonal**). If $\bar{A}\bullet\bar{B} = AB\cos\theta = 0$ and both $A \neq 0$ and $B \neq 0$, then $\cos\theta = 0$ which means that $\theta = 90°$.

Example:

$$\bar{P} = 3\hat{x} - 5\hat{y} + 4\hat{z} \quad \text{and} \quad \bar{Q} = 2\hat{x} + 2\hat{y} + \hat{z}$$

The magnitude of a vector can be zero only if each and every component is equal to zero. In this case neither vector has a magnitude of zero. Further we have that

$$\bar{P}\bullet\bar{Q} = (3\hat{x} - 5\hat{y} + 4\hat{z})\bullet(2\hat{x} + 2\hat{y} + \hat{z}) = 6 - 10 + 4 = 0$$